22 The Physiology and Pathophysiology of the Kidneys in Aging

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GENERAL OVERVIEW OF THE BIOLOGY OF AGING

All, or nearly all, biologic organisms exhibit the phenomenon of aging. The few exceptions are scientific curiosities.[1] Many speculations and hypotheses concerning the biologic basis of aging have been advanced, but as yet none have provided a universal explanation for the aging process.[2] The life span of humans is regarded as limited at some maximum term that few approach.[1–3] Aging is an inevitable consequence of life. The rate of aging varies considerably, even among identical members of the same species,[1] an indication of the relatively minor role of heredity per se in determination of the life span of species. It has been estimated, from studies of monozygotic twins, that only 20% to 35% of the life span can be attributed to heredity (chromosomal or mitochondrial).[4] Thus, most theories of aging consider heredity, the environment, and chance to play variable roles in determination of the observed life span.[1] Metabolic rates and the balance of energy demand and supply appear to have a strong influence on life span.[5]

The fundamental biologic pathways accounting for the diverse manifestations of aging have been the subject of intense investigation for many decades. Much progress has been made; however, many gaps in our knowledge of this process still remain,[6–31] including the process of renal aging. It is generally regarded that aging is a composite effect of disordered gene function, environmental influences, and chance.[1] At the cellular level, degeneration and faulty gene repair are core mechanisms of aging.[1] The net result is cellular and organ senescence. The rate of aging appears to be partially controlled by genetic pathways and biochemical processes. The explanation for the conservation of these processes that occur after maximal reproductive success is largely lacking.[1,3,4] As such, it is possible that many aspects of the aging process have not been evolutionarily conserved. Because of these factors, it seems likely that the biologic events responsible for aging may differ among species, complicating the study of aging enormously—studies of aging in experimental animals may therefore have limited relevance to human aging.

The major hallmarks of aging are genomic instability, epigenetic alterations, mitochondrial dysfunction, dysregulated nutrient sensing, telomere attrition, loss of protein homeostasis, stem cell exhaustion, accumulation of senescent cells, oxidation and glycation of tissue proteins, and altered intercellular communication (reviewed by Sturmlechner and coworkers[27] and López-Otín and associates[32]).

A detailed description of these processes underlying aging is beyond the scope of this brief introductory review. Age-dependent accumulation of stochastic damage to critical molecular pathways seems to be a dominant driver of aging.[1] The generation of ATP to provide energy to sustain cellular function is key for life, and this process is systematically altered in aging.[33]

Energy production, via mitochondrial dysfunction, regularly declines with age,[1,33] possibly due to stochastic damage to mitochondrial DNA and inefficient repair. Oxidative damage to DNA has been postulated to be one of the root causes of aging.[5,34] Mitochondrial inefficiency can lead to impaired cellular autophagy and cellular senescence.[35–38] Alternate hypotheses of defects in oxidative metabolism leading to a shift from aerobic to anaerobic metabolism are also possible.[5] A mismatch occurs in aging between lowered energy demand and excess supply. Caloric restriction has been found to be one of the most powerful experimental means of retarding the aging process.[32,39] Sirtuins, nicotinamide adenine dinucleotide (NAD^+)-dependent proteins, which deacetylate crucial enzymes in the oxidative metabolic pathway, may be responsible for this effect.[10,11,32,40,41] Sirtuin production diminishes with age and promotion of sirtuin production, as by the administration of mammalian target of rapamycin (mTOR) inhibitors, which mimic caloric restriction, can prolong life span.[32] This is known as the "bioenergetic theory of aging."[2] Whether a genetic program (the aging clock) defines the decline in bioenergetics with aging is unclear.[42] However, as stated above, it is well known that the rate of aging varies considerably, even among genetically identical humans and other species.[1] Epigenetic alterations and DNA methylation may account for some of these variations.[43]

Aging is also associated with shortening of the telomeres in nuclear DNA due to changes in the activity of telomerase, an enzyme essential for maintaining telomere length in dividing cells.[2,44,45] Gradual loss of telomeres leads to cessation of cell division after a maximum number of cell divisions, known as the *Hayflick limit*, and induces senescence, ultimately leading to cell death.[46] Attrition of telomeres with aging may explain in part the association of a higher risk for cancer in older people. Diseases of premature aging, such as progeria or Cockayne syndrome, can be caused by specific mutations (laminin in progeria), and are associated with marked telomere shortening or defects in DNA repair.[47,48] Indeed, telomere length and urinary 8-oxo-7,8-dihydroguanosine excretion are good biomarkers of aging.[49]

The individual organ systems of the human body exhibit phenomena connected to the overall aging process, also at variable rates. These organ-based manifestations of aging include loss of skin elasticity, decreases in hair pigmentation, slowing of nerve impulses, decreased mineral density of bone, decreased compliance of major vessels, decreased forced expiratory lung volume, decreased muscle mass, reduced metabolic rate, and many others. The kidneys share in these inevitable biologic consequences of aging, as will be detailed in this chapter. Numerous reviews and treatises covering the general topic of renal aging have been published over the past 4 decades. They represent a rich source of collateral reading on this fascinating subject.[6–31] It seems likely that the complex processes operating in aging at the organism level also play important roles in the manifestations of organ-based senescence, such as that displayed the kidneys.

The genetic component of renal aging has been analyzed by genome-wide association studies and transcriptomics,[50–52] and several candidate loci and factors have been identified. Increased oxidative or glycative stress,[53,54] reduced Klotho generation,[15,55–58] enhanced fibrosis,[59,60] increased capillary rarefaction,[9] and increased activation of the angiotensin II type 1 receptor[61–63] may all play important roles in renal aging. Klotho deficiency promotes inflammation.[55,56] Toxins such as D-serine[64] might be involved in the development of fibrosis in aging. Experimental evidence (in mice) has suggested that the level of fructokinase activity may be essential for renal aging.[65] In addition, endostatin and transglutaminase activity might also be involved in the renal fibrosis of aging according to studies in mice, but both these effects might be species specific.[66] The factors postulated to be involved in renal aging are summarized in Box 22.1.

Box 22.1 Some Factors Postulated to be Involved in Renal Aging

↓Sirtuin 1/6 (a histone deacetylase enzyme)
↓Klotho expression (and Wnt signaling)
↓Antioxidant production,↑oxidant activity
↓ Energy demand: ↑energy supply (mitochondrial dysfunction)
↑Telomere shortening
↑ DNA damage repair
↑ DNA methylation
↑Angiotensin II receptor signaling (via Wnt)
↑Cell cycle arrest (GI, via P16ink)
↓Autophagy
↑Fibrosis (transforming growth factor beta [TGF-β]–mediated)
↓Elimination of senescent cells
↓Insulin-like growth factor 1 (IGF-1) signaling
↓Proliferator-activated receptor-γ (PPARγ) activity
↑Capillary rarefaction
↑Podocyte apoptosis or detachment (podocytopenia—absolute and/or relative to capillary surface area)
↑Vascular sclerosis and glomerular ischemia
↑Advanced glycation end products
↑D-Serine toxicity
↑ Endostatin, transglutaminase activation

GI, Gastrointestinal.

HEALTHY AGING AND AGE-RELATED COMORBIDITY

The aging process not only leads to subtle and cumulative alterations in cellular and organ function, but it also predisposes an individual to certain age-related diseases, such as atherosclerosis, hypertension, diabetes, cancer, osteoporosis, and dementia. Disentanglement of these disease states from phenomena that might be called "healthy" aging can be very challenging. Healthy aging might be defined as the state that is universal, or nearly so, in all aging subjects, whereas age-related comorbidities affect only some of the aging population and involve processes that are disease-specific. For example, a decline in bone density or forced expiratory lung volume is characteristic of aging; defining a threshold for a disease-associated decline in these functions requires comparison with the expected changes with healthy aging. Type 2 diabetes prevalence increases with aging, and the aging process leads to beta cell exhaustion in the pancreatic islet cells. Therefore, aging per se predisposes some older subjects to develop overt diabetes.[67] The best examples of healthy aging are individuals selected for the donation of one kidney for renal transplantation, but even this healthiest of the healthy may not be entirely normal because they also may be mildly obese, have treated mild hypertension, or have covert diseases not easily identifiable, even in an exhaustive clinical, laboratory, and imaging-based pretransplantation evaluation. Studies of older apparently healthy adults in the community are inevitably an admixture of healthy aging and aging with comorbidity, sometimes clinically unrecognized.[68] In this context, proteomic analyses of low-molecular-weight proteins in urine obtained from supposedly healthy subjects have suggested resemblances between healthy aging and chronic kidney disease (CKD), but these findings might be the consequence of overlapping comorbidities (covert and overt) with aging per se.[69] In this chapter, we will use the healthy adult living kidney donor as the prototype for normal aging, recognizing some of the pitfalls in this assumption. A detailed discussion on the approach to management of the kidney disease in older adults can be found in Chapter 84.

DIFFERENCES BETWEEN HUMANS AND OTHER ANIMALS

Although many of the fundamental cellular processes of biologic aging are evolutionarily conserved, disparities at the organ system level may exist between species concerning the anatomic and functional consequences of aging. For example, many animal species (e.g., murine species) continue to grow throughout their life span, whereas humans cease growing after attaining maturity. The growth regulatory genetic program might be evolutionarily conserved among mammals, but the pace of growth is modulated in larger animals, including humans.[70] Metabolic demand may therefore differ among aging humans and aging experimental animals. Such differences can have profound effects on the organ-based manifestations of aging, including the kidneys. Thus, one needs to be cautious about inferring mechanisms of organ aging in humans from studies of experimental animals or lower organisms. The circumstances of birth, unique to humans, and in utero organogenesis can also have effects later in life, as will be discussed later.[71]

ANATOMIC CHANGES OF THE KIDNEYS WITH AGING

MACROSCOPIC CHANGES

As presently understood, aging is a universal and inevitable biologic process that is associated with macroscopic and microscopic structural changes in the kidneys, likely to be a causal association. Advances in the understanding of macroscopic structural changes have been made, first using ultrasound and more recently using computed tomography (CT) and magnetic resonance imaging (MRI).

KIDNEY SIZE AND VOLUME

Older studies that investigated kidney size with one- or two-dimensional measurements of kidney length or area have noted some degree of decrease with age.[72,73] Emamian and colleagues obtained ultrasound measurements of the kidney in 665 adult volunteers between 30 and 70 years of age and found that smaller kidney volumes correlated with older age.[74] Gourtsoyiannis and colleagues analyzed CT scans from 360 patients with no kidney disease to estimate kidney parenchymal thickness.[75] They found that for each decade of older age, kidney parenchymal thickness decreased about 10%. Another CT study in 1040 asymptomatic adults found that the factors associated with larger kidney size were male gender, taller height, and larger body mass index (BMI); whereas age and renal artery stenosis were associated with smaller kidney size.[76] Another study of 1056 patients showed similar findings and provided evidence that atherosclerosis accelerated the kidney size decline with older age.[77] However, a study of 225 adult healthy potential kidney donors did not find a statistically significant decrease in kidney size with older age, although the findings may have been limited by the age range and sample size of the study.[78]

A large MRI study assessed total kidney volume in 1852 subjects from the Framingham Heart Study.[79] The authors reported that the average decline in the total kidney volume in both genders was 16.3 cm^3/decade, with a stronger decline beyond 60 years of age, and a stronger decline in men than in women. In addition, using a subset of 196 healthy women and 112 apparently healthy men, this study determined the gender-specific upper and lower 10th percentile thresholds for the total kidney volume. In the multivariable models, the kidney volume above the 90th percentile was associated with younger age (odds ratio [OR], 0.67), whereas kidney volume below the 10th percentile was associated with older age (OR, 1.67).

KIDNEY CORTEX AND MEDULLA

Wang and colleagues have studied the CT scans of 1344 potential living kidney donors up to 75 years of age. They found a stable kidney parenchymal volume in those up to 50 years of age and a decline thereafter.[80] This study also obtained the volumes of the kidney cortex (average, 73% of parenchymal volume) and medulla (average, 27% of parenchymal volume) separately. An increasing medullary volume with age seems to attenuate some of the loss of total kidney parenchymal volume with age due to decreasing cortical volume (Fig. 22.1). Besides increased medullary volume, with age masking some of the kidney volume decline with age,

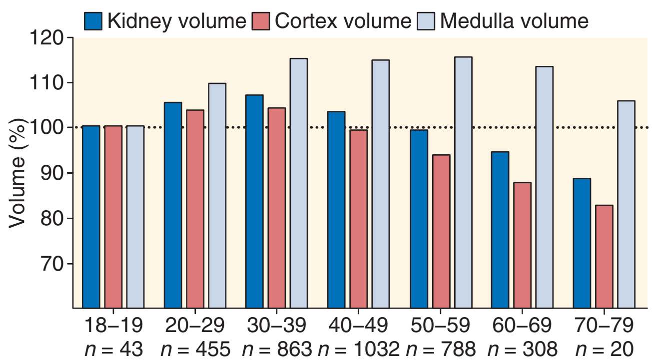

Fig. 22.1 Total kidney, cortical, and medullary volumes among 3509 potential kidney donors in the Aging Kidney Anatomy study (expansion of previously reported findings). Among all donors, older than 40 years, cortical volume decreases and medullary volume increases, making total kidney volume relatively constant until 50 years of age. Beyond that, medullary volume no longer increases, leading to a decrease in total kidney volume from the decreasing cortical volume. Findings were compared proportional to the respective volumes in the youngest age group (18–19 years). (From Wang X, Vrtiska TJ, Avula RT, et al. Age, kidney function, and risk factors associate differently with cortical and medullary volumes of the kidney. *Kidney Int.* 2014;85:677–685.)

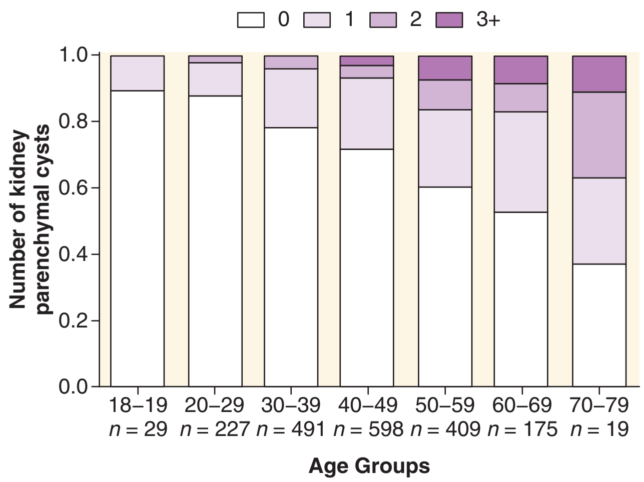

Fig. 22.2 Number of cortical and medullary simple cysts 5 mm or more by age among 1948 potential living kidney donors. In the figure fraction, *white* represents no cysts, whereas *dark purple* represents the fraction of three or more cysts. Intermediate fractions on a *light purple* scale represent the presence of one or two cysts.

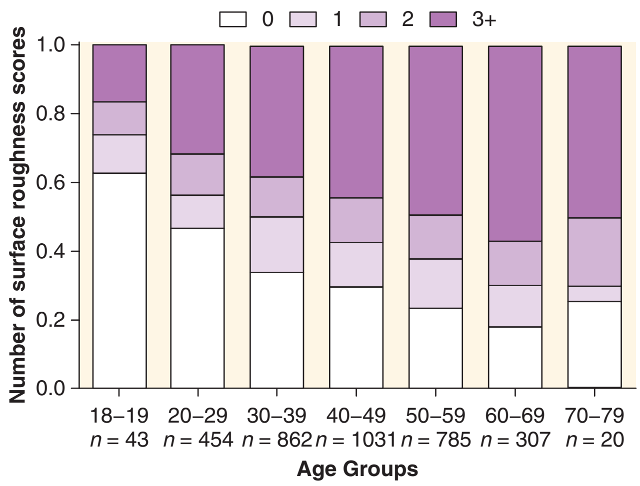

Fig. 22.3 Kidney surface roughness increasing with age among 3502 potential kidney donors in the Aging Kidney Anatomy study (expansion of previously reported findings). Based on the involved proportion of kidney surface with roughness (lumpy irregular rather than smooth surface), scores were given from 0 to 3 for each kidney and then averaged between kidneys. A score of 0 indicates no roughness; 1, roughness up to 25%; 2, roughness of 26% to 50%; and 3, roughness >50% of the kidney surface. (From Denic A, Alexander MP, Kaushik V, et al. Detection and clinical patterns of nephron hypertrophy and nephrosclerosis among apparently healthy adults. *Am J Kidney Dis.* 2016; 68:58–67.)

other studies have also suggested that an increase in renal sinus fat with age also masks some of the observed total kidney volume decline with older age.[74,75]

KIDNEY CYSTS

Kidney cysts are relatively common, of tubular diverticuli origin,[81] and their frequency and size increase with older age (Fig. 22.2).[82,83] There has been increased detection of cysts due to technologic advancements with imaging modalities, so it is important to consider cyst size thresholds when counting cysts.[82] A study of 1948 potential kidney donors demonstrated that even in this predominantly healthy population, cortical and medullary cysts larger than or equal to 5 mm were more frequent in older men and were associated with larger body surface area, albuminuria, and hypertension.[82] Moreover, this study generated upper reference limits (97.5th percentile) for the number of cysts in both genders by age. The upper reference limit for the number of cysts in both kidneys of 18 to 29-year-olds is one for both men and women, but increases to ten in men and four in women older than 60 years. Besides simple kidney cysts, parapelvic cysts, hyperdense cysts, angiomyolipomas, and cysts or tumors suspicious for cancer are also more frequent with older age.[82] Parapelvic cysts are thought to have a lymphatic origin[84] and increase with age, but are not associated with hypertension or albuminuria.

OTHER STRUCTURAL CHANGES

Other kidney parenchymal changes that become more prevalent with older age in ostensibly healthy adults include calcifications, focal cortical scars, fibromuscular dysplasia, and renal artery atherosclerosis without stenosis.[85] Of these, the two findings most strongly associated with age are atherosclerosis of renal arteries and focal cortical scarring. In donors younger than 30 years, the prevalence of atherosclerosis and focal scarring were 0.4% and 1.5%, respectively. However, in donors older than 60 years, the prevalence of atherosclerosis was nearly 25%, and the prevalence of focal scarring was 8%. Focal scarring also contributes to an increase in a kidney surface roughness with age (Fig. 22.3).[86]

MICROSCOPIC CHANGES

GLOMERULI

Number of Glomeruli

Several different approaches have been used to count the total number of glomeruli per kidney, with similar findings. On average, humans have 900,000 glomeruli/kidney, but with significant variability, ranging from as low as 200,000 up to 2.7 million glomeruli/kidney.[87,88] Lower glomerular number counts are likely the result of both the reduction in nephrogenesis stemming from intrauterine events and the progressive loss of glomeruli with age. Counts of the glomerular number have been obtained from autopsy studies, where complete sectioning of the kidney can be performed.[89] However, many autopsy studies have not distinguished normal nonsclerotic glomeruli from globally sclerotic glomeruli in their counts. Furthermore, the comorbid state in autopsy studies (often unknown, with sudden unexpected deaths) may increase global glomerulosclerosis beyond that expected from aging alone.

Living kidney donors provide a unique opportunity to determine how the glomerular number relates to aging. A study in 1638 living kidney donors who were carefully screened to confirm health estimated numbers of nonsclerotic and globally sclerotic glomeruli using the predonation CT scan and the implantation biopsy.[88] The youngest kidney donors (18–29 years of age) had a mean number of nonsclerotic glomeruli of about 990,000 and 17,000 globally sclerotic glomeruli/kidney. The oldest kidney donors (70–75 years) had about 520,000 nonsclerotic glomeruli and 142,000 globally

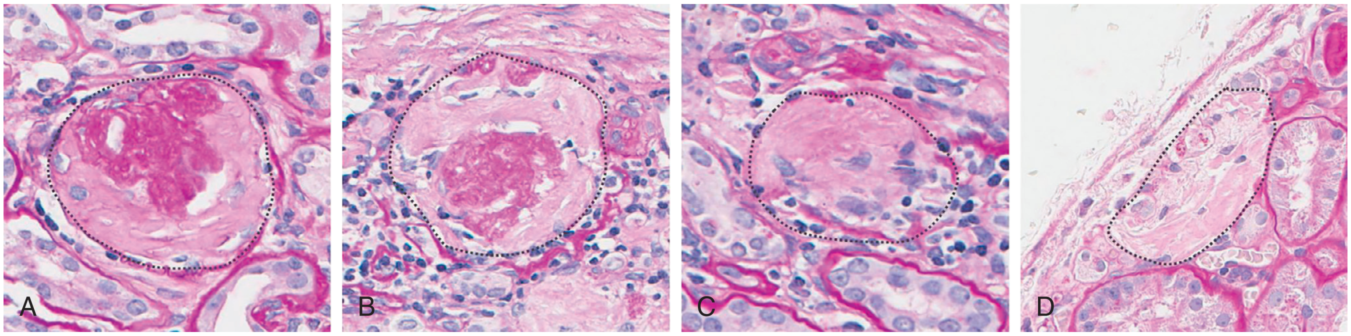

Fig. 22.4 Examples of globally sclerosed glomeruli (GSG) in different stages of their involution. (A, B) GSG are easily discernible in the biopsy sample. (C, D) GSG that may be overlooked because they progressively atrophy and lose a discernible capsule. All GSG are traced with a *black dashed line.*

sclerotic glomeruli per kidney. The 48% loss in nonsclerotic glomeruli, with only a 15% increase in globally sclerotic glomeruli numbers, supports the hypothesis that globally sclerotic glomeruli are eventually completely reabsorbed or significantly atrophied so that they can no longer be clearly detected on tissue sections examined by light microscopy (Fig. 22.4). This is in agreement with an older study by Hayman and associates, who postulated that "scars of destroyed glomeruli disappear without leaving recognizable traces."[90] The concept of missing or reabsorbed glomeruli is very important because a regular pathology report of the percentage of glomerulosclerosis on a renal biopsy may substantially underappreciate the true age-related loss of glomeruli.

Clinical Relevance 1

The reabsorption of globally sclerosed glomeruli with age may lead to an underappreciation of the older kidneys according to renal biopsy assessment.

Despite significant methodologic differences, studies have reported strikingly similar rates of glomerular loss per kidney per year—6200 in living kidney donors[88] and 6800 in autopsy studies.[89]

Table 22.1 Expected Number of Globally Sclerotic Glomeruli per Biopsy Section[a]

Age Group (years)	Total Number of Glomeruli Found per Biopsy Section							
	1	2	3–4	5–8	9–16	17–32	33–48	49–64
18–29	0.5	0.5	0.5	0.5	1	1	1	1
30–34	0.5	0.5	0.5	0.5	1	1	1	1.5
35–39	0.5	0.5	0.5	0.5	1	1.5	2	2
40–44	0.5	0.5	0.5	1	1	2	2.5	3
45–49	0.5	0.5	1	1	1.5	2	3	4
50–54	1	1	1	1.5	2	3	5	5
55–59	1	1	1.5	1.5	2	3.5	4.5	6
60–64	1	1.5	1.5	2	2.5	4	5.5	7
65–69	1	2	2	2.5	3	4.5	6.5	8
70–74	1	2	2.5	3	4	5.5	7.5	9
75–77	1	2	2.5	3	4	6	8	9.5

[a]As predicted by the total number of glomeruli and age from the Aging Kidney Anatomy study.

Adapted from Kremers WK, Denic A, Lieske JC, et al. Distinguishing age-related from disease-related glomerulosclerosis on kidney biopsy: the Aging Kidney Anatomy study. *Nephrol Dial Transplant.* 2015;30:2034–2039.

Glomerulosclerosis

An increasing percentage of global glomerulosclerosis is a feature of an aging kidney that has been demonstrated in autopsy and living kidney donor studies.[89,91,92] An autopsy study of 58 kidneys (48 from adults) from a mixed population of Australian Aborigines and non-Aborigines, US Caucasians, and African-Americans has shown a range of 0% to 23% with global glomerulosclerosis and a strong association with older age.[89] A study of 1203 living kidney donors has confirmed these findings and showed an increasing prevalence of focal and global glomerulosclerosis (FGGS), but not focal and segmental glomerulosclerosis (FSGS) with older age.[91] Whereas the prevalence of any global glomerulosclerosis was only 19% in the youngest kidney donors (age 18–29 years), the prevalence was 82% among the 11 oldest donors (70–77 years of age). In a study of 1847 of 2052 living kidney donors who were normotensive across three centers, the upper 95th percentile reference limit of glomerulosclerosis expected for age was determined (Table 22.1).[93] For example, for a 25-year-old who has a biopsy with 24 glomeruli, up to one globally sclerotic glomeruli on biopsy would be reasonable before suspecting glomerulosclerosis from disease. However, for a 76-year-old who has a biopsy with 24 glomeruli, up to six globally sclerotic glomeruli on biopsy would be reasonable before suspecting glomerulosclerosis from disease. In this study, among the 5% of donors in whom the number of globally sclerotic glomeruli was higher than the 95th percentile expected for age, a greater prevalence of hypertension and interstitial fibrosis and a higher percentage of ischemic-appearing glomeruli (pericapsular fibrosis, capsular thickening, and capillary loop wrinkling) on a biopsy were observed.[93] The association between the abnormal number of globally sclerotic glomeruli for age and ischemic glomeruli remained, even after hypertensive donors were excluded. Interestingly, use of these age-specific reference limits for globally sclerotic glomeruli helps identify which patients with nephrotic syndrome will develop progressive CKD.[94]

Certain morphologic findings of globally sclerotic glomeruli should also be taken into consideration; for example, solidification forms of glomerulosclerosis are always pathologic and are never simply an age-related finding.[95–97]

> ***Clinical Relevance 2***
> Glomerulosclerosis associated with aging is focal and global, in contrast to focal, segmental, and solidification forms of glomerulosclerosis occurring with disease.

The obsolescent type of morphology of global glomerulosclerosis is characterized by the filling of Bowman's space with collagenous material, accompanied by retraction of the capillary tuft. This is the type of global glomerulosclerosis obsereved in normal aging kidneys. Alternatively, the solidified type of glomerulosclerois is assocated with and commonly seen in hypertensive nephrosclerosis found in black people, with or without high-risk alleles on the APOL1 locus.[98] The obsolescent-type global glomerulosclerosis is presumed to reflect an ischemic process. FSGS is not associated with healthy aging in humans, unlike murine species. The lesion of FSGS in growing rats is likely to be mediated by podocyte injury and detachment. FSGS lesions in humans, therefore almost always indicates an underlying pathologic disease process and not simple renal senescence.[95–97]The possession of two risk alleles for APOL1 appears to enhance age-related global glomerulosclerosis in individuals of (West) African ancestry.[99,100] This phenomenon might help explain in part the higher risk of black people for the development of hypertension and GFR decline at an earlier age among black people compared with white people.[101]

Glomerular Hypertrophy

Entire nephrons hypertrophy in diabetes and obesity,[102–105] conditions that often become more prevalent with older age. Nephron hypertrophy is also seen in subjects with low nephron endowment at birth.[106] Although hypertrophy occurs in the glomerular and tubular segments of a nephron unit,[107] glomerulomegaly is much easier to appreciate on visual inspection of a kidney biopsy than tubular enlargement. The relationship between glomerular size and older age needs to account for the underlying study population (e.g., age-related comorbidites such as diabetes and obesity or nephron endowement at birth) and whether glomerular size was estimated from only nonsclerotic glomeruli or both nonsclerosed and the smaller globally sclerosed glomeruli that become more common with age.

Many studies have reported conflicting results, likely related to these issues, with some showing an increase in glomerular size with age[108–110] and others showing a decrease in glomerular size with age.[111,112] Studies that were limited to carefully screened healthy living kidney donors and only studied nonsclerosed glomeruli have found no change in glomerular size with aging.[86,88,113] Nonsclerosed glomerular volume in patients with renal tumors who underwent radical nephrectomy was stable until 75 years of age, similar to findings in living kidney donors.[114] However, beyond 75 years of age, nonsclerosed glomerular volume decreased due to a higher proportion of smaller, ischemic-appearing glomeruli. Given this lack of increase in glomerular volume, albuminuria would not be expected with aging alone in humans because albuminuria occurs with glomerular hypertrophy via a disorganized glomerular structure that is unable to prevent protein leaking efficiently.[110] Glomerular enlargement is also observed in blacks with two risk alleles at the APOL1 locus, which rather suggests low nephron endowment (see later) and/or accelerated loss of nephrons with aging.[99–101] Increased glomerular volume (glomerular hypertrophy) and glomerulosclerosis in aging subjects is predicted by comorbidity, such as obesity, diabetes, hypertension, and proteinuria. Such hypertrophy is also associated with male gender, taller height, and a family history of end-stage kidney disease (ESKD).[105,113]

TUBULES

Tubular Hypertrophy

Notably, glomeruli only comprise about 4% of the total cortical volume.[88] The remaining parenchymal volume is largely comprised of renal tubules and tubular enlargement.[107–115] Although glomerulomegaly most likely does not occur with healthy aging (or in the absence of APOL1 risk alleles in individuals of African origin), in humans tubular enlargement with older age is evident with increased area of tubular profiles.[86] Increase in tubular size as well as atrophy and disappearance of globally sclerotic glomeruli disperses the remaining glomeruli further apart from each other, thereby decreasing glomerular density with aging.[116] However, the relationship between glomerular (nonsclerosed and globally sclerosed) density varies, depending on how much global glomerulosclerosis is present in the section biopsied. In biopsy sections with less than 10% global glomerulosclerosis, the glomerular density is decreased with age, whereas in biopsy sections with more than 10% global glomerulosclerosis, the glomerular density is increased with age.[116] In regions of the renal cortex without significant glomerulosclerosis, age-related tubular enlargement disperses glomeruli further apart, decreasing their density. However, in regions with significant glomerulosclerosis, the overall atrophy of tubules and glomeruli brings the glomeruli closer together, thereby increasing their density (Fig. 22.5).

Tubular Diverticuli

Darmady and colleagues found in an autopsy study that tubular diverticuli increase with older age.[110] This finding parallels two other age-related findings in healthy adults. First, simple parenchymal cysts, which likely originate from these diverticuli,[81] become more frequent with older age. Second, the mean profile tubular area increases with older age.[86] Enlargement of tubules, with upregulation of growth factors, may contribute to the development of diverticuli, with the eventual formation of cysts.[117,118]

ARTERIES AND ARTERIOLES

Arteriosclerosis and arteriolar hyalinosis are two common findings in the kidney biopsy of normal adults that become

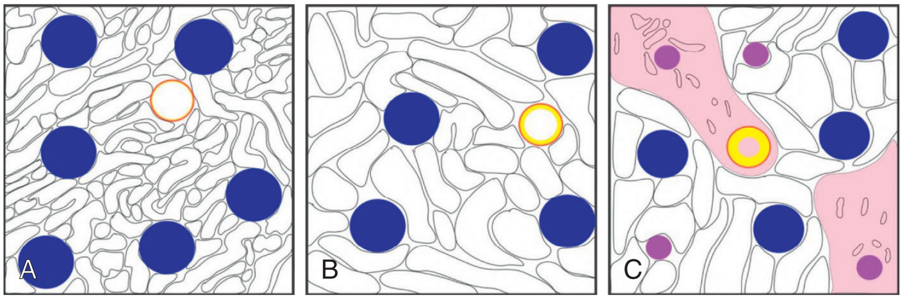

Fig. 22.5 Schematic illustration of how the percentage of glomerulosclerosis influences glomerular density. (A) Example of a young individual with a certain glomerular density *(blue solid circle)*, normal tubules and normal artery *(orange circle),* with minimal to no intimal thickening *(yellow circle).* (B) In an older person in whom the biopsy shows less than 10% glomerulosclerosis, tubules become larger and disperse the glomeruli apart, and their density decreases. (C) In older persons in whom there is more than 10% glomerulosclerosis *(pink solid circles)*, the associated interstitial fibrosis and tubular atrophy *(light pink areas)* bring the glomeruli closer together, thus increasing their density. Arteriosclerosis with intimal thickening *(thicker yellow line)* is also pronounced in these regions in older adults.

more prevalent with age.[86] Arteriosclerosis (luminal stenosis by intimal thickening) with aging may cause ischemic injury, leading to a lower cortical to medullary volume ratio, even in healthy adults.[86] Arteriolar hyalinosis increases with age and is associated with renal artery atherosclerosis in healthy adults, consistent with studies that associate arterolar hyalinosis with renal atherosclerosis in hypertensive patients.[119–121] Older age and hypertension may lead to global glomerulosclerosis and nephrosclerosis via reduced blood flow due to ischemic injury from the narrowing of small arteries and arterioles.[122] Notably, lower total nephron number/kidney is associated with arteriosclerosis, even after adjusting for age and gender.[88]

NEPHROSCLEROSIS AND INTERSTITIAL FIBROSIS

The term *nephrosclerosis* describes a microstructural biopsy pattern of global glomerulosclerosis, arteriosclerosis, and interstitial fibrosis with tubular atrophy. Nephrosclerosis is notably seen with hypertension,[123] but is also described in healthy older kidney donors without hypertension or only mild hypertension.[91] The nephrosclerosis appearance is thought to be due to increased intimal thickening of small arteries in the kidneys (arteriosclerosis, arteriolosclerosis), leading to glomerular ischemia, with the wrinkling of capillary tufts, thickening of the basement membrane, and progressive pericapsular fibrosis.[123,124] Simultaneously, proteinaceous material accumulates in Bowman's space, likely due to a combination of a disturbed balance between formation and breakdown of the glomerular extracellular matrix[125] and podocyte depletion,[91,109,125] probably due to perturbations in local mechanical forces.[126] Eventually, ischemic glomerular tufts fully collapse, forming globally sclerosed glomeruli. As a consequence of glomerulosclerosis, accompanying tubules atrophy, with an accumulation of surrounding interstitial fibrosis.

All four features of nephrosclerosis—global glomerulosclerosis, arteriosclerosis, interstitial fibrosis, and tubular atrophy—increase with older age and, when combined together (at equal weighting) they represent a so-called nephrosclerosis score (Fig. 22.6).[91] If we define the nephrosclerosis as presence of at least two of the previously mentioned abnormalities, then from Fig. 22.6 we can see that the youngest donors (18–19 years of age) do not have detectable nephrosclerosis. The prevalence of nephrosclerosis

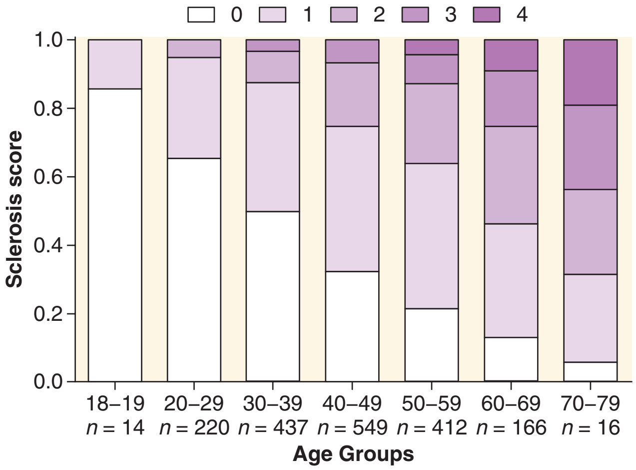

Fig. 22.6 The nephrosclerosis score increases with age among 1814 living kidney donors from the Aging Kidney Anatomy study (expansion of previously reported findings). The total nephrosclerosis score is obtained by adding the individual scores of histologic abnormalities: any global glomerulosclerosis, any arteriosclerosis, interstitial fibrosis greater than 5%, and any tubular atrophy. *White bars* in all age groups represent a score of 0 (no abnormalities present), and a *dark purple bar* represents a score of 4 (presence of all four pathologic abnormalities). Three intermediate scores are presented with different shades of *purple color*. (From Rule AD, Amer H, Cornell LD, et al. The association between age and nephrosclerosis on renal biopsy among healthy adults. *Ann Intern Med.* 2010;152:561–567.)

in 20- to 29-year-old donors is 5% and rises to 69% among the oldest donors.

Thus, the number and size of the nephrons are the primary determinants of the kidney cortical volume. A combination of age-related glomerulosclerosis with corresponding interstitial fibrosis and tubular atrophy is responsible for the observed loss of cortical volume with healthy aging seen in living kidney donors. This nephrosclerosis typically starts in glomeruli in the superficial cortical zones. Because the corresponding tubules of these superficial glomeruli contribute more to the cortical volume, their atrophy with age-related nephrosclerosis leads to a decrease in cortical volume with aging. At the same time, there is tubular hypertrophy of the remaining nephrons, particularly those in the medulla. This observation explains why, despite a 48% decrease in nephron number from ages 18 to 76 years, there is only a 17% decrease in cortical volume and only a 12% decrease in kidney volume.[74,75]

One might expect that with age-related nephrosclerosis, there would be compensatory hypertrophy of the remaining, still functional glomeruli. However, studies in living kidney donors have not confimed this hypothesis.[116] A possible explanation may be that with healthy human aging, there is a decreased metabolic demand for glomerular function, so that the loss of glomeruli does not cause the enlargement of the remaining glomeruli. Notably, comorbidities that can become more prevalent in old age, such as obesity and diabetes, with albuminuria, do associate with glomerular hypertrophy.[116] Also, low nephron endowment at birth, or disorders that accelarate nephron loss with advancing age, can contibute to the glomerulomegaly observed in adulthood.

FUNCTIONAL CHANGES OF THE KIDNEYS WITH AGING

These include changes in the GFR and other functional changes.

GLOMERULAR FILTRATION RATE

WHOLE-KIDNEY GLOMERULAR FILTRATION RATE

In a series of pioneering studies begun over 60 years ago, Davies and Shock examined 70 healthy adults between the ages of 24 and 89 years and, in a cross-sectional analysis, showed (by urinary inulin clearance) a linear GFR decline in subjects older than 30 years.[127] The GFR in the oldest group was found to be 46% less as compared with the GFR in the youngest group. Several decades later, Lindeman and colleagues carried out a longitudinal study (the first of its kind) by following 254 mostly healthy adult individuals, although some had diabetes, for up to 14 years.[128] They found that the mean annual decline in GFR, estimated by 24-hour urinary creatinine clearance, was 7.5 mL/min per decade; however, one-third of the subjects had no decrease in kidney function, and a small subset actually had an increase. In addition to measurement error explaining this finding, a rise in GFR with age might also be explained by hyperfiltration due to the comorbidities (e.g., obesity, diabetes) that become more common with aging.[129] This longitudinal rate of GFR decline is similar to the cross-sectional decline of 6.3 mL/min per decade obtained from a study of potential kidney donors (GFR measured by iothalamate clearance).[91] Among 4500 potential kidney donors in the expanded analysis from the Aging Kidney Anatomy study (Fig. 22.7), the average decline per decade for estimated GFR was 7.4 mL/min, whereas for measured GFR (iothalamate clearance) it was 6.1 mL/min/1.73 m^2. The estimated GFR by the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equation underestimates the GFR in healthy populations, such as potential kidney donors, because it was developed using mostly CKD patients.[130,131]

Although Fig. 22.7 shows mean estimated GFR (eGFR) and measured GFR, there is variability within each age group (Table 22.2).[132] It can be appreciated that both upper and lower reference limits also decline with aging. Similar findings in healthy donors have been reported by Pottel and colleagues.[133] Other longitudinal studies of the decline in renal function (GFR or creatinine clearance) with aging have shown generally similar findings to those of Lindeman and associates,[128] including the relative stability of renal function for extended periods in some otherwise healthy aging subjects.[134–136] In addition, in these studies, the rate of decline of renal function appeared to increase with each passing decade in those older than about 30 to 40 years.

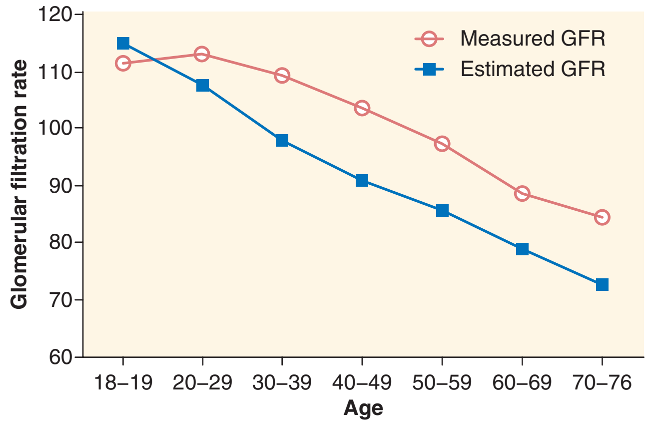

Fig. 22.7 Age-related glomerular filtration rate (*GFR*; mean values) decline among 4500 potential kidney donors in the Aging Kidney Anatomy study (expansion of previously reported findings). Decline in the estimated GFR (CKD-EPI study equation; see text) is calculated to be 7.4 mL/min/decade and 6.1 mL/min/decade for the measured GFR. (From Rule AD, Amer H, Cornell LD, et al. The association between age and nephrosclerosis on renal biopsy among healthy adults. *Ann Intern Med.* 2010;152:561–567.)

Table 22.2 Reference Values for Estimated and Measured Glomerular Filtration Rate in Potential Kidney Donors[a]

Age Group (years)	No.	Estimated Glomerular Filtration Rate (GFR) (Measured GFR) Fifth Percentile	Median	Mean	95th Percentile
18–19	46	88 (87)	114 (107)	115 (111)	145 (144)
20–29	584	81 (84)	108 (111)	108 (113)	130 (149)
30–39	1090	75 (82)	98 (106)	98 (109)	118 (147)
40–49	1364	69 (77)	91 (102)	91 (104)	110 (135)
50–59	1001	65 (72)	85 (94)	85 (97)	103 (133)
60–69	379	61 (64)	78 (87)	79 (88)	97 (118)
70–76	36	55 (57)	71 (85)	73 (84)	91 (113)

[a]Expansion of previously reported findings.
From Murata K, Baumann NA, Saenger AK, et al. Relative performance of the MDRD and CKD-EPI equations for estimating glomerular filtration rate among patients with varied clinical presentations. *Clin J Am Soc Nephrol.* 2011;6:1963–1972.

SINGLE-NEPHRON GLOMERULAR FILTRATION RATE

Several studies have found that a higher whole-kidney GFR correlates with larger kidneys or kidney cortical volume.[79,80,86] A study of 1520 healthy living kidney donors also investigated how the whole-kidney GFR correlated with the microstructural components of cortical volume. After age and gender adjustments, a higher whole-kidney GFR was associated with larger nephrons (larger glomeruli and tubules), whereas a lower whole-kidney GFR did not associate with any measure of nephrosclerosis, including global glomerulosclerosis, interstitial fibrosis with tubular atrophy, and arteriosclerosis.[86] The age-related nephron loss due to increased nephrosclerosis is followed by a parallel whole-kidney GFR decline.[88] To disentangle whole-kidney GFR from the nephron number, further single-nephron GFR (snGFR) studies are required.

Although in vivo measurements of snGFR through micropuncture are possible and have been done in animal studies,[137,138] a direct measure of snGFR (by micropuncture) is not feasible or safe in humans. However, dividing the whole-kidney GFR by the number of functional (nonsclerosed) glomeruli in both kidneys allows for an estimation of the mean snGFR.[139] The snGFR remains stable with age because the loss of nephrons parallels the loss in the total GFR.[88] Thus, the effect of age on snGFR is minimal, consistent with other physiologic characteristics, including height and gender, whereas obesity, family history of ESKD, and nephrosclerosis exceeding that expected for age are associated with an increased snGFR.[139,140] There may be higher snGFR among the oldest donors (70–75 years of age), but this finding may be confounded by donor selection factors (e.g., the requirement of whole-kidney GFR >80 mL/min/1.73 m^2) in this age group, which may only allow those with some degree of hyperfiltration to donate and be available to study with a kidney biopsy.[139,140]

Although other studies of snGFR in living human subjects are lacking, there have been studies assessing the single-nephron ultrafiltration coefficient (snK_f) in individuals similar to living kidney donors.[141–143] The snK_f represents the filtering capacity of the glomerulus, as determined by the surface area and permeability of the glomerular filtration barrier. In these studies, snK_f was estimated from electron microscopy measures of the glomeruli on kidney biopsy.[141] The snGFR could be calculated by multiplying snK_f by the perfusion pressure across the glomerular filtration barrier.[144] Thus, it is possible that the relationship of clinical characteristics to snGFR may be similar to that of snK_f. Consistent with the findings of a relatively stable snGFR across the age spectrum described above, snK_f did not show significant differences between younger and older living kidney donors.[141,145] Adaptive hyperfiltration does occur in the aging kidney following unilateral nephrectomy[146]; however, either due to an increase in Kf, glomerular capillary filtration pressure, or increased glomerular plasma flow or some combination of the three, the extent of the increase in snGFR (and whole-kidney GFR [wkGFR]) is reduced compared with younger persons. The definition of hyperfiltration based on measurements of the whole-kidney GFR is best determined by an age-adapted measured GFR (not eGFR), uncorrected for body surface area. However, such definitions do not distinguish between a higher snGFR and higher nephron number.[147]

OTHER FUNCTIONAL CHANGES

RENAL PLASMA FLOW

Renal plasma flow (RPF) represents the volume of plasma delivered to both kidneys per unit of time. The RPF can be calculated from the amount of plasma that is cleared of para-aminohippurate (PAH) per unit of time. PAH is a compound with a nearly 100% extraction through the kidneys. A Swedish study of 122 potential kidney donors, ages 21 to 67, has shown that similar to the GFR, the RPF declines with older age.[140] Older age influences the degree of renal blood flow change as a response to exercise,[148,149] attenuates the renal hemodynamic response to atrial natriuretic peptide[150] and the degree of vasodilation.[151] One study has postulated that higher levels of the endogenous nitric oxide (NO) inhibitor asymmetric dimethylarginine (ADMA) lead to a reduced RPF and hypertension with older age.[152] Animal studies have provided some evidence that ADMA is associated with tubulointerstitial ischemia and fibrosis,[153] glomerular capillary loss, and glomerulosclerosis.[154] In a study of 19 healthy individuals, divided into three age groups—young, middle-aged, and older—an infusion of dopamine and amino acids was used to test the effects of maximal vasodilation on RPF and NO levels. Whereas both RPF and NO levels increased in young and middle-aged individuals, they did not change in the older group.[155] This suggests that due to advanced age-related vascular changes in older adults, renal blood vessels are less responsive to maximal vasodilation.

FILTRATION FRACTION

The filtration fraction (FF) represents the proportion of fluid entering the kidneys that reaches into the renal tubules—that is, it is a ratio between the GFR and RPF. Normally, the value for FF is about 20%. However, in older age, atherosclerosis and arteriosclerosis of the arteries and arterioles reduces blood flow to kidneys. Thus, the FF may increase, particularly in those of very advanced age.[140]

RENAL RESERVE IN AGING

The phenomenon of renal reserve refers to an acute increase in GFR (usually by 20% or more from basal values) when subjects are fed oral protein loads (typically cooked red meat) or infused intravenously with certain amino acids.[155–159] The mechanisms underlying such physiologic changes is not fully understood, but appears to be a consequence of vasodilatation, an increase in glomerular plasma flow, and an increase in the snGFR.[160] See Chapter 3. Growth hormone may be involved, but this is controversial[155–157] The efficiency of renal reserve is blunted to some degree by aging per se, most often in the very older.[155] Renal functional reserve is well maintained up to about age 80 years in otherwise healthy men and women.[161] Diminished levels of sex hormones may play a role in the diminished renal reserve seen with aging (see later). The diurnal variation in GFR is also blunted with aging.[162]

Aging in indigenous Kuna Indians (Panama), who subsist on a low-protein, high–dark chocolate diet and remain largely free of hypertension and cardiovascular disease (CVD), even at an advanced age, is associated with a steady decline in GFR and RFP with aging, similar to that observed in westernized countries.[163] These findings are not consistent with a hypothesis that rising blood pressures (hypertension) are causally related to the decline in GFR with aging.

SEXUAL DIMORPHISM IN AGING

In a series of studies, mainly focused in murine species, Baylis and coworkers developed the concept of sexual dimorphism in renal aging[164–168] (also see review by Gava and associates[169]). Females develop less age-related decline in renal function, perhaps due to a renoprotective action of estrogens. In contrast, males have a greater loss of renal function with age, perhaps due to the adverse effects of androgens. The mechanisms involved are complex, but seem to implicate disturbances of the NO synthesis pathways, and a lower rate of NO production might be influenced by sex steroids.[165,166] A role for the renin-angiotensin-aldosterone system (RAAS) has also been described.[168] Interestingly, in aging experimental animals, there was no observed rise in glomerular capillary pressure or glomerulomegaly in males or females, castrated or not. This suggests that the age-dependent sexually dimorphic loss of renal function does not appear to be dependent on hemodynamic changes or glomerular hypertrophy, similar to that observed in humans.[168] Age-dependent sexual dimorphic effects on renal function might be related to increased body size in males and therefore may be conditional to a disparity between energy demand and supply.

Pathophysiologic Explanations for Aging-Associated Nephrosclerosis

Podocentric Versus Ischemic Hypotheses (Animal and Human Studies). As stated earlier, it has been shown that renal blood flow declines with older age.[170] Studies in rats have shown that in addition to the reduction in blood flow, renal arterioles in older rats have altered sensitivity to NO[171] and angiotensin II.[172] However, the main limitation of animal studies is the virtual absence of arteriosclerosis, thus limiting studies of age-related ischemic changes. Studies in human kidneys have demonstrated that age-related arteriolosclerosis and intimal and medial hypertrophy in intrarenal arteries resemble those observed in extrarenal arteries.[173,174] These pathologic changes are especially common in interlobular arteries and are frequently observed in renal biopsies from normal individuals without cardiovascular disease. Although it is unclear what causes the vascular changes in intrarenal arteries, Martin and Sheaff have postulated that they are directly linked to global glomerulosclerosis, which occurs first in the superficial cortical layers and is followed by local interstitial fibrosis and tubular atrophy.[124] Subsequently, as a compensatory mechanism, the deeper glomeruli hypertrophy and, over time, undergo hyperfiltration injury, ultimately developing glomerulosclerosis. More recently, studies in healthy living kidney donors have corroborated the hypothesis of Martin and Sheaff that age-related glomerulosclerosis is primarily of vascular origin due to observed ischemic changes in the glomeruli, intracapsular fibrosis,[175] and increased prevalence of arteriosclerosis.[91] Moreover, implantation renal biopsies of healthy kidney donors can also be ischemic appearing, presumably still functional glomeruli, with a wrinkling of capillary loops, thickened basement membrane, and mild intracapsular fibrosis. All these described findings may be associated with ischemia and gradually lead to progressive shrinkage of the glomerular tufts and collagen deposition in Bowman's space, finally ending with complete global glomerulosclerosis.

Alternatively, podocyte depletion can also give rise to the glomerulosclerosis of aging. The podocytes are highly differentiated and specialized epithelial cells in a glomerulus, with limited capacity for cell division and turnover.[176,177] The two main reasons that lead to podocyte depletion, in absolute or relative terms, are actual loss of podocytes (cell death or detachment from the tuft) and glomerulomegaly—each podocyte subsequently has to cover the increased filtration surface area. The notion that podocyte dysbiosis may be the culprit for glomerulosclerosis is not new. Three decades ago, Kriz and colleagues[178] postulated that podocyte depletion and a subsequent bare glomerular basement membrane is the first step that leads to focal and segmental glomerulosclerosis. Studies in rats have provided evidence that reduced numbers of podocytes lead to glomerulosclerosis.[179,180] Using puromycin aminonucleoside (PAN) treatment as a model for podocyte oxidative injury, one study has linked glomerulosclerosis and podocyte number depletion.[179] Others have shown increased glomerulosclerosis (usually of the focal and segmental variety) when podocytes fail to follow glomerular growth as a result of molecular growth signalling[180] or may be due to podocyte hypertrophic stress associated with glomerulomegaly in rats without calorie restriction.[181] Interestingly, calorie restriction abolished glomerulomegaly, podocyte hypertrophy/stress and pydocyte loss with resulting glomerulosclerosis.[181] Using a transgenic approach in rats, the same group has also developed evidence that the reduced number of podocytes alone is sufficient to cause glomerulosclerosis.[182]

Taken together, all these animal (murine) studies favor the concept that acquired or congenital podocytopenia (absolute or relative) is a critical driving force for the development of glomerulosclerosis, particularly of the focal and segmental variety (see review by Kriz and associates[178]). However, the biology of aging in experimental animals differs from that of humans (see above), and these differences must be taken into account when translating pathophysiologic concepts from animals to humans.

More recently, human kidney biopsies have been used to study podocytes; these have found that the pattern of global glomerulosclerosis with aging is somewhat different in studies not limited to living kidney donors.[108] A study of 89 kidney biopsies from living kidney donors, deceased kidney donors, and normal poles of radical nephrectomy specimens performed for tumor, found that podocyte depletion is identified as a potential culprit for age-related glomerulosclerosis.[108] The authors counted podocytes, measured their size and density, and demonstrated a decline in podocyte density with older age due to a reduced count of podocytes per glomerulus and larger glomerular volume. In particular, younger individuals had more than a threefold higher podocyte density compared with individuals older than 70 years (>300 podocytes/10^6 μm^3 vs. <100 podocytes/10^6 μm^3). In the older individuals, not only was the podocyte density lower, but their detachment rate was much higher, and podocytes showed molecular evidence of cell stress. Moreover, in some glomeruli with significant podocyte detachment, the authors observed binucleated podocytes as evidence of failed mitotic attempts and glomerular capillary wrinkling, tuft collapse, and pericapsular fibrosis. All these findings led the authors to propose a hypothesis (the podocentric hypothesis) for the glomerular natural history with aging, extending from glomerular hypertrophy to tuft collapse to glomerulosclerosis.[108] However, studies limited to living donors did not show glomerular hypertrophy with aging,[86] suggesting that age-related comorbidities rather than aging alone may explain these findings. In addition, it is noteworthy that increasing albuminuria is not a feature of normal healthy aging.[91] Thus, the podocentric hypothesis may not be relevant to human renal aging.

CONTRIBUTION OF COMORBIDITIES TO RENAL AGING

Aging is commonly accompanied by comorbidities that can have an independent impact on renal structure and function. These include obesity, diabetes, and nephron endowment at birth. For example, obesity and/or diabetes (type 2) can lead to glomerulomegaly, glomerular hyperfiltration, and albuminuria (see Chapter 51). In part, these comorbidities contribute to the rising prevalence of increased albuminuria with aging seen in cross-sectional epidemiologic studies.[68] Hypertension, commonly seen in older adults, can also modify the underlying normal pattern of renal senescence seen with aging, as discussed earlier.

It also seems likely that nephron endowment at birth will have a modifying effect on the pace of normal physiologic renal aging. Nephron underendowment at birth, due to fetal dysmaturity, will be accompanied by early single-nephron hyperfiltration and glomerular enlargement, which can lead to maladaptive glomerular injury, podocytopenia, glomerulosclerosis and perhaps an acceleration of the normal rate of nephron loss.[183] The combination of glomerulopenia (low nephron endowment at birth) and physiologic renal senescence could, at least theoretically, lead to features of CKD (reduced GFR and/or albuminuria) observed in later life. This postulated phenomenon has not been well studied, so these effects are largely speculative. Reduced glomerular density and glomerular hypertrophy are signs of nephron underendowment and have been shown to have an adverse effect on the progression of many renal diseases, including some that may affect older adults.[184,185] Low birth weight, fetal dysmaturity, and impaired nephrogenesis can contribute to the burden of kidney disease globally, especially in older adults and in subjects with primary or secondary glomerular diseases (see Chapter 21).[185]

FLUID, ELECTROLYTE, AND ACID-BASE HOMEOSTASIS IN AGING

SODIUM HOMEOSTASIS

A study in 89 healthy subjects has demonstrated that older age significantly affects the kidney's capacity to reabsorb filtered sodium. Following the initiation of sodium restriction, subjects older than 60 years needed nearly twice as much time to reduce sodium excretion than subjects younger than 30 years (31 vs. 17.6 hours, respectively).[186] The observed difference may be due to the significantly reduced sodium reabsorption in the distal tubules, which was found to be nearly 30% less in older people versus younger controls.[187] Sodium homeostasis can also be influenced by age-related changes in levels and responses to renin and aldosterone, hormones that regulate sodium conservation (discussed further in Chapter 14). Although plasma renin activity and aldosterone levels are lower in older adults, they are not associated with changes in fluid or electrolyte metabolism.[188] These changes increase older adults' susceptibility to sodium retention in disease states or with the use of medications that alter renin release. More recently, the slower response to sodium restriction observed in older adults was reproduced by using angiotensin-converting enzyme (ACE) inhibitors and blocking the RAAS.[189]

One group studied the natriuretic response in normal individuals of different ages after volume expansion and volume contraction. In the 24-hour period following a 2-L infusion of normal saline, the natriuresis was slower in older individuals.[190] It has been hypothesized that the higher prevalence of hypertension in older subjects may be due in part to this attenuated natriuretic response with aging. There is also a reduction in the natriuretic response to saline loading in older versus younger kidney donors after donation.[189]

Atrial natriuretic peptide (ANP) is a hormone secreted by atrial myocytes, and one of its many functions is to control sodium excretion at the tubular level. With older age, the tubular response to ANP is reduced. ANP inhibits sodium reabsorption from the luminal side and induces hyperfiltration and suppression of renin release, all of which together lead to natriuresis, diuresis, and lowering of blood pressure. Normally, ANP is cleared rapidly from plasma; however, if degradation enzymes or clearance receptors are blocked, the half-life can be prolonged.[191] Studies have shown that the plasma ANP levels are several times higher in older individuals, and this may be a compensatory effect of the reduced response at the receptor level. This hypothesis is in agreement with two studies in which sodium excretion plateaued after an ANP infusion in older subjects,[192,193] whereas in younger subjects sodium excretion continued to increase with increasing ANP dose.[192] Of importance, it appears that ANP production does not change with older age; therefore the observed increase of ANP levels in older adults results from reduced metabolic clearance.[194,195] Moreover, others have shown that in response to a saline load, plasma levels of ANP increase more robustly in older compared with younger individuals.[196,197]

The role of altered renal sodium handling in the pathogenesis of age-related elevation of systemic arterial pressure is a topic of great contemporary interest (see also Chapter 46; reviewed by Frame and Wainford[198]). Clearly vascular factors, such as reduced compliance of major arterial vessels, and neurohumoral alterations, such as increased sympathetic nervous system activity, play important roles in the elevation of systolic blood pressure with aging. The sensitivity of blood pressure changes consequent to salt administration increases with age.[199] Animal studies have also suggested increased intrarenal angiotensin II signaling in age-related blood pressure increases.[200] Experimental studies in rodents has also shown decreased expression of the sodium potassium chloride cotransporter (NKCC2) in the loop of Henle with aging[201]; the activity of the thiazide-sensitive sodium chloride cotransporter (NCC) in the distal tubule may be altered in aging, but this is controversial.[198]

The expression of the epithelial sodium channel (ENaC) can be reduced in aging rodents.[202] These studies collectively indicate that dysregulation of sodium handling by the aging nephron can be involved in age-related blood pressure elevation. Integration of this knowledge into the complex array of biologic factors in systems affecting blood pressure (e.g., aortic compliance, sympathetic nervous system activity, RAAS) will be challenging.[198] The role of reduced nephron mass (glomerulopenia) that accompanies physiologic aging cannot be ignored as a participant in blood pressure changes with aging.

POTASSIUM HOMEOSTASIS

Potassium homeostasis is regulated by several different mechanisms; specifically, in the kidney, the potassium excretion rate is altered depending on the intake. The role of the kidney in potassium homeostasis is very important because normally around 90% of the dietary potassium intake is excreted in the urine and only around 10% in the gastrointestinal tract. The normal internal potassium distribution is also regulated by hormones, such as insulin, catecholamines, and aldosterone. In general, after filtering through the glomerulus, potassium is primarily reabsorbed via the proximal tubule and thick ascending limb of Henle such that only a small amount should normally reach the distal tubules (discussed further in Chapter 17).[203]

Total body potassium (TBK) declines with older age with a reduction in intracellular stores that accompany the decline in muscle mass (sarcopenia) seen with normal aging. A study of 188 healthy volunteers has shown that the annual rate of TBK decline is 7.2 mg/kg in women and 9.2 mg/kg in men,[204] which increases in both genders after 60 years of age.[205] The observed decline in TBK may also be explained by the lower renin and aldosterone levels,[206] as well as the blunted response to aldosterone in older adults (see later).[207] A study of 43 healthy older individuals demonstrated a relative decrease in fractional potassium excretion in relation to creatinine clearance.[208] This finding is consistent with animal studies in which potassium excretion was less efficient in old rats fed a high-potassium diet.[209] The same group subsequently showed that the transtubular potassium gradient is reduced in older compared with young subjects, implying reduced efficiency in potassium excretion in older adults.[210] Insulin-mediated potassium homeostasis in human subjects is not influenced by the age of the subject.[211]

Older age is usually accompanied by comorbidities, and older adults often take medications that interfere with potassium excretion. Examples are potassium-sparing diuretics, renin inhibitors, ACE inhibitors, angiotensin receptor blockers, heparin, calcineurin inhibitors, sodium channel blockers, and nonsteroidal antiinflammatory drugs (NSAIDs). Care is needed when starting these medications for older patients, given their increased propensity to develop hyperkalemia.

Clinical Relevance 3

Older adults are at increased risk of hyperkalemia, even given their reduced total body potassium stores and altered tubular potassium transport.

MAGNESIUM HOMEOSTASIS

It is estimated that about 99% of magnesium is stored in cells, predominantly in bones and muscles, and only about 1% is in extracellular fluid, including plasma. Normally, the kidneys reabsorb 96% of filtered magnesium[212] and regulate magnesium homeostasis (see Chapter 18).

Because plasma and intracellular magnesium homeostasis are very finely regulated,[213] total plasma magnesium is maintained remarkably stable with increasing age in healthy individuals.[214,215] However, a study of 36 healthy older individuals has demonstrated a significant subclinical deficit of magnesium that was not detected by serum magnesium levels.[216] Another more recent study has confirmed this finding, and older subjects (>65 years) had significantly reduced serum levels of both ionized and total magnesium.[214]

The published literature is scarce with regard to normal age-related changes in magnesium homeostasis in the kidneys. Changes in magnesium homeostasis are likely due to diseases that frequently occur in older age, use of therapeutic agents that alter its metabolism, and previously discussed age-related kidney functional decline.[214] The microstructural changes that lead to functional decline in older kidneys may also result in reduced tubular reabsorption of magnesium, thus causing hypomagnesemia. In addition to the age-related changes in the kidney, other comorbidities and drugs (e.g., loop diuretics, digitalis) in older adults can further worsen the magnesium deficiency.[214] On the other hand, hypermagnesemia occurs very rarely, even in the disease setting, including in those with severe acute kidney injury (AKI), CKD, or ESKD.[217] Interestingly, low dietary magnesium intake in older adults may also contribute to a faster rate of decline in the GFR, potentially exacerbating a proinflammatory milieu, endothelial dysfunction, and promotion of vascular calcification.[218]

CALCIUM HOMEOSTASIS

In normal adults, every day approximately 10 g of calcium is filtered through the glomeruli, and 98% to 99% is reabsorbed, so that net calcium excretion during a 24-hour period ranges from 100 to 200 mg. Most filtered calcium reabsorption (≈60%–70%) occurs in the proximal tubules, around 20% in the loop of Henle (thick ascending limb), around 5% to 10% in the distal convoluted tubule, and only around 5% in the collecting ducts (see also Chapter 18).

There have been several studies of calcium homeostasis with aging. A study of rats in three different age groups has shown that calcium excretion and reabsorption are equally maintained in all three groups, which may be linked to observed age-related renal hypertrophy in older rats.[219] However, other animal studies and studies in humans have shown that there are age-related changes in renal calcium homeostasis, such as decreased tubular reabsorption and/or the renal response to parathyroid hormone (PTH).[220–223] Renal and intestinal calbindins are proteins with a critical role in active transcellular calcium transport, and their expression decreases with age.[224–226] There is also an age-related decrease in the capacity of 1,25-dihydroxyvitamin D_3 (1,25-$(OH)_2D_3$) to stimulate calcium reabsorption, increasing PTH levels in both rats and humans.[227,228] A study in mice has found that an age-related reduction in TRPV5 and TRPV6 (renal and duodenal calcium intracellular transporters, respectively) lead to impaired renal and duodenal calcium reabsorption, consistent with the increased calciuria observed in older mice.[229] Alpha-Klotho (α-Klotho) is a gene associated with accelerated aging when mutated.[230] It encodes a protein that is predominantly expressed in tissues involved with calcium homeostasis. A study in mice has demonstrated that alpha-klotho binds to a subunit of sodium-potassium adenosine triphosphatase (Na^+-K^+-ATPase),[231] suggesting that the age-related decline in klotho protein could contribute to reduced sodium potassium ATPase sensing of low calcium levels. Serum-soluble α-Klotho (sKlotho) levels decline with advancing age in humans, independent of the GFR.[55]

There is also an age-related decrease in intestinal calcium reabsorption, which is linked with reduced 1α-hydroxylase activity, 1,25-$(OH)_2D_3$ levels, and increased basal PTH levels. A study of 22 healthy men has found that older individuals have a twofold higher serum PTH level, with no difference in blood ionized calcium. Moreover, older men also had a higher threshold for PTH release; however, this was not associated with a decrease in blood ionized calcium or 1,25-$(OH)_2D_3$.[232] This finding suggests that the relationship between PTH and calcium in older age is changed so that regardless of calcium concentrations, PTH levels are higher. An interesting study has compared healthy older individuals (≥75 years of age) to patients with CKD and a similar GFR and found that CKD patients have higher serum levels and fractional excretion of calcium, suggesting greater calcium wasting in these patients.[233] Concentrations of vitamin D–dependent calcium-binding proteins also decrease with older age, which is associated with the change in intestinal calcium absorption.[209,224] Another study of healthy young and older men sought to determine whether older age affects renal responsiveness to PTH infusion. The authors found that the renal response to PTH infusion and renal vitamin D levels were equivalent in both groups, with the only difference being that the production of vitamin D is somewhat delayed in older adults.[234]

PHOSPHORUS HOMEOSTASIS

More than 99% of body phosphorus is stored in the bones and less than 1% is in the serum in the form of inorganic phosphate (Pi). The maintenance of a narrow range of serum Pi levels is critical for numerous cellular functions, including energy metabolism and bone formation, or as a constituent of phospholipids and nucleic acids. Decline in the GFR with older age leads to reduced levels of non–protein-bound (free) serum calcium, but increased levels of serum phosphate.[235] In response to this, PTH synthesis increases, which in turn decreases the number of sodium phosphate cotransporters in the proximal tubules, thus leading to increased phosphaturia. High serum phosphate levels also stimulate the production of fibroblast growth factor 23 (FGF23), which has been suggested to be involved in regulating normal serum phosphate levels during kidney functional decline and suppressing 1,25 $(OH)_2D_3$ formation. FGF23 levels have been found to increase with declining GFR but not with age per se.[55]

In addition to the hormonal regulation of phosphate levels in aging, animal studies have demonstrated a PTH-independent reduction in the intrinsic tubular capacity to reabsorb phosphate.[236] In this study, kidneys of older thyroparathyroidectomized animals responded to pharmacologic dose of PTH to a lesser extent than younger animals.[236] Others have demonstrated that kidneys in aged animals show impaired phosphate transport in renal tubules in response to a low-phosphate diet.[237,238] It has been proposed that the reason for this impaired phosphate transport is decreased fluidity of a luminal brush border membrane due to the age-related increase in cholesterol and sphingomyelin,[238] which then negatively affects sodium phosphate cotransporter activity directly.[239] The same group later showed that acute and chronic changes in cholesterol differentially affected sodium phosphate cotransporter activity.[240] Whereas acute changes in cholesterol modulate the apical membrane expression of sodium phosphate cotransporter (posttranslational regulation), chronic changes modulate the abundance of this cotransporter (translational regulation).[240] Consistent with in vivo studies, an in vitro study of cultured renal tubular cells harvested from young and adult rats has shown reduced phosphate uptake and adaptation to a phosphate-free medium only in the adult renal cells.[241] Another study has revealed that the reason for the age-related decline in tubular phosphate reabsorption and tubular adaptation to a low dietary phosphate is the lower expression of the renal sodium phosphate cotransporters on the apical brush border membrane.[242]

VITAMIN D HOMEOSTASIS

Older persons commonly exhibit vitamin D insufficiency or deficiency, often in association with a subtle state of chronic inflammation. Serum levels of 25-hydroxyvitamin D (25- [OH]D) are inversely associated with biomarkers of inflammation in older adults, such as C-reactive protein and interleukin 6 (IL-6).[243] It is not known whether these changes are due to primary changes in vitamin D metabolism or to a secondary effect of inflammation. Vitamin D absorption from dietary sources may not be impaired with aging, but reduced dietary intake can contribute to low serum 25-(OH) D levels in aging individuals.[244] The reduced production of 7-deoxyhydrocholestrol in the aging epidermis with ultraviolet exposure is likely involved in the decreased production of pre–vitamin D_3 (cholecalciferol) in older adults.[245] Serum 1,25-$(OH)_2D$ (calcitriol) levels decline in aging, at least in part due to the decline in the GFR and impaired hydroxylation of 25-(OH)D mentioned above.[244] Changes in vitamin D sufficiency or insufficiency in older subjects can modulate left ventricular geometry and participate in the risk of left ventricular hypertrophy in hypertensive older subjects.[246]

It is important to keep in mind the age-related changes of 1,25-$(OH)_2D$ metabolism on intestinal phosphate absorption (described earlier), because several studies have shown that dietary vitamin D supplementation improves renal and intestinal phosphate absorption in vitamin D–deficient animals.[247–249] Regarding the previously mentioned role of cholesterol in the fluidity of the apical brush membrane, it appears that deprivation or administration of vitamin D alters the levels of several fatty acids in this membrane.[250] Therefore, age-related changes in 1,25-$(OH)_2D$ production may potentially have an effect on phosphate reabsorption by modulating lipid content in the apical brush membrane.

Thus, aging has diverse effects on calcium, phosphorus, and vitamin D metabolism. These include a decrease in intestinal absorption of calcium, impaired vitamin D production in the skin, dietary vitamin D deficiency, impaired production of 1,25-$(OH)_2D_3$, increased PTH secretion, lower serum Klotho levels, higher FGF23 levels, and alterations in renal tubular handling of calcium and phosphorus.

ACID-BASE HOMEOSTASIS

The kidneys also play a major role in the regulation of acid-base homeostasis. Normally, during the 24-hour period, kidneys reabsorb around 4500 mmol of filtered bicarbonate. Kidneys have a large capacity to excrete the endogenously generated proton (H^+) by generating ammonia (NH_3) and its excretion (NH_4^+; see Chapter 16). Older adults are more prone to develop acid-base dysregulation for several reasons. First, age-related structural and functional changes in the kidney lead to reduced adaptive responsiveness to dietary and/or environmental changes. Second, the age-related decline in the GFR reduces the capacity of the kidney to buffer metabolic changes and excrete the excess H^+ load.[251] Along with age-related kidney functional decline, there is also a reduced capacity for bicarbonate conservation and generation, which, combined with the stable endogenous H^+ production, may lead to metabolic acidosis. This process is enhanced and worrisome in patients with concomitant CKD.[252] A community-based observational study of healthy subjects aged 6 to 75 years has found that the capacity for H^+ excretion is similar from childhood to young adulthood, but is significantly reduced in older age.[253]

Ammonium excretion decreases with age. Administration of an acute acid load in 26 healthy men of various ages has demonstrated that despite a prompt increase in acid excretion in all of them, the older subjects excreted a much lower percentage of ingested acid.[254] Interestingly, because reduced acid excretion was equivalent to reduced inulin clearance, acid excretion normalized per the GFR was similar in both young and old subjects. Additional studies in four young and four older subjects who, in addition to acid load, received glutamine, showed equivalent responses of young and old kidneys to glutamine.[254] Taken together, it has been suggested that the age-related decrease in H^+ excretion is due to reduced renal tubular mass, rather than tubular dysfunction.[254] A study of young and old rats who received an acid load was consistent with the human study.[255] After rats received the acid load, total H^+ excretion was reduced in older rats—50% the first day and 25% the second day. Additional insights from this study were the increased activity of the sodium-hydrogen exchanger and decreased phosphate transport in isolated proximal tubules in both age groups.[255]

Overall, these studies suggest that older age appears not to alter the physiologic regulation of acid-base homeostasis but does lead to reduced and diminished responses. The age-related perturbations of acid-base homeostasis seem to influence electrolyte balance. A Swedish study of 85 healthy older individuals has shown an association between net acid excretion and potassium-adjusted magnesuria, irrespective of magnesium intake.[256] Another study of 384 older individuals, followed over 3 years, found that a potassium-rich, alkaline residue (vegetable-rich) diet, which results in increased potassium excretion, was associated with a higher percentage of lean body mass.[257] The protective effect of an alkaline residue diet on muscle mass in older adults is consistent with the opposing effect of an acid residue (red meat-rich) diet. A study of nearly 10,000 participants in the general population has found that a higher dietary acid load is associated with lower serum bicarbonate concentrations; the association is much stronger in middle-aged and older than younger individuals.[258] Consistent with this, a study of older patients with CKD has found that oral sodium bicarbonate supplementation corrects metabolic acidosis, improves serum albumin levels, and decreases whole-body protein degradation.[259] Potassium bicarbonate supplementation in postmenopausal women neutralized the endogenous acid, improved calcium and phosphate balance, and reduced bone reabsorption.[260] This was confirmed in a more recent double-blind study of older men and women—in which particularly bicarbonate and not potassium—had a protective effect on bone reabsorption and calcium excretion.[261] Another recent randomized, double-blind, placebo-controlled study of 52 healthy older men and women has found that oral potassium citrate neutralizes the dietary acid load and has a long-lasting effect to reduce calciuria, without affecting gastrointestinal calcium absorption.[262] Taken together, the results of several studies are consistent with the hypothesis that a higher alkaline residue (vegetarian) diet prevents bone loss in older individuals.

WATER HOMEOSTASIS

Water homeostasis is regulated by a high-gain feedback mechanism that involves the hypothalamus, neurohypophysis, and kidneys.[263,264] In kidneys, the water and sodium from the glomerular filtrate are reabsorbed in tubules through water channel aquaporins (AQPs) and sodium cotransporters.[265–267] The tubular reabsorption of water primarily depends on the driving force (high interstitial osmolality in the deeper medullary zones) and osmotic equilibrium of water across the tubular epithelia (high osmotic water permeability of the membrane). The large majority of glomerular filtrate is reabsorbed in the proximal tubules and descending thin limbs of the loop of Henle.[266,268] The next tubular segments (thin and thick ascending limbs and distal convoluted tubules) are relatively water-impermeable.[269,270] Finally, the main parts of the nephron where vasopressin-based regulation of body water homeostasis occurs are the connecting tubule and collecting duct[271–273] (see also Chapter 15). Both maximum urinary concentration and diluting capacity are decreased by aging,[274] leading to a higher risk for hypernatremia and hyponatremia, respectively.

Findley has proposed an age-related dysfunction of the hypothalamic-renal axis based on clinical observations of increased vasopressin secretion in older age.[275] With older age, maximal urine concentrating ability is particularly reduced.[274] Compared with younger individuals, older adults have about a 50% reduced capacity to conserve water and solutes. Due to the combined effect of age-related changes in body composition, progressive microstructural changes in the kidney, and changes in plasma osmolality and fluid volume, some older adults are more prone to developing disturbances in water homeostasis. One of the major changes in body composition is an increase in total fraction of body fat by 5% to 10% and an equivalent decrease in total body water, so that an older man on average has 7 to 8 fewer liters of total body water than a young man with the same body weight.[276] The obvious consequence is that in the event of acute loss, or overload of body water, a more severe change in osmolality will occur in older men. A study comparing plasma osmolality in older and younger individuals before and after a similar extent of water deprivation has confirmed this presumption.[277] Alterations in water and sodium balance frequently lead to hypo- or hypernatremia associated with hypo- or hypervolemia in older adults.[278,279] In addition, dysfunction in water homeostasis may also occur due to abnormal expression and trafficking of AQPs and solute transporters. For example, an animal study has shown that the AQP2 transporter involved in urine-concentrating ability is downregulated in the medulla of older rats.[280] This molecular mechanism is consistent with a reduced urine-concentrating ability in older adults. Water homeostasis is also influenced by the previously described microstructural changes that occur with the aging kidney. These changes occurring with healthy aging do not pose an acute threat or have an impact while an individual is healthy. However, in the event of stress, acute disease, volume load, or dehydration, the combined effects of age-related loss of renal mass (i.e., functional reserve) and the change in body composition may cause a significant disruption in water and solute homeostasis.[281,282]

One of the consequences of the age-related GFR decline is an increase in filtrate reabsorption in the proximal tubules and decreased fluid delivery to the distal diluting tubules, resulting in a reduced diluting capacity of the kidney.[283] This is evidenced by a reduced ability to excrete a free water load[277] or reduced maximal free water clearance in older adults.[283] In addition to the reduced diluting capacity, older kidneys also lose the capacity to conserve water during states of dehydration.[284] Interestingly, another study has demonstrated that the reduced capacity of water conservation occurs regardless of high plasma vasopressin levels; thus, this is most likely due to intrinsic renal factors.[285] Taken together, therefore, these age-related changes may have significant clinical implications, such as worsening of dehydration in older adults in the setting of vomiting, diarrhea, or reduced water and food intake.

Vasopressin secretion, the renal response to vasopressin, and thirst control are also affected by aging. Vasopressin secretion in the hypothalamus is under the delicate control of osmoreceptors in and around the organum vasculosum of the lamina terminalis and the anterior wall of the third ventricle in the brain. Most studies have found that basal vasopressin concentrations in the healthy older are usually higher than in younger controls.[276] One study has reported no age-related differences in basal vasopressin levels,[286] whereas another study has found that basal vasopressin may actually be lower in older individuals.[287] Nevertheless, most studies about water homeostasis in aging have shown that compared with younger individuals, older adults have a greater increase of vasopressin secretion per unit change in plasma osmolality, and this is consistent with the higher osmoreceptor sensitivity in older adults.[288]

Vasopressin regulates renal water excretion by controlling the abundance of the water channel AQP2 and its insertion into the apical membrane of epithelial cells in the distal nephron and collecting tubules.[289] In the membrane, AQP2 proteins form channels that allow reabsorption of water molecules from the lumen of the collecting ducts into the medullary interstitium driven by the medullary osmotic gradient. Given that vasopressin levels are mostly found to be higher in older adults, a potential secretory defect in the pituitary gland is improbable and cannot explain the age-related reduced renal response to vasopressin. Animal studies have offered potential explanations for the reduced renal response to vasopressin, which include reduced expression of vasopressin receptors in the collecting ducts and impaired second messenger response to vasopressin receptor signaling. One study has demonstrated that despite a normal vasopressin secretory response, there is an age-related reduction in renal concentrating ability after moderate water restriction.[290] This occurred due to the lower expression of AQP2 water channels in older rats and an impaired ability to upregulate AQP2 production. Other animal studies have proposed that reduced vasopressin receptor signaling significantly hinders the medullary concentrating gradient required for urine concentration.[289,290]

Normally, stimulation of thirst osmoreceptors in the hypothalamus signals the higher cerebral cortex to develop a conscious perception of thirst and water-seeking behavior. Aging also affects thirst control.[291] One study has demonstrated that water-deprived older men do not show a subjective thirst increase or mouth dryness.[287] Moreover, when access to water was allowed, compared with younger participants, older adults consumed less water and were unable to restore serum and plasma osmolality to predeprivation levels.[287] This finding suggests that older adults have a reduced thirst response to osmotic changes. It has been proposed that they may have a higher osmolal set point for thirst—that is, for a given plasma osmolality, older adults have a reduced degree of perceived thirst, thus leading to less water intake.[292] Maximum diluting capacity is mildly impaired in older adults, largely a reflection of the reduced GFR.[293] The minimum urinary osmolality (Uosm) after a water load averages about 90 mOsm/kg H_2O in subjects older than 70 years, whereas in younger subjects the value is about 50 mOsm/kg H_2O. If solute excretion is reduced (e.g., with protein or salt restriction), the mild impairment of diluting capacity can expose older patients to water excess syndromes (hyponatremia), with only modest intakes of free water. For example, at a Uosm of 90 mOsm/kg H_2O (maximum dilution in older adults) and a daily osmolal excretion of 360 mOsm/day (about half of normal levels), the maximal electrolyte free water intake would be about 4 L/day. Any free water intake in excess of this value would lead to dilutional hyponatremia.

Thus, aging undoubtedly influences water homeostasis in many different ways. Water conservation (urinary concentrating ability) is primarily affected, although a mild form of impaired diluting capacity may also be present. This is important to keep in mind, not only during the diagnostic process, but also while taking care of older adults and planning for different clinical, surgical, or pharmacologic interventions. Alterations in solute intake (e.g., protein or sodium) in older adults can also have profound effects on water homeostasis.

RENAL ENDOCRINE CHANGES WITH AGING

RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM

Aging is associated with a decline in plasma renin activity (PRA), serum aldosterone levels, and urinary aldosterone excretion rates, accompanied by reduced responses to actions that stimulate the RAAS, such as upright posture or salt depletion.[62,63] This state is known as *aging-associated hyporeninemic hypoaldosteronism*. Both renal renin formation and release are reduced in aging.[294] In addition, activity of the RAAS has been linked to aging phenomena in a pathogenic fashion.[295] Defects in renal NO generation can be linked to reduced activity of the RAAS in aging.[10] The activity of the RAAS can also be linked to canonical Wnt signaling,[10] and both can be dysregulated in the aging process.[8] Although a decline in activity of the RAAS regularly occurs with aging (independently of gender), neither a change in renal perfusion pressure or delivery of solute to the macula densa appears to be involved.[10] Increased ANP levels, reduced sympathetic nervous system activity, or both can be seen with aging and might be involved in the perturbations of the RAAS seen with aging.[10] Reduced renal renin generation with aging appears to be a posttranslational phenomenon due to impaired release of stored renin.[10] Trivial changes in ACE activity and angiotensinogen conversion occur in aging.[10] The sensitivity of the glomerular afferent and efferent vessels to angiotensin II appear to increase with aging.[10] Treatment of animals (rats) with an age-dependent decline in renal function and proteinuria can reduce proteinuria and glomerulosclerosis, but this does not have any effect on glomerulomegaly or the snGFR.[8,10] Distinguishing the effects of aging per se and superimposed disease states (e.g., low nephron endowment) on the RAAS can be very difficult.

Aldosterone production and serum and urinary aldosterone levels are lower in older adults compared with younger persons,[296] and there may also be reduced aldosterone responsivity to elevations in serum potassium levels.[297] In addition, there are age-related changes in adrenal aldosterone synthetase–producing cells, causing islands or clusters of aldosterone-producing cells in the adrenal cortex of aging humans. These may be precursors of aldosterone-secreting adenomas in some cases.[296]

ERYTHROPOIETIN

Serum levels of erythropoietin (EPO) levels tend to increase with age, even in nonanemic subjects.[297,298] This might be in compensation for increased (subclinical) blood loss, accelerated red blood cell turnover (decreased erythrocyte half-life), or increased resistance of red cell precursors to the effect of EPO.[298] Testosterone deficiency might be involved in the latter phenomenon.[299] A mild anemia (hemoglobin <12 g/dL) is fairly common in aging, but this varies by geography.[300] In about one-third of these subjects, a nutritional deficiency (e.g., iron, folate, vitamin B_{12}) is responsible, in one-third, a concomitant chronic disease is the culprit (e.g., CKD, cancer, infection), but in one-third no precise cause can be determined.[300] Independent of anemia, high spontaneous EPO levels in older adults are associated with an increased risk of congestive heart failure, but not myocardial infarction or progressive CKD.[301] The mechanisms responsible for this association are not well understood, but could be due to tissue hypoxia in low-perfusion states associated with heart failure.

IMPLICATIONS OF NORMAL PHYSIOLOGY ON RENAL FUNCTION

This has effects on the diagnosis and prognosis of chronic kidney disease in aging adults.

DIAGNOSIS

The diagnosis of CKD has been codified by the Kidney Disease Outcomes Quality Initiative (KDOQI) and Kidney Disease: Improving Global Outcomes (KDIGO) clinical practice guideline initiatives,[302,303] published in 2002 and 2013, respectively, as well as others from more regional formulations, such as NICE in the United Kingdom and CARI in Australia.[304,305] These classification and categorization recommendations rely principally on the GFR (estimated or measured, in mL/min/1.73 m^2) and albuminuria (usually the urinary albumin-to-creatinine ratio [uACR], in mg/g or mg/mmol] in spot urine samples; see also Chapters 23 and 30). However, other signs of renal injury, such as abnormal urinalyses (e.g., hematuria), imaging, or renal biopsy findings could lead to a diagnosis of CKD, even with a normal GFR (even adjusted for age) or uACR. Abnormalities in these biomarkers must persist for at least 3 months to qualify as a valid indicator of CKD, a requirement that is frequently not met in many epidemiologic studies that are carried out only once. When these classification schemes were developed, it was arbitrarily decided that any adult subject, 20 years of age or older, with a persisting GFR of less than 60 mL/min/1.73 m^2, could be classified as having CKD (category 3A, 3B, 4, or 5), irrespective of any other signs of renal injury, including abnormal urinalysis, albuminuria, imaging, or renal pathology. Thus, a persisting GFR of 45 to 59 mL/min/1.73^2 could lead to a diagnosis of CKD (category 3A) in an adult. This guideline created a conundrum because values of the GFR in this range can also be observed in healthy older adults. The lower limit of normal for the measured GFR (mGFR) in a healthy adult 60 to 69 years of age is about 55 mL/min/1.73 m^2 and, for an adult 70 to 79 years of age, it is about 49 mL/min/1.73 m^2.[133] Significant overlap is present between the threshold for classifying CKD (independent of age) and the normal values for GFR in older adults. The lower value for the GFR in older adults is a manifestation of a physiologic process of nephron loss unaccompanied by compensatory hyperfiltration in the residual nephrons (see above), so it is difficult to conflate these findings as a disease. Not surprisingly, a substantial and continuing debate has arisen concerning the utility of an absolute threshold for defining CKD, based on the GFR alone, without any modification for age.[306–310] Because abnormal albuminuria is not consistent with normal, healthy human aging, this controversy largely focuses on possible overdiagnosis in older adults of the G3A/a1 category of CKD (GFR, 45 to 59 mL/min/1.73 m^2 and uACR of <30 mg/g [3 mg/mmol]) according to KDIGO.[311] The GFR may fall below 45 mL/min/1.73 m^2 in subjects older than 80 years, but it can be very difficult to determine if such subjects are entirely healthy because they frequently have concomitant disorders, such as cancer, heart failure, dietary protein deficiency, and sarcopenia. This can complicate evaluation of the GFR (mGFR or eGFR). In addition, significant comorbidity and biochemical abnormalities are commonly observed in individuals older than 60 years of age, compounding the difficulty of ascertaining kidney health based on the GFR alone.[312,313]

Typically, the GFR values used in categorizing CKD are estimates rather than measured values. This use of estimating formulas for the GFR (eGFR) results in another conundrum in the diagnosis of CKD in aging—namely, the imprecision and variability in eGFR formulas for evaluating the true GFR in older adults[312] (discussed further in Chapters 59 and 84). In a population-wide study of community-living adults, Ebert and coworkers[312] found the prevalence of CKD categories 3, 4, and 5 (eGFR <60 mL/min/1.73 m^2) in subjects 70 to 79 years of age to range between 16% and 52%, depending on the formula used to estimate the GFR. The corresponding value for subjects 80 to 89 years of age were 42% to 84%.

The lowest values for CKD prevalence were found using the CKD-EPI creatinine equation; the highest values were found using the Berlin Initiative Study (BIS)-1 creatinine equation.[312] In subjects older than 70 years of age, the BIS-1 creatine equation is a more precise estimate of mGFR compared with the CKD-EPI equation, at least in a European, largely white population.[314] Estimating formulas for GFR when applied at the population level generally give high values for CKD prevalence (≈11%– 14%),[315] especially when the formulas were derived from a largely CKD cohort. Much lower values for CKD are observed when eGFR formulas are derived from a normal healthy cohort[316]

The use of cystatin C–based GFR estimating equations (alone or in combination with creatinine-based equations) has been advanced as a tool to aid in the diagnosis of CKD. In 2012, KDIGO suggested measuring eGFR-creatinine-cystatin C and eGFR-cystatin C in adults (including older adults) when the eGFR-creatinine is 45 to 59 mL/min/1.73 m^2 (and no other markers of CKD are present). If eGFR-cystatin C or eGFR-creatinine-cystatin C are less than 60 mL/min/1.73 m^2, the diagnosis of CKD is confirmed.[303] The degree to which these suggestions specifically apply to older adults has not been rigorously tested and may depend on the form of the eGFR-cystatin C or eGFR-creatinine-cystatin C equation. For example, in older adults, the BIS-2 cystatin C equation is a more accurate estimate of mGFR than the CKD-EPI-cystatin C equation.[314] Other eGFR equations, such as the FAS-creatinine, FAS-cystatin C equation, or Lund-Malmo creatinine equation need further evaluation for improving the precision and accuracy of the diagnosis of CKD in older adults using the eGFR alone, but early studies have shown great promise for the utility of these equations in an older population.[317]

It must also be recalled that cystatin C is a mid-molecular-weight (13.3 kD) serine proteinase inhibitor involved in the inflammatory response.[318] It is normally filtered and then completely destroyed by tubular reabsorption and degradation. Very little cystatin C is present in normal urine. Thus, production rates of cystatin C are difficult to assess and can vary in many states (e.g., obesity, diabetes, thyroid disease, inflammation) commonly found in older adults. Like creatinine, there are many non-GFR determinants of its serum concentration, the variable used in calculation of the eGFR.[319,320] In older adults, loss of muscle mass (sarcopenia) may lower serum creatinine values relative to the measured GFR and give rise to an overestimate of mGFR by eGFR creatinine equations. However, serum cystatin C levels tend not be affected greatly by muscle mass or gender, and not at all by ancestry. In contrast, in older adults with chronic inflammation (of any cause), obesity, diabetes, and the metabolic syndrome may alter cystatin C production, leading to underestimation of the mGFR. The combination of eGFR-creatinine-cystatin C may give a slightly more accurate assessment of the true GFR in population-based studies,[321] but there will still be much individual variation.[322–324] Tracking eGFR-cystatin C with metabolic factors that can contribute to CVD complicates its suitability as a biomarker for CVD risk related to the GFR in older adults.[325] In addition, endothelial injury consequent to inflammation or capillary hypertension might further impair the transglomerular permeability coefficient for cystatin C (the shrunken pore theory),[326] giving rise to reduced cystatin C eGFR values relative to the inulin or iohexol mGFR.[326] Furthermore, a low serum albumin level (perhaps reflecting subtle degrees of chronic inflammation) is associated with reduced eGFR-creatinine in older frail subjects.[327]

Direct comparisons of a gold standard method of measuring the GFR and estimating formulas for the GFR in aging subjects are relatively uncommon, but both the BIS[314] and Reykjavik older cohort studies[322] have provided much valuable information. In the latter study, the Lund-Malmo and full age spectrum (FAS) creatinine equation have somewhat higher accuracy than the CKD-EPI-creatinine formula. The CKD-EPI, Lund-Malmo, and FAS creatinine-based equations were roughly equivalent for the detection of a mGFR less than 60 mL/min/1.73 m^2. All formulas incorporating a cystatin C measurement exhibited consistent improvements in accuracy compared with the corresponding creatinine-based equations. The relevance of these findings to the diagnosis of CKD in individual older subjects needs further study, but they clearly demonstrate some of the pitfalls in using creatinine-based eGFR formulas for the older adult population.

Thus, the effects of physiologic renal aging on the GFR confound its use as the defining characteristic of CKD in older adults because CKD is a diagnosis made most frequently in older adults with category 3A CKD (GFR, 45–59 mL/min/1.73 m^2), usually with normal albuminuria (category A1; uACR <30 mg/g [3 mg/mmol]). This conundrum raises an issue of possible overdiagnosis and mislabeling of CKD, both at the individual and population-wide level, that distorts evaluation of the true societal burden of CKD. When added to the false-positive identification of CKD in single epidemiologic studies (see above), this leads to the possibility that CKD is not as common as it is alleged to be, and calls for revising the definition of CKD in older adults should be taken seriously. The vagaries of estimating equations for the GFR in older adults cannot be ignored.[328]

PROGNOSIS

In the KDIGO classification of CKD, the prognosis for adverse events (all-cause mortality, ESKD, doubling of serum creatinine level, CVD [e.g., ischemic heart disease, congestive heart failure, stroke]) is the third dimension of the scheme. This is usually displayed as a multicolored heat map of the rising risk of events—green, no added risk; red, high risk—often determined from large-scale epidemiologic studies.[302,303] The risk values are usually stated as relative risk or odds ratio compared with some arbitrary control group, often those with eGFR greater than 60 mL/min/1.73 m^2 and no signs of renal injury (e.g., normal uACR). These prognosis- based heat maps are commonly assembled from data acquired from single time point epidemiologic studies, so that the persistence of the CKD-defining abnormality (eGFR, uACR, or both) is not confirmed. When a more rigorous assessment of chronicity is pursued, it has been noted that such single time point studies commonly overestimate the prevalence of CKD by as much as 30% due to false-positives.[329]

The use of the eGFR to evaluate prognosis is also confounded by the fact that an age variable is included in all adult estimating formulas to adjust for the effects of age per se on the synthesis and production biomarker (creatinine or cystatin C). The age coefficients in these formulas are optimized for estimating the mGFR, not prognostication for all-cause mortality or ESKD risk. However, when the values obtained by such estimates are applied to prognostication, the age variable becomes important, because age, independent of the GFR, has a marked influence on some of the risks evaluated, such as mortality. If an estimating equation using one or more biomarkers that accurately assess GFR, without a requirement for the age variable, were developed, the accuracy of the risk prediction might be improved and made comparable to that found with the true GFR. It should be stressed that although the number and size of studies using the mGFR as the dependent variable in prognostic evaluation are relatively small in comparison to the large-scale, eGFR-based epidemiologic studies, the patterns linking the GFR to outcome are similar. These studies describe a pattern of a threshold (nonlinear) relationship, in which the threshold varies according to age.[330] With albuminuria, the risks varies in a log-linear fashion with the uACR, independent of age, and without a threshold (above normal values). When one compares the relative risk of all-cause mortality using a reference value for the GFR that approximates the normal young adult mean value (≈107 mL/min/1.73 m^2), then the minimum relative risk for an all-cause mortality event is above an eGFR of 75 mL/min/1.73 m^2 for subjects 18 to 54 years of age, but above 45 mL/min/1.73 m^2 for those older than 75 years.[331] The relative risk of all-cause mortality relative to a declining eGFR is blunted by advancing age, but the absolute rates of all-cause mortality remain high with aging. This analysis strongly suggests that with a prognosis-dominated matrix for creating a GFR threshold for CKD, an age-stratified approach is desirable. It is noteworthy that the remaining life expectancy at any age older than about 35 years is not materially affected until the eGFR falls below about 45 mL/min/1.73 m^2.[332–334] Adding an age stratification to the existing scheme for identifying true CKD and its associated risks has proven to be a challenging task. Simple adjustments, such as altering the threshold of eGFR to <45 mL/min/1.73 m^2 for subjects without abnormal albuminuria who are older than 65 years, may be useful, but result in unintended consequences, like the so-called birthday paradox (when a 64-year-old reaches a 65th birthday, CKD is "cured").

Other methods of age stratification for CKD diagnosis have been advanced that may have greater practicality in large health care systems for improved triage.[335] Creation of multiple age-stratified heat maps using the eGFR and estimates of prognosis for all-cause mortality may also be helpful.[336] Whatever the solution may be to this ongoing conundrum of defining CKD and its risks in aging subjects, it is imperative also to recognize that the specific formula for identifying the eGFR as the variable introduces a complexity of its own due to the variability of bias, precision, and accuracy among the numerous estimating formulas and biomarkers used, especially when applied to older adults. For example, the eGFR-creatinine formulas do not add much to the prediction of CVD (and thereby all-cause mortality) risk in older adults, whereas eGFR-cystatin C formulas do, despite not being superior to eGFR-creatinine for estimating the mGFR.[325] Both formulas can incrementally increase the accuracy of CVD risk prediction (see later).

CARDIOVASCULAR DISEASE AND DECLINE IN THE GLOMERULAR FILTRATION RATE WITH AGING

There is little doubt that a progressive decline in the GFR is associated with an increased risk of fatal or nonfatal CVD, including atherosclerotic CVD and congestive heart failure (CHF), sudden cardiac death, nonvalvular atrial fibrillation, stroke, and peripheral vascular disease. Cross-sectional, population-based epidemiologic studies cannot easily determine whether these associations are causal or not. In fact, the existence of CVD may itself result in a decline in the GFR (e.g., severe CHF, atherosclerotic renovascular diseases, ischemic nephropathy), thus confounding the directional nature of the observed associations. Aging itself can be a mechanism for some of the associations between a decline in the GFR and CVD, but this can be adjusted for by examining the CVD prevalence in older subjects with a better-preserved GFR. However, the fact that the GFR declines regularly with aging requires that the comparator group used for such adjustments have GFR values (on average) similar to those predicted for by normal physiologic aging. Teasing out age-related GFR decline from pathologic GFR decline in an aging individual can be difficult, but the demonstration of other biomarkers of renal injury, such as the presence of abnormal albuminuria or abnormal imaging, can be very helpful (abnormal albuminuria does not occur with healthy aging).[91] The threshold for a GFR-related augmentation of CVD risk is likely to be age-dependent. In studies of older individuals, the risk of excess CVD events seem to appear at an eGFR less than 45 mL/min/1.73 m^2 (category 3B CKD),[333] but this will likely be modified by the nature and severity of comorbidities present, such as dyslipidemia, diabetes, or long-standing hypertension. A decreased GFR can influence the risk of CVD, independent of the traditional risk factors (e.g., obesity, smoking, dyslipidemia, diabetes, hypertension), but the magnitude of the additional risk for CVD events imposed by a reduced GFR in the range of 45 to 59 mL/min/1.73 m^2 is small in older adults. For this reason, a reduced GFR is not commonly included in CVD risk prediction scoring systems (e.g., Framingham Risk Score, American College of Cardiology/American Heart Association [ACC/AHA] pooled risk prediction model).[337]

As discussed in detail in other chapters (see Chapters 40 and 54), more advanced forms of CKD can augment the CVD risk by numerous mechanisms, including endothelial cell or vascular wall injury from uremic toxins, vascular ossification, myocardial hypertrophy and pathologic remodeling, chronic volume expansion, chronic inflammation, uremic dyslipidemia (proatherogenic high-density lipoprotein [HDL], hypertriglyceridemia), arterial hypertension, and thrombotic microangiopathy. The fundamental phenomenon of aging can interact with these pathologic processes in numerous ways that are very difficult to disentangle unambiguously.

END-STAGE RENAL DISEASE AND AGING

A requirement for treatment by renal replacement modalities (renal replacement therapy [RRT]; dialysis and/or transplantation) is not uncommon in older adults. According to the US Renal Data System (USRDS) Annual Report in 2017, the age-specific annual incidence rate for RRT for subjects 75 years of age or older was about 1600 per million population (pmp) and, for those 65 to 74 years of age, was about 1250 pmp in 2015 (www.usrds.org). Both these values have been slowly declining (for uncertain reasons) since about 2010 but are still well above the average (all-age) RRT incident rate of 378 pmp/year. As has been the case for many years, males outnumber females for incident counts of RRT by about 20% to 30%. For pre-ESKD CKD, the male-to-female ratio is exactly the opposite (www.usrds.org). However, this may in part be artifactual due to the gender coefficient in eGFR creatinine equations, which give women lower eGFR values, and the lower muscle mass in women than men, which increases their ACR. The precursors of ESKD are the earlier stages (categories) of CKD, some of which progress, at varying rates, to ESKD. An important consideration of the development of RRT requiring ESKD is the competing risk of death, usually from a CVD or neoplasia event. This risk is magnified in older adults so that at an advanced age, any patient with CKD category 3 (eGFR, 30–59 mL/min/1.73 m^2) is more likely to die before reaching ESKD.[338]

In a landmark study, Eriksen and Ingebretsen conducted a 10-year, population- based study in Tromso, Norway, including subjects (median age, 75 years; interquartile range [IQR], 67.7–80.4 years) with confirmed CKD category 3.[340] About 70% of the subjects experienced some decline in the eGFR over time, somewhat slower in females than in males. The 10-year cumulative risk of reaching treated ESKD was 4% (greater in men than in women), whereas the cumulative risk of death was 51% (greater in men than in women). The prognosis of CKD and the eventual likelihood of requiring RRT was highly gender-dependent and greatly influenced by the competing risk of death. In another key study, O'Hare and coworkers[338] studied 209,622 US veterans (97% male; mean age, 73 ± 9 years) with category 3 to 5 CKD followed for a mean of 3.2 years. Although the rates of death and ESKD were inversely related to the eGFR value at baseline irrespective of age, at comparable levels of eGFR, older age was associated with higher death rates and a lower rate of ESKD treatment. The threshold of eGFR below which the rate of ESKD exceeded the risk of death was about 15 mL/min/1.73 m^2 for subjects 65 to 84 years of age. As with the Eriksen and Ingebretsen study,[340] the rate of decline of the eGFR was slower in the older subjects. The effect of gender could not be examined in this study. Nevertheless, both these studies clearly indicated that both age and gender are important modifying factors for the transition from CKD (category 3–5) to treated ESKD.

The impact of age on the association of eGFR and albuminuria on outcomes of mortality and treated ESKD were also studied in an exhaustive analysis of 2,051,244 participants from 33 general population or high-risk CVD cohorts and 13 CKD cohorts by Hallan and colleagues in the CKD Prognosis Consortium.[330] In most of these cohorts, the eGFR values were determined only once, and CKD was not confirmed by the duration criteria, leading to the possibility of a high false-positive rate for the identification of CKD. In addition, the analysis used a single reference value of 80 mL/min/1.73 m^2 for the eGFR. The study showed a roughly comparable hazard ratio for the risk of ESKD with declining eGFR at all ages, but the average absolute rate of ESKD declined substantially with age as a function of the declining eGFR, consistent with the observations discussed above. The relative and absolute risks for ESKD both increased as the uACR rose above 10 mg/g, but the magnitude of these risks were somewhat blunted in much older adults (>75 years). Although a threshold of about 60 mL/min/1.73 m^2 was noted for defining an increased relative risk of ESKD at all ages, the choice of a single reference comparator of 80 mL/min/1.73 m^2 (rather than the lowest risk group for age as the reference comparator) might have influenced the findings, as discussed above in relationship to the impact of the eGFR on all-cause mortality at varying ages). It is noteworthy that the absolute rate of ESKD in the very old (>75 years) did not increase above baseline values until the eGFR was well below 45 mL/min/1.73 m^2.

In a longitudinal study conducted by Shardlow and colleagues[341,342] in the United Kingdom, older subjects (mean age, 73 years) with category 3 CKD (mean eGFR, 53 mL/min/1.73 m^2) had a low prevalence of abnormal albuminuria (16%). After 5 years of additional follow-up, they had a low prevalence of ESKD (0.2%), progression to a higher stage of CKD occurred in 17.7%, remissions of CKD developed in 19.3%, and stable renal function was observed in 34.1%. Albuminuria at diagnosis was a key risk factor for progression. In this study, 14.2% died before the 5-year follow-up and 14.8% were lost to follow-up.

ACUTE KIDNEY INJURY AND DISEASES OF THE KIDNEY AND URINARY TRACT IN AGING

A strong relationship exists between the incidence and prevalence of AKI and age.[343] The pathophysiologic mechanisms underlying this association are complex and multifactorial (discussed further in Chapter 28). The physiologic reduction of nephron number and GFR with advancing age, discussed above, alters the pharmacokinetics of many water-soluble drugs. This can lead to an accumulation of nephrotoxic levels of agents in body fluids, leading to AKI.[344] Dosage modifications, primarily in the interval between dosing, rather than in loading dosage, are required to minimize such risks.[198] Agents that affect glomerular hemodynamics, such as NSAIDs or RAAS inhibitors, can cause a reduction in the GFR in aging subjects that is more profound than in younger subjects, particularly when combined with some degree of extracellular volume depletion. Older subjects also may have a comorbidity that exposes them to an episode of AKI, such as CHF, diabetes, urosepsis, and cancer, and require more interventions that may predispose to AKI, such as surgery and angiography. As discussed later, aging is also associated with an increased prevalence of diseases of the kidney and urinary tract that may directly cause AKI.

DISEASES OF THE KIDNEY AND URINARY TRACT IN AGING

The occurrence of specific diseases of the kidneys and urinary tract can be influenced markedly by the age of a subject. These diseases are discussed in detail in Chapters 36, 37, and 41 and will not be elaborated further here. Certain glomerular and vascular diseases have a predilection to affect older adults (Box 22.2).[31,345–347] Benign prostatic hypertrophy is common in older men and can result in lower urinary tract obstruction and CKD, sometimes irreversible. Obstructive uropathy can also be the consequence of ureteric involvement in gynecologic cancer in older women and bladder dysfunction associated with neurologic diseases and diabetes.

Box 22.2 Kidney and Urinary Tract Diseases That Occur More Commonly in Older Adults

Glomerular and Vascular Diseases

Crescentic glomerulonephritis due to systemic or renal-limited antineutrophil cytoplasmic autoantibody (ANCA)–associated vasculitis
Membranous nephropathy (primary and cancer-related)
Monoclonal immunoglobulin deposition diseases (AL amyloidosis and nonamyloid types)
Diabetic nephropathy
Fibrillary glomerulonephritis
Anti–glomerular basement membrane (GBM) disease (mainly females)
Atheroembolic disease
Atheromatous renovascular stenosis

Tubulointerstitial Diseases

Acute kidney injury (toxic or ischemic cause)
Lymphomatous infiltration
Renal cell cancer

Urinary Tract Disorders

Lower urinary tract obstruction due to prostate disease
Ureteric obstruction from cancer (cervical cancer in females)
Transitional cell carcinoma of the bladder
Prostate cancer

CAN RENAL AGING BE MODIFIED?

As the molecular mechanisms of organ senescence are gradually revealed, it is natural to ask about their potential for modification.[348–351] This is a difficult question because the biologic processes are so complex and are active over an extended time period. In addition, it is necessary to make a clear distinction between the effects of interventions directed at diseases associated with aging (e.g., better glycemic control in diabetes, avoidance of obesity, control of hypertension) and the modification of the fundamental aging process itself. So far, no elixir of life or "fountain of youth" has been discovered, but several promising avenues have been identified. Caloric restriction, mTOR inhibition, and RAAS blockade all retard aging across many animal species,[352] but their effects in humans are less well understood for obvious reasons. It is possible that all three of these approaches converge on mitochondrial energy production to slow aging. The angiotensin II type I receptor may play a pivotal role in renal senility,[352] independent of angiotensin II, perhaps via a sirtuin-linked mechanism.[352,353] However, because normal healthy aging, with the steady attrition of glomeruli, is not usually accompanied by albuminuria or a rise in the snGFR (see above), the prospects of RAAS inhibition as a means of retarding the rate of renal aging are not promising. Caloric restriction seems to cause a reduction in renal fibrosis (via mTOR inhibition and 5'-adenosine monophosphate-activated protein kinase [AMPK] activation).[35,353] Calorie restriction and mTOR inhibition activates sirtuin 1, which has antiaging properties. Other agents, such as the biguanide metformin, may also have antiaging effects via an action on the AMPK/mTOR pathway.[354] Telomere shortening can be modulated by telomerase activity, but whether the manipulation of telomere length will be safe and effective in delaying organ senescence is unknown. Modulating Klotho expression pharmacologically might have beneficial effects, via reduced oxidative stress. Elimination of senescent cells (senolytic therapy) by targeting apoptosis of senescent cells (TASCs) via disruption of the FOX04 peptide-p53 interaction, shows great promise in improving renal function in aging mice.[355,356] Resveratrol, a polyphenol present in many pigmented vegetables and fruits, can ameliorate aging as a calorie restriction mimetic, possibly by improving the metabolic profile and inhibiting cyclic adenosine monophosphate (cAMP) phosphodiesterases and the AMPK pathway.[357] Because an imbalance of energy demand and energy supply may underlie aging, a reprogramming of metabolism from anaerobic glycolysis to aerobic glycolysis might be an effective strategy.[5] Antifibrosis regimens, such as with the transforming growth factor beta (TGF-β)/smad pathway inhibition, BMP-7 administration, or the use of other agents, represents a promising avenue of research,[339,358] especially in the domain of renal aging. Translating studies carried out in lower animals (e.g., *Caenorhabditis elegans*, mice, rats) to the human organism will be tedious, time-consuming, and fraught with obstacles. Armed with new tools, however, such as nephron number quantification, it may be possible to test some of the most promising strategies directly in humans. Hopefully, in the not too distant future, renal aging will not be inevitable, even though immortality will of course continue to be an elusive goal.

Complete reference list available at ExpertConsult.com.

KEY REFERENCES

2. Robert L, Fulop T. Aging: facts and theories. *Interdiscip Top Gerontol.* 2014;39:1–215.
7. Epstein M. Aging and the kidney. *J Am Soc Nephrol.* 1996;7(8):1106–1122. Review.
23. Bitzer M, Wiggins J. Aging biology in the kidney. *Adv Chronic Kidney Dis.* 2016;23(1):12–18. doi:10.1053/j.ackd.2015.11.005. Review.
30. Choudhury D, Levi M. Aging and kidney Disease. In: *Brenner and Rector, the Kidney.* 10th ed. New York: Elsevier; 2015:727–751.
60. Yang HC, Fogo AB. Fibrosis and renal aging. *Kidney Int Suppl (2011).* 2014;4(1):75–78. Review.
63. Conti S, Cassis P, Benigni A. Aging and the renin-angiotensin system. *Hypertension.* 2012;60(4):878–883. [Epub 2012 Aug 27]; Review.
68. Inker LA, Okparavero A, Tighiouart H, et al. Midlife blood pressure and late-life GFR and albuminuria: an elderly general population cohort. *Am J Kidney Dis.* 2015;66(2):240–248. doi:10.1053/j.ajkd.2015.03.030. [Epub 2015 May 16].
71. Luyckx VA, Brenner BM. Birth weight, malnutrition and kidney-associated outcomes–a global concern. *Nat Rev Nephrol.* 2015;11(3):135–149. doi:10.1038/nrneph.2014.251. [Epub 2015 Jan 20]; Review.
86. Denic A, Alexander MP, Kaushik V, et al. Detection and clinical patterns of nephron hypertrophy and nephrosclerosis among apparently healthy adults. *Am J Kidney Dis.* 2016;68:58–67.
88. Denic A, Lieske JC, Chakkera HA, et al. The substantial loss of nephrons in healthy human kidneys with aging. *J Am Soc Nephrol.* 2017;28:313–320.
91. Rule AD, Amer H, Cornell LD, et al. The association between age and nephrosclerosis on renal biopsy among healthy adults. *Ann Intern Med.* 2010;152:561–567.
106. Thomson SC, Vallon V, Blantz RC. Kidney function in early diabetes: the tubular hypothesis of glomerular filtration. *Am J Physiol Renal Physiol.* 2004;286:F8–F15.
109. Hodgin JB, Bitzer M, Wickman L, et al. Glomerular aging and focal global glomerulosclerosis: a podometric perspective. *J Am Soc Nephrol.* 2015;26:3162–3178.

114. Denic A, Mathew J, Nagineni VV, et al. Clinical and pathology findings associate consistently with larger glomerular volume. *J Am Soc Nephrol*. 2018;29(7):1960–1969. [Epub 2018 May 22]; Erratum in: *J Am Soc Nephrol*. 2018;29(9):2445.
116. Rule AD, Semret MH, Amer H, et al. Association of kidney function and metabolic risk factors with density of glomeruli on renal biopsy samples from living donors. *Mayo Clin Proc*. 2011;86:282–290.
124. Martin JE, Sheaff MT. Renal ageing. *J Pathol*. 2007;211:198–205.
128. Lindeman RD, Tobin J, Shock NW. Longitudinal studies on the rate of decline in renal function with age. *J Am Geriatr Soc*. 1985;33:278–285.
133. Pottel H, Delanaye P, Weekers L, et al. Age-dependent reference intervals for estimated and measured glomerular filtration rate. *Clin Kidney J*. 2017;10(4):545–551. doi:10.1093/ckj/sfx026. [Epub 2017 Apr 28].
135. Cohen E, Nardi Y, Krause I, et al. A longitudinal assessment of the natural rate of decline in renal function with age. *J Nephrol*. 2014;27(6):635–641.
139. Denic A, Mathew J, Lerman LO, et al. Single-nephron glomerular filtration rate in healthy adults. *N Engl J Med*. 2017;376:2349–2357.
145. Hoang K, Tan JC, Derby G, et al. Determinants of glomerular hypofiltration in aging humans. *Kidney Int*. 2003;64(4):1417–1424.
147. Chakkera HA, Denic A, Kremers WK, et al. Comparison of high glomerular filtration rate thresholds for identifying hyperfiltration. *Nephrol Dial Transplant*. 2018;doi:10.1093/ndt/gfy332. [Epub ahead of print].
160. De Moor B, Vanwalleghem JF, Swennen Q, et al. Haemodynamic or metabolic stimulation tests to reveal the renal functional response: requiem or revival? *Clin Kidney J*. 2018;11(5):623–654.
164. Reckelhoff JF, Samsell L, Dey R, et al. The effect of aging on glomerular hemodynamics in the rat. *Am J Kidney Dis*. 1992;20(1):70–75.
175. Glassock RJ, Rule AD. The implications of anatomical and functional changes of the aging kidney: with an emphasis on the glomeruli. *Kidney Int*. 2012;82:270–277.
198. Frame AA, Wainford RD. Mechanisms of altered renal sodium handling in age-related hypertension. *Am J Physiol Renal Physiol*. 2018;315(1):F1–F6.
205. Kyle UG, Genton L, Hans D, et al. Age-related differences in fat-free mass, skeletal muscle, body cell mass and fat mass between 18 and 94 years. *Eur J Clin Nutr*. 2001;55:663–672.
230. Kuro-o M, Matsumura Y, Aizawa H, et al. Mutation of the mouse klotho gene leads to a syndrome resembling ageing. *Nature*. 1997; 390:45–51.
233. Musso CG, Juarez R, Vilas M, et al. Renal calcium, phosphorus, magnesium and uric acid handling: comparison between stage III chronic kidney disease patients and healthy oldest old. *Int Urol Nephrol*. 2012;44:1559–1562.
246. Ameri P, Canepa M, Milaneschi Y, et al. Relationship between vitamin D status and left ventricular geometry in a healthy population: results from the Baltimore Longitudinal Study of Aging. *J Intern Med*. 2013;273(3):253–262.
258. Amodu A, Abramowitz MK. Dietary acid, age, and serum bicarbonate levels among adults in the United States. *Clin J Am Soc Nephrol*. 2013;8:2034–2042.
274. Sands JM. Urine concentrating and diluting ability during aging. *J Gerontol A Biol Sci Med Sci*. 2012;67:1352–1357.
281. Beck LH. The aging kidney. Defending a delicate balance of fluid and electrolytes. *Geriatrics*. 2000;55:26–28, 31–32.
297. Mulkerrin E, Epstein FH, Clark BA. Aldosterone responses to hyperkalemia in healthy older humans. *J Am Soc Nephrol*. 1995;6(5): 1459–1462.
298. Ershler WB, Sheng S, McKelvey J, et al. Serum erythropoietin and aging: a longitudinal analysis. *J Am Geriatr Soc*. 2005;53(8):1360–1365.
307. Glassock R, Delanaye P, El Nahas M. An age-calibrated classification of chronic kidney disease. *JAMA*. 2015;314(6):559–560.
308. Levey AS, Inker LA, Coresh J. Chronic kidney disease in older people. *JAMA*. 2015;314(6):557–558.
312. Ebert N, Jakob O, Gaedeke J, et al. Prevalence of reduced kidney function and albuminuria in older adults: the Berlin Initiative Study. *Nephrol Dial Transplant*. 2017;32(6):997–1005.
315. Glassock RJ, Warnock DG, Delanaye P. The global burden of chronic kidney disease: estimates, variability and pitfalls. *Nat Rev Nephrol*. 2017;13(2):104–114.
317. Pottel H, Delanaye P, Schaeffner E, et al. Estimating glomerular filtration rate for the full age spectrum from serum creatinine and cystatin C. *Nephrol Dial Transplant*. 2017;32(3):497–507.
323. Meeusen JW, Rule AD, Voskoboev N, et al. Performance of cystatin C- and creatinine-based estimated glomerular filtration rate equations depends on patient characteristics. *Clin Chem*. 2015;61(10):1265–1272.
324. Fan L, Levey AS, Gudnason V, et al. Comparing GFR estimating equations using cystatin C and creatinine in older individuals. *J Am Soc Nephrol*. 2015;26(8):1982–1989.
325. Nerpin E, Ingelsson E, Risérus U, et al. The combined contribution of albuminuria and glomerular filtration rate to the prediction of cardiovascular mortality in older men. *Nephrol Dial Transplant*. 2011;26(9):2820–2827.
326. Grubb A, Lindström V, Jonsson M, et al. Reduction in glomerular pore size is not restricted to pregnant women. Evidence for a new syndrome: 'Shrunken pore syndrome'. *Scand J Clin Lab Invest*. 2015;75(4):333–340.
330. Hallan SI, Matsushita K, Sang Y, et al. Age and association of kidney measures with mortality and end-stage renal disease. *JAMA*. 2012;308(22):2349–2360.
331. Denic A, Glassock RJ, Rule AD. Structural and functional changes with the aging kidney. *Adv Chronic Kidney Dis*. 2016;23(1):19–28.
332. Gansevoort RT, Correa-Rotter R, Hemmelgarn BR, et al. Chronic kidney disease and cardiovascular risk: epidemiology, mechanisms, and prevention. *Lancet*. 2013;382(9889):339–352.
336. Warnock DG, Delanaye P, Glassock RJ. Risks for all-cause mortality: stratified by age, estimated glomerular filtration rate and albuminuria. *Nephron*. 2017;136(4):292–297.
338. O'Hare AM, Choi AI, Bertenthal D, et al. Age affects outcomes in chronic kidney disease. *J Am Soc Nephrol*. 2007;18(10):2758–2765.
341. Shardlow A, McIntyre NJ, Fluck RJ, et al. Chronic kidney disease in primary care: outcomes after five years in a prospective cohort study. *PLoS Med*. 2016;13(9):e1002128.
346. Jin B, Zeng C, Ge Y, et al. The spectrum of biopsy-proven kidney diseases in elderly Chinese patients. *Nephrol Dial Transplant*. 2014;29(12):2251–2259.
351. Kanasaki K, Kitada M, Koya D. Pathophysiology of the aging kidney and therapeutic interventions. *Hypertens Res*. 2012;35(12):1121–1128.

REFERENCES

1. Finch CE, Kirkwood TBL. *Chance, Development and Aging*. Oxford: Oxford University Press; 2000.
2. Robert L, Fulop T. Aging: facts and theories. *Interdiscip Top Gerontol*. 2014;39:1–215.
3. Vijg J, Le Bourg E. Aging and the Inevitable Limit to Human Life Span. *Gerontology*. 2017;63(5):432–434. doi:10.1159/000477210. [Epub 2017 May 17]; PubMed PMID: 28511176.
4. van den Berg N, Beekman M, Smith KR, et al. Historical demography and longevity genetics: back to the future. *Ageing Res Rev*. 2017;38:28–39. doi:10.1016/j.arr.2017.06.005. [Epub 2017 Jul 5]; Review. PubMed PMID: 28689042.
5. Feng Z, Hanson RW, Berger NA, et al. Reprogramming of energy metabolism as a driver of aging. *Oncotarget*. 2016;7(13):15410–15420. doi:10.18632/oncotarget.7645. Review. PubMed PMID: 26919253. PubMed Central PMCID: PMC4941250.
6. Kaysen GA, Myers BD. The aging kidney. *Clin Geriatr Med*. 1985; 1(1):207–222. Review. PubMed PMID: 3913500.
7. Epstein M. Aging and the kidney. *J Am Soc Nephrol*. 1996;7(8):1106–1122. Review. PubMed PMID: 8866401.
8. Baylis C, Schmidt R. The aging glomerulus. *Semin Nephrol*. 1996; 16(4):265–276. Review. PubMed PMID: 8829265.
9. Thomas SE, Anderson S, Gordon KL, et al. Tubulointerstitial disease in aging: evidence for underlying peritubular capillary damage, a potential role for renal ischemia. *J Am Soc Nephrol*. 1998;9(2):231–242. PubMed PMID: 9527399.
10. Baylis C, Corman B. The aging kidney: insights from experimental studies. *J Am Soc Nephrol*. 1998;9(4):699–709. Review. PubMed PMID: 9555673.
11. Schlanger L. Kidney senesceence. Geriatric nephrology curriculum. *Am Soc Nephrol*. 2009;1–7.
12. Weinstein JR, Anderson S. The aging kidney: physiological changes. *Adv Chronic Kidney Dis*. 2010;17(4):302–307. doi:10.1053/j.ackd.2010.05.002. Review. PubMed PMID: 20610357. PubMed Central PMCID: PMC2901622.
13. Rodríguez-Castro EM, Córdova HR. Aging and the kidney. *Bol Asoc Med P R*. 2011;103(3):57–62. Review. PubMed PMID: 23210336.
14. Perico N, Remuzzi G, Benigni A. Aging and the kidney. *Curr Opin Nephrol Hypertens*. 2011;20(3):312–317. doi:10.1097/MNH.0b013e328344c327. Review. PubMed PMID: 21358327.
15. John GB, Cheng CY, Kuro-o M. Role of Klotho in aging, phosphate metabolism, and CKD. *Am J Kidney Dis*. 2011;58(1):127–134. doi:10.1053/j.ajkd.2010.12.027. [Epub 2011 Apr 15]; Review. PubMed PMID: 21496980. PubMed Central PMCID: PMC3191324.
16. Sataranatarajan K, Feliers D, Mariappan MM, et al. Molecular events in matrix protein metabolism in the aging kidney. *Aging Cell*. 2012;11(6):1065–1073. doi:10.1111/acel.12008. [Epub 2012 Oct 19]; PubMed PMID: 23020145. PubMed Central PMCID: PMC5812369.
17. Wiggins JE. Aging in the glomerulus. *J Gerontol A Biol Sci Med Sci*. 2012;67(12):1358–1364. doi:10.1093/gerona/gls157. [Epub 2012 Jul 25]; Review. PubMed PMID: 22843670. PubMed Central PMCID: PMC3636672.
18. Karam Z, Tuazon J. Anatomic and physiologic changes of the aging kidney. *Clin Geriatr Med*. 2013;29(3):555–564. doi:10.1016/j.cger.2013.05.006. Review. PubMed PMID: 23849007.
19. Bolignano D, Mattace-Raso F, Sijbrands EJ, et al. The aging kidney revisited: a systematic review. *Ageing Res Rev*. 2014;14:65–80. doi:10.1016/j.arr.2014.02.003. [Epub 2014 Feb 15]; Review. PubMed PMID: 24548926.
20. Wang X, Bonventre JV, Parrish AR. The aging kidney: increased susceptibility to nephrotoxicity. *Int J Mol Sci*. 2014;15(9):15358–15376. doi:10.3390/ijms150915358. Review. PubMed PMID: 25257519. PubMed Central PMCID: PMC4200815.
21. Schmitt R, Susnik N, Melk A. Molecular aspects of renal senescence. *Curr Opin Organ Transplant*. 2015;20(4):412–416. doi:10.1097/MOT.0000000000000214. Review. PubMed PMID: 26126196.
22. O'Sullivan ED, Hughes J, Ferenbach DA. Renal aging: causes and consequences. *J Am Soc Nephrol*. 2017;28(2):407–420. doi:10.1681/ASN.2015121308. [Epub 2016 Nov 15]; Review. PubMed PMID: 28143966. PubMed Central PMCID: PMC5280008.
23. Bitzer M, Wiggins J. Aging biology in the kidney. *Adv Chronic Kidney Dis*. 2016;23(1):12–18. doi:10.1053/j.ackd.2015.11.005. Review. PubMed PMID: 26709058.
24. Schmitt R, Melk A. Molecular mechanisms of renal aging. *Kidney Int*. 2017;92(3):569–579. doi:10.1016/j.kint.2017.02.036. [Epub 2017 Jul 18]; Review. PubMed PMID: 28729036.
25. Gekle M. Kidney and aging - A narrative review. *Exp Gerontol*. 2017;87(Pt B):153–155. doi:10.1016/j.exger.2016.03.013. [Epub 2016 Mar 24]; Review. PubMed PMID: 27032877.
26. Sobamowo H, Prabhakar SS. The kidney in aging: physiological changes and pathological implications. *Prog Mol Biol Transl Sci*. 2017;146:303–340. doi:10.1016/bs.pmbts.2016.12.018. [Epub 2017 Feb 13]; Review. PubMed PMID: 28253989.
27. Sturmlechner I, Durik M, Sieben CJ, et al. Cellular senescence in renal ageing and disease. *Nat Rev Nephrol*. 2017;13(2):77–89. doi:10.1038/nrneph.2016.183. [Epub 2016 Dec 28]; Review. PubMed PMID: 28029153.
28. Shiels PG, McGuinness D, Eriksson M, et al. The role of epigenetics in renal ageing. *Nat Rev Nephrol*. 2017;13(8):471–482. doi:10.1038/nrneph.2017.78. [Epub 2017 Jun 19]; Review. PubMed PMID: 28626222.
29. Hommos MS, Glassock RJ, Rule AD. Structural and functional changes in human kidneys with healthy aging. *J Am Soc Nephrol*. 2017;28(10):2838–2844. doi:10.1681/ASN.2017040421. [Epub 2017 Aug 8]; Review. PubMed PMID: 28790143. PubMed Central PMCID: PMC5619977.
30. Choudhury D, Levi M. Aging and kidney Disease. In: *Brenner and Rector, the Kidney*. 10th ed. New York: Elsevier; 2015:727–751.
31. Nunez JFM, Cameron JS, Oreopoulos DG, eds. *The Aging Kidney in Health and Disease*. New York: Springer; 2008.
32. López-Otín C, Blasco MA, Partridge L, et al. The hallmarks of aging. *Cell*. 2013;153(6):1194–1217. doi:10.1016/j.cell.2013.05.039. Review. PubMed PMID: 23746838. PubMed Central PMCID: PMC3836174.
33. Chistiakov DA, Sobenin IA, Revin VV, et al. Mitochondrial aging and age-related dysfunction of mitochondria. *Biomed Res Int*. 2014;2014:238463. doi:10.1155/2014/238463. [Epub 2014 Apr 10]; Review. PubMed PMID: 24818134. PubMed Central PMCID: PMC4003832.
34. Davies JMS, Cillard J, Friguet B, et al. The Oxygen Paradox, the French Paradox, and age-related diseases. *Geroscience*. 2017;39(5–6):499–550. doi:10.1007/s11357-017-0002-y. [Epub 2017 Dec 21]; Review. PubMed PMID: 29270905. PubMed Central PMCID: PMC5745211.
35. Dong D, Cai GY, Ning YC, et al. Alleviation of senescence and epithelial-mesenchymal transition in aging kidney by short-term caloric restriction and caloric restriction mimetics via modulation of AMPK/mTOR signaling. *Oncotarget*. 2017;8(10):16109–16121. doi:10.18632/oncotarget.14884. PubMed PMID: 28147330. PubMed Central PMCID: PMC5369951.
36. Takabatake Y, Kimura T, Takahashi A, et al. Autophagy and the kidney: health and disease. *Nephrol Dial Transplant*. 2014;29(9):1639–1647. doi:10.1093/ndt/gft535. [Epub 2014 Feb 10]; Review. PubMed PMID: 24520117.
37. Lenoir O, Tharaux PL, Huber TB. Autophagy in kidney disease and aging: lessons from rodent models. *Kidney Int*. 2016;90(5):950–964. doi:10.1016/j.kint.2016.04.014. [Epub 2016 Jun 18]; Review. PubMed PMID: 27325184.
38. Han X, Tai H, Wang X, et al. AMPK activation protects cells from oxidative stress-induced senescence via autophagic flux restoration and intracellular NAD(+) elevation. *Aging Cell*. 2016;15(3):416–427. doi:10.1111/acel.12446. [Epub 2016 Feb 18]; PubMed PMID: 26890602. PubMed Central PMCID: PMC4854918.
39. Jankauskas SS, Pevzner IB, Andrianova NV, et al. The age-associated loss of ischemic preconditioning in the kidney is accompanied by mitochondrial dysfunction, increased protein acetylation and decreased autophagy. *Sci Rep*. 2017;7:44430. doi:10.1038/srep44430. PubMed PMID: 28294175. PubMed Central PMCID: PMC5353572.
40. Chuang PY, Cai W, Li X, et al. Reduction in podocyte SIRT1 accelerates kidney injury in aging mice. *Am J Physiol Renal Physiol*. 2017;313(3):F621–F628. doi:10.1152/ajprenal.00255.2017. [Epub 2017 Jun 14]; PubMed PMID: 28615249. PubMed Central PMCID: PMC5625108.
41. Huang W, Liu H, Zhu S, et al. Sirt6 deficiency results in progression of glomerular injury in the kidney. *Aging (Albany NY)*. 2017;9(3):1069–1083. doi:10.18632/aging.101214. PubMed PMID: 28351995. PubMed Central PMCID: PMC5391219.
42. Lowe D, Horvath S, Raj K. Epigenetic clock analyses of cellular senescence and ageing. *Oncotarget*. 2016;7(8):8524–8531. doi:10.18632/

oncotarget.7383. PubMed PMID: 26885756. PubMed Central PMCID: PMC4890984.
43. Marioni RE, Shah S, McRae AF, et al. DNA methylation age of blood predicts all-cause mortality in later life. *Genome Biol*. 2015;16:25. doi:10.1186/s13059-015-0584-6. PubMed PMID: 25633388. PubMed Central PMCID: PMC4350614.
44. Rizvi S, Raza ST, Mahdi F. Telomere length variations in aging and age-related diseases. *Curr Aging Sci*. 2014;7(3):161–167. Review. PubMed PMID: 25612739.
45. De Vusser K, Pieters N, Janssen B, et al. Telomere length, cardiovascular risk and arteriosclerosis in human kidneys: an observational cohort study. *Aging (Albany NY)*. 2015;7(10):766–775.
46. Shay JW, Wright WE. Hayflick, his limit, and cellular ageing. *Nat Rev Mol Cell Biol*. 2000;1(1):72–76. doi:10.1038/35036093. PubMed PMID: 11413492.
47. Ullrich NJ, Gordon LB. Hutchinson-Gilford progeria syndrome. *Handb Clin Neurol*. 2015;132:249–264. doi:10.1016/B978-0-444-62702-5.00018-4. Review. PubMed PMID: 26564085.
48. Ben Chehida A, Ghali N, Ben Abdelaziz R, et al. Renal involvement in 2 siblings with Cockayne syndrome. *Iran J Kidney Dis*. 2017;11(3):253–255. PubMed PMID: 28575888.
49. Gan W, Liu XL, Yu T, et al. Urinary 8-oxo-7,8-dihydroguanosine as a Potential Biomarker of Aging. *Front Aging Neurosci*. 2018;10:34. doi:10.3389/fnagi.2018.00034. eCollection 2018. PubMed PMID: 29535624. PubMed Central PMCID: PMC5835306.
50. Noordmans GA, Hillebrands JL, van Goor H, et al. A roadmap for the genetic analysis of renal aging. *Aging Cell*. 2015;14(5):725–733. doi:10.1111/acel.12378. [Epub 2015 Jul 29]; Review. PubMed PMID: 26219736. PubMed Central PMCID: PMC4568960.
51. Rodwell GE, Sonu R, Zahn JM, et al. A transcriptional profile of aging in the human kidney. *PLoS Biol*. 2004;2(12):e427. [Epub 2004 Nov 30]; PubMed PMID: 15562319. PubMed Central PMCID: PMC532391.
52. Noordmans GA, Huang Y, Savage H, et al. Genetic analysis of intracapillary glomerular lipoprotein deposits in aging mice. *PLoS ONE*. 2014;9(10):e111308. doi:10.1371/journal.pone.0111308. eCollection 2014. PubMed PMID: 25353171. PubMed Central PMCID: PMC4213026.
53. Lim JH, Kim EN, Kim MY, et al. Age-associated molecular changes in the kidney in aged mice. *Oxid Med Cell Longev*. 2012;2012:171383. doi:10.1155/2012/171383. [Epub 2012 Dec 30]; PubMed PMID: 23326623. PubMed Central PMCID: PMC3544311.
54. Inagi R. Glycative stress and glyoxalase in kidney disease and aging. *Biochem Soc Trans*. 2014;42(2):457–460. doi:10.1042/BST20140007. Review. PubMed PMID: 24646260.
55. Koyama D, Sato Y, Aizawa M, et al. Soluble αKlotho as a candidate for the biomarker of aging. *Biochem Biophys Res Commun*. 2015;467(4):1019–1025. doi:10.1016/j.bbrc.2015.10.018. [Epub 2015 Oct 14]; PubMed PMID: 26462468.
56. Zeng Y, Wang PH, Zhang M, et al. Aging-related renal injury and inflammation are associated with downregulation of Klotho and induction of RIG-I/NF-κB signaling pathway in senescence-accelerated mice. *Aging Clin Exp Res*. 2016;28(1):69–76. doi:10.1007/s40520-015-0371-y. [Epub 2015 May 19]; PubMed PMID: 25986237.
57. Kalaitzidis RG, Duni A, Siamopoulos KC. Klotho, the Holy Grail of the kidney: from salt sensitivity to chronic kidney disease. *Int Urol Nephrol*. 2016;48(10):1657–1666. doi:10.1007/s11255-016-1325-9. [Epub 2016 May 23]; Review. PubMed PMID: 27215557.
58. Drew DA, Katz R, Kritchevsky S, et al. Association between soluble Klotho and change in kidney function: the health aging and body composition study. *J Am Soc Nephrol*. 2017;28(6):1859–1866. doi:10.1681/ASN.2016080828. [Epub 2017 Jan 19]; PubMed PMID: 28104822. PubMed Central PMCID: PMC5461794.
59. Humphreys BD. Mechanisms of renal fibrosis. *Annu Rev Physiol*. 2018;80:309–326. doi:10.1146/annurev-physiol-022516-034227. [Epub 2017 Oct 25]; PubMed PMID: 29068765.
60. Yang HC, Fogo AB. Fibrosis and renal aging. *Kidney Int Suppl (2011)*. 2014;4(1):75–78. Review. PubMed PMID: 26312154. PubMed Central PMCID: PMC4536965.
61. Uneda K, Wakui H, Maeda A, et al. Angiotensin II type 1 receptor-associated protein regulates kidney aging and lifespan independent of angiotensin. *J Am Heart Assoc*. 2017;6(8):pii: e006120. doi:10.1161/JAHA.117.006120. PubMed PMID: 28751545. PubMed Central PMCID: PMC5586453.
62. Yoon HE, Choi BS. The renin-angiotensin system and aging in the kidney. *Korean J Intern Med*. 2014;29(3):291–295. doi:10.3904/kjim.2014.29.3.291. [Epub 2014 Apr 29]; Review. PubMed PMID: 24851061. PubMed Central PMCID: PMC4028516.
63. Conti S, Cassis P, Benigni A. Aging and the renin-angiotensin system. *Hypertension*. 2012;60(4):878–883. doi:10.1161/HYPERTENSIONAHA.110.155895. [Epub 2012 Aug 27]; Review. PubMed PMID: 22926952.
64. Okada A, Nangaku M, Jao TM, et al. D-serine, a novel uremic toxin, induces senescence in human renal tubular cells via GCN2 activation. *Sci Rep*. 2017;7(1):11168. doi:10.1038/s41598-017-11049-8. PubMed PMID: 28894140. PubMed Central PMCID: PMC5593843.
65. Roncal-Jimenez CA, Ishimoto T, Lanaspa MA, et al. Aging-associated renal disease in mice is fructokinase dependent. *Am J Physiol Renal Physiol*. 2016;311(4):F722–F730. doi:10.1152/ajprenal.00306.2016. [Epub 2016 Jul 27]; PubMed PMID: 27465991. PubMed Central PMCID: PMC5142232.
66. Lin CH, Chen J, Zhang Z, et al. Endostatin and transglutaminase 2 are involved in fibrosis of the aging kidney. *Kidney Int*. 2016;89(6):1281–1292. doi:10.1016/j.kint.2016.01.030. [Epub 2016 Apr 14]; PubMed PMID: 27165830. PubMed Central PMCID: PMC4868664.
67. Bucala R. Diabetes, aging, and their tissue complications. *J Clin Invest*. 2014;124(5):1887–1888. doi:10.1172/JCI75224. [Epub 2014 May 1]; Review. PubMed PMID: 24789881. PubMed Central PMCID: PMC4001559.
68. Inker LA, Okparavero A, Tighiouart H, et al. Midlife blood pressure and late-life GFR and albuminuria: an older general population cohort. *Am J Kidney Dis*. 2015;66(2):240–248. doi:10.1053/j.ajkd.2015.03.030. [Epub 2015 May 16]; PubMed PMID: 25987258. PubMed Central PMCID: PMC4516633.
69. Zürbig P, Decramer S, Dakna M, et al. The human urinary proteome reveals high similarity between kidney aging and chronic kidney disease. *Proteomics*. 2009;9(8):2108–2117. doi:10.1002/pmic.200800560. PubMed PMID: 19296547. PubMed Central PMCID: PMC2768386.
70. Delaney A, Padmanabhan V, Rezvani G, et al. Evolutionary conservation and modulation of a juvenile growth-regulating genetic program. *J Mol Endocrinol*. 2014;52(3):269–277. doi:10.1530/JME-13-0263. Print 2014 Jun. PubMed PMID: 24776848. PubMed Central PMCID: PMC4051439.
71. Luyckx VA, Brenner BM. Birth weight, malnutrition and kidney-associated outcomes–a global concern. *Nat Rev Nephrol*. 2015;11(3):135–149. doi:10.1038/nrneph.2014.251. [Epub 2015 Jan 20]; Review. PubMed PMID: 25599618.
72. Friedenberg MJ, Walz BJ, McAlister WH, et al. Roentgen size of normal kidneys: computers analysis of 1,286 cases. *Radiology*. 1965;84:1022–1030.
73. McLachlan M, Wasserman P. Changes in sizes and distensibility of the aging kidney. *Br J Radiol*. 1981;54:488–491.
74. Emamian SA, Nielsen MB, Pedersen JF, et al. Kidney dimensions at sonography: correlation with age, sex, and habitus in 665 adult volunteers. *AJR Am J Roentgenol*. 1993;160:83–86.
75. Gourtsoyiannis N, Prassopoulos P, Cavouras D, et al. The thickness of the renal parenchyma decreases with age: a CT study of 360 patients. *AJR Am J Roentgenol*. 1990;155:541–544.
76. Glodny B, Unterholzner V, Taferner B, et al. Normal kidney size and its influencing factors - a 64-slice MDCT study of 1.040 asymptomatic patients. *BMC Urol*. 2009;9:19.
77. Bax L, van der Graaf Y, Rabelink AJ, et al. Influence of atherosclerosis on age-related changes in renal size and function. *Eur J Clin Invest*. 2003;33:34–40.
78. Johnson S, Rishi R, Andone A, et al. Determinants and functional significance of renal parenchymal volume in adults. *Clin J Am Soc Nephrol*. 2011;6:70–76.
79. Roseman DA, Hwang SJ, Oyama-Manabe N, et al. Clinical associations of total kidney volume: the Framingham Heart Study. *Nephrol Dial Transplant*. 2016.
80. Wang X, Vrtiska TJ, Avula RT, et al. Age, kidney function, and risk factors associate differently with cortical and medullary volumes of the kidney. *Kidney Int*. 2014;85:677–685.
81. Baert L, Steg A. Is the diverticulum of the distal and collecting tubules a preliminary stage of the simple cyst in the adult? *J Urol*. 1977;118:707–710.
82. Rule AD, Sasiwimonphan K, Lieske JC, et al. Characteristics of renal cystic and solid lesions based on contrast-enhanced computed tomography of potential kidney donors. *Am J Kidney Dis*. 2012;59:611–618.

83. Eknoyan G. A clinical view of simple and complex renal cysts. *J Am Soc Nephrol.* 2009;20:1874–1876.
84. Schwarz A, Lenz T, Klaen R, et al. Hygroma renale: pararenal lymphatic cysts associated with renin-dependent hypertension (Page kidney). Case report on bilateral cysts and successful therapy by marsupialization. *J Urol.* 1993;150:953–957.
85. Lorenz EC, Vrtiska TJ, Lieske JC, et al. Prevalence of renal artery and kidney abnormalities by computed tomography among healthy adults. *Clin J Am Soc Nephrol.* 2010;5:431–438.
86. Denic A, Alexander MP, Kaushik V, et al. Detection and clinical patterns of nephron hypertrophy and nephrosclerosis among apparently healthy adults. *Am J Kidney Dis.* 2016;68:58–67.
87. Hoy WE, Ingelfinger JR, Hallan S, et al. The early development of the kidney and implications for future health. *J Dev Orig Health Dis.* 2010;1:216–233.
88. Denic A, Lieske JC, Chakkera HA, et al. The substantial loss of nephrons in healthy human kidneys with aging. *J Am Soc Nephrol.* 2017;28:313–320.
89. Hoy WE, Douglas-Denton RN, Hughson MD, et al. A stereological study of glomerular number and volume: preliminary findings in a multiracial study of kidneys at autopsy. *Kidney Int Suppl.* 2003;S31–S37.
90. Hayman JM, Martin JW, Miller M. Renal function and the number of glomeruli in the human kidney. *Arch Intern Med.* 1939;64:69–83.
91. Rule AD, Amer H, Cornell LD, et al. The association between age and nephrosclerosis on renal biopsy among healthy adults. *Ann Intern Med.* 2010;152:561–567.
92. Vazquez Martul E, Veiga Barreiro A. Importance of kidney biopsy in graft selection. *Transplant Proc.* 2003;35:1658–1660.
93. Kremers WK, Denic A, Lieske JC, et al. Distinguishing age-related from disease-related glomerulosclerosis on kidney biopsy: the Aging Kidney Anatomy study. *Nephrol Dial Transplant.* 2015;30:2034–2039.
94. Hommos MS, Zeng C, Liu Z, et al. Global glomerulosclerosis with nephrotic syndrome; the clinical importance of age adjustment. *Kidney Int.* 2018;93(5):1175–1182. doi:10.1016/j.kint.2017.09.028. [Epub 2017 Dec 19]; PubMed PMID: 29273332. PubMed Central PMCID: PMC5911429.
95. Fogo AB. Causes and pathogenesis of focal segmental glomerulosclerosis. *Nat Rev Nephrol.* 2015;11:76–87.
96. Jennette JC, Charles L, Grubb W. Glomerulomegaly and focal segmental glomerulosclerosis associated with obesity and sleep-apnea syndrome. *Am J Kidney Dis.* 1987;10:470–472.
97. Bailey RR, Lynn KL, Burry AF, et al. Proteinuria, glomerulomegaly and focal glomerulosclerosis in a grossly obese man with obstructive sleep apnea syndrome. *Aust N Z J Med.* 1989;19:473–474.
98. Marcantoni C, Ma LJ, Federspiel C, et al. Hypertensive nephrosclerosis in African Americans versus Caucasians. *Kidney Int.* 2002;62(1):172–180.
99. Hoy WE, Hughson MD, Kopp JB, et al. APOL1 risk alleles are associated with exaggerated age-related changes in glomerular number and volume in African-American adults: an autopsy study. *J Am Soc Nephrol.* 2015;26(12):3179–3189. doi:10.1681/ASN.2014080768. [Epub 2015 Jun 2]; PubMed PMID: 26038529. PubMed Central PMCID: PMC4657832.
100. Meyer TW, Lenihan CR. Glomerular Effects of Age and APOL1. *J Am Soc Nephrol.* 2015;26(12):2901–2903. doi:10.1681/ASN.2015040459. [Epub 2015 Jun 2]; PubMed PMID: 26038527. PubMed Central PMCID: PMC4657849.
101. Grams ME, Rebholz CM, Chen Y, et al. Race, APOL1 risk, and eGFR decline in the general population. *J Am Soc Nephrol.* 2016;27(9):2842–2850. doi:10.1681/ASN.2015070763. [Epub 2016 Mar 10]; PubMed PMID: 26966015. PubMed Central PMCID: PMC5004654.
102. Tsuboi N, Utsunomiya Y, Kanzaki G, et al. Low glomerular density with glomerulomegaly in obesity-related glomerulopathy. *Clin J Am Soc Nephrol.* 2012;7:735–741.
103. Tsuboi N, Koike K, Hirano K, et al. Clinical features and long-term renal outcomes of Japanese patients with obesity-related glomerulopathy. *Clin Exp Nephrol.* 2013;17:379–385.
104. Tsuboi N, Utsunomiya Y, Hosoya T. Obesity-related glomerulopathy and the nephron complement. *Nephrol Dial Transplant.* 2013;28(suppl 4):iv108–iv113.
105. Chakkera HA, Chang YH, Thomas LF, et al. Obesity correlates with glomerulomegaly but is not associated with kidney dysfunction early after donation. *Transplant Direct.* 2015;1:1–6.
106. Thomson SC, Vallon V, Blantz RC. Kidney function in early diabetes: the tubular hypothesis of glomerular filtration. *Am J Physiol Renal Physiol.* 2004;286:F8–F15.
107. Abdi R, Slakey D, Kittur D, et al. Heterogeneity of glomerular size in normal donor kidneys: impact of race. *Am J Kidney Dis.* 1998;32:43–46.
108. Fulladosa X, Moreso F, Narvaez JA, et al. Estimation of total glomerular number in stable renal transplants. *J Am Soc Nephrol.* 2003;14:2662–2668.
109. Hodgin JB, Bitzer M, Wickman L, et al. Glomerular aging and focal global glomerulosclerosis: a podometric perspective. *J Am Soc Nephrol.* 2015;26:3162–3178.
110. Darmady EM, Offer J, Woodhouse MA. The parameters of the ageing kidney. *J Pathol.* 1973;109:195–207.
111. Nyengaard JR, Bendtsen TF. Glomerular number and size in relation to age, kidney weight, and body surface in normal man. *Anat Rec.* 1992;232:194–201.
112. Tan JC, Busque S, Workeneh B, et al. Effects of aging on glomerular function and number in living kidney donors. *Kidney Int.* 2010;78:686–692.
113. Hayslett JP, Kashgarian M, Epstein FH. Functional correlates of compensatory renal hypertrophy. *J Clin Invest.* 1968;47:774–799.
114. Denic A, Mathew J, Nagineni VV, et al. Clinical and pathology findings associate consistently with larger glomerular volume. *J Am Soc Nephrol.* 2018;29(7):1960–1969. doi:10.1681/ASN.2017121305. [Epub 2018 May 22]; Erratum in: *J Am Soc Nephrol.* 2018;29(9):2445. PubMed PMID: 29789431. PubMed Central PMCID: PMC6050922.
115. Elsherbiny HE, Alexander MP, Kremers WK, et al. Nephron hypertrophy and glomerulosclerosis and their association with kidney function and risk factors among living kidney donors. *Clin J Am Soc Nephrol.* 2014;9:1892–1902.
116. Rule AD, Semret MH, Amer H, et al. Association of kidney function and metabolic risk factors with density of glomeruli on renal biopsy samples from living donors. *Mayo Clin Proc.* 2011;86:282–290.
117. Liu JS, Ishikawa I, Horiguchi T. Incidence of acquired renal cysts in biopsy specimens. *Nephron.* 2000;84:142–147.
118. Konda R, Sato H, Hatafuku F, et al. Expression of hepatocyte growth factor and its receptor C-met in acquired renal cystic disease associated with renal cell carcinoma. *J Urol.* 2004;171:2166–2170.
119. Burchfiel CM, Tracy RE, Chyou PH, et al. Cardiovascular risk factors and hyalinization of renal arterioles at autopsy. The Honolulu Heart Program. *Arterioscler Thromb Vasc Biol.* 1997;17:760–768.
120. Kubo M, Kiyohara Y, Kato I, et al. Risk factors for renal glomerular and vascular changes in an autopsy-based population survey: the Hisayama study. *Kidney Int.* 2003;63:1508–1515.
121. Ninomiya T, Kubo M, Doi Y, et al. Prehypertension increases the risk for renal arteriosclerosis in autopsies: the Hisayama Study. *J Am Soc Nephrol.* 2007;18:2135–2142.
122. Hughson MD, Puelles VG, Hoy WE, et al. Hypertension, glomerular hypertrophy and nephrosclerosis: the effect of race. *Nephrol Dial Transplant.* 2014;29:1399–1409.
123. Robbins S, Kumar V. *Robbins and Cotran Pathologic Basis of Disease.* 8th ed. Philadelphia: Saunders/Elsevier; 2010.
124. Martin JE, Sheaff MT. Renal ageing. *J Pathol.* 2007;211:198–205.
125. Mancilla E, Avila-Casado C, Uribe-Uribe N, et al. Time-zero renal biopsy in living kidney transplantation: a valuable opportunity to correlate predonation clinical data with histological abnormalities. *Transplantation.* 2008;86:1684–1688.
126. Kriz W, Lemley KV. A potential role for mechanical forces in the detachment of podocytes and the progression of CKD. *J Am Soc Nephrol.* 2015;26(2):258–269. doi:10.1681/ASN.2014030278. [Epub 2014 Jul 24]; Review. PubMed PMID: 25060060. PubMed Central PMCID: PMC4310663.
127. Davies DF, Shock NW. Age changes in glomerular filtration rate, effective renal plasma flow, and tubular excretory capacity in adult males. *J Clin Invest.* 1950;29:496–507.
128. Lindeman RD, Tobin J, Shock NW. Longitudinal studies on the rate of decline in renal function with age. *J Am Geriatr Soc.* 1985;33:278–285.
129. Eriksen BO, Lochen ML, Arntzen KA, et al. Subclinical cardiovascular disease is associated with a high glomerular filtration rate in the nondiabetic general population. *Kidney Int.* 2014;86:146–153.
130. Rule AD, Larson TS, Bergstralh EJ, et al. Using serum creatinine to estimate glomerular filtration rate: accuracy in good health and in chronic kidney disease. *Ann Intern Med.* 2004;141:929–937.

131. Murata K, Baumann NA, Saenger AK, et al. Relative performance of th e MDRD and CKD-EPI equations for estimating glomerular filtration rate among patients with varied clinical presentations. *Clin J Am Soc Nephrol.* 2011;6:1963–1972.
132. Rule AD, Gussak HM, Pond GR, et al. Measured and estimated GFR in healthy potential kidney donors. *Am J Kidney Dis.* 2004;43(1):112–119. Erratum in: *Am J Kidney Dis.* 2004;44(6):1126. *Am J Kidney Dis.* 2005;46(1):170. PubMed PMID: 14712434.
133. Pottel H, Delanaye P, Weekers L, et al. Age-dependent reference intervals for estimated and measured glomerular filtration rate. *Clin Kidney J.* 2017;10(4):545–551. doi:10.1093/ckj/sfx026. [Epub 2017 Apr 28]; PubMed PMID: 28852494. PubMed Central PMCID: PMC5570001.
134. Baba M, Shimbo T, Horio M, et al. Longitudinal Study of the Decline in Renal Function in Healthy Subjects. *PLoS ONE.* 2015;10(6):e0129036. doi:10.1371/journal.pone.0129036. eCollection 2015. Erratum in: *PLoS One.* 2015;10(8):e0135992. PubMed PMID: 26061083. PubMed Central PMCID: PMC4464887.
135. Cohen E, Nardi Y, Krause I, et al. A longitudinal assessment of the natural rate of decline in renal function with age. *J Nephrol.* 2014;27(6):635–641. doi:10.1007/s40620-014-0077-9. [Epub 2014 Mar 19]; PubMed PMID: 24643437.
136. Muntner P. Longitudinal measurements of renal function. *Semin Nephrol.* 2009;29(6):650–657. doi:10.1016/j.semnephrol.2009.07.010. PubMed PMID: 20006797.
137. Brenner BM, Troy JL, Daugharty TM, et al. Dynamics of glomerular ultrafiltration in the rat. II. Plasma-flow dependence of GFR. *Am J Physiol.* 1972;223:1184–1190.
138. Ott CE, Marchand GR, Diaz-Buxo JA, et al. Determinants of glomerular filtration rate in the dog. *Am J Physiol.* 1976;231:235–239.
139. Denic A, Mathew J, Lerman LO, et al. Single-nephron glomerular filtration rate in healthy adults. *N Engl J Med.* 2017;376:2349–2357.
140. Berg UB. Differences in decline in GFR with age between males and females. Reference data on clearances of inulin and PAH in potential kidney donors. *Nephrol Dial Transplant.* 2006;21:2577–2582.
141. Tan JC, Busque S, Workeneh B, et al. Effects of aging on glomerular function and number in living kidney donors. *Kidney Int.* 2010;78(7):686–692. doi:10.1038/ki.2010.128. [Epub 2010 May 12]; PubMed PMID: 20463656. PubMed Central PMCID: PMC3353650.
142. Lenihan CR, Busque S, Derby G, et al. Longitudinal study of living kidney donor glomerular dynamics after nephrectomy. *J Clin Invest.* 2015;125(3):1311–1318. doi:10.1172/JCI78885. [Epub 2015 Feb 17]; PubMed PMID: 25689253. PubMed Central PMCID: PMC4362245.
143. Lenihan CR, Busque S, Derby G, et al. The association of predonation hypertension with glomerular function and number in older living kidney donors. *J Am Soc Nephrol.* 2015;26(6):1261–1267. doi:10.1681/ASN.2014030304. [Epub 2014 Dec 18]; PubMed PMID: 25525178. PubMed Central PMCID: PMC4446872.
144. Chen YR, Sung K, Tharin S. Symptomatic anterior cervical osteophyte causing dysphagia: case report, imaging, and review of the literature. *Cureus.* 2016;8(2):e473. doi:10.7759/cureus.473. PubMed PMID: 27004150. PubMed Central PMCID: PMC4779080.
145. Hoang K, Tan JC, Derby G, et al. Determinants of glomerular hypofiltration in aging humans. *Kidney Int.* 2003;64(4):1417–1424. PubMed PMID: 12969161.
146. Saxena AB, Myers BD, Derby G, et al. Adaptive hyperfiltration in the aging kidney after contralateral nephrectomy. *Am J Physiol Renal Physiol.* 2006;291(3):F629–F634. [Epub 2006 Mar 8]; PubMed PMID: 16525160.
147. Chakkera HA, Denic A, Kremers WK, et al. Comparison of high glomerular filtration rate thresholds for identifying hyperfiltration. *Nephrol Dial Transplant.* 2018;doi:10.1093/ndt/gfy332. [Epub ahead of print].
148. Kenney WL, Ho CW. Age alters regional distribution of blood flow during moderate-intensity exercise. *J Appl Physiol.* 1995;79:1112–1119.
149. Ho CW, Beard JL, Farrell PA, et al. Age, fitness, and regional blood flow during exercise in the heat. *J Appl Physiol.* 1997;82:1126–1135.
150. Mulkerrin EC, Brain A, Hampton D, et al. Reduced renal hemodynamic response to atrial natriuretic peptide in older volunteers. *Am J Kidney Dis.* 1993;22:538–544.
151. Fuiano G, Sund S, Mazza G, et al. Renal hemodynamic response to maximal vasodilating stimulus in healthy older subjects. *Kidney Int.* 2001;59:1052–1058.
152. Kielstein JT, Bode-Boger SM, Frolich JC, et al. Asymmetric dimethylarginine, blood pressure, and renal perfusion in older subjects. *Circulation.* 2003;107:1891–1895.
153. Matsumoto Y, Ueda S, Yamagishi S, et al. Dimethylarginine dimethylaminohydrolase prevents progression of renal dysfunction by inhibiting loss of peritubular capillaries and tubulointerstitial fibrosis in a rat model of chronic kidney disease. *J Am Soc Nephrol.* 2007;18:1525–1533.
154. Ueda S, Yamagishi S, Matsumoto Y, et al. Involvement of asymmetric dimethylarginine (ADMA) in glomerular capillary loss and sclerosis in a rat model of chronic kidney disease (CKD). *Life Sci.* 2009;84:853–856.
155. Esposito C, Plati A, Mazzullo T, et al. Renal function and functional reserve in healthy older individuals. *J Nephrol.* 2007;20:617–625.
156. Musso CG, Reynaldi J, Martinez B, et al. Renal reserve in the oldest old. *Int Urol Nephrol.* 2011;43(1):253–256. doi:10.1007/s11255-010-9769-9. [Epub 2010 Jul 1]; PubMed PMID: 20593305.
157. Kleinman KS, Glassock RJ. Glomerular filtration rate fails to increase following protein ingestion in hypothalamo-hypophyseal-deficient adults. Preliminary observations. *Am J Nephrol.* 1986;6(3):169–174. PubMed PMID: 3740125.
158. Dullaart RP, Meijer S, Marbach P, et al. Renal reserve filtration capacity in growth hormone deficient subjects. *Eur J Clin Invest.* 1992;22(8):562–568. PubMed PMID: 1425863.
159. Nair KS, Pabico RC, Truglia JA, et al. Mechanism of glomerular hyperfiltration after a protein meal in humans. Role of hormones and amino acids. *Diabetes Care.* 1994;17(7):711–715. PubMed PMID: 7924782.
160. De Moor B, Vanwalleghem JF, Swennen Q, et al. Haemodynamic or metabolic stimulation tests to reveal the renal functional response: requiem or revival? *Clin Kidney J.* 2018;11(5):623–654.
161. Fliser D, Zeier M, Nowack R, et al. Renal functional reserve in healthy older subjects. *J Am Soc Nephrol.* 1993;3(7):1371–1377.
162. Sunaga K, Sudoh T, Fujimura A. Lack of diurnal variation in glomerular filtration rates in older adults. *J Clin Pharmacol.* 1996;36(3):203–205. PubMed PMID: 8690813.
163. Hollenberg NK, Rivera A, Meinking T, et al. Age, renal perfusion and function in island-dwelling indigenous Kuna Amerinds of Panama. *Nephron.* 1999;82(2):131–138. PubMed PMID: 10364705.
164. Reckelhoff JF, Samsell L, Dey R, et al. The effect of aging on glomerular hemodynamics in the rat. *Am J Kidney Dis.* 1992;20(1):70–75. PubMed PMID: 1621681.
165. Erdely A, Greenfeld Z, Wagner L, et al. Sexual dimorphism in the aging kidney: effects on injury and nitric oxide system. *Kidney Int.* 2003;63(3):1021–1026. PubMed PMID: 12631083. PubMed Central PMCID: PMC2793681.
166. Baylis C. Sexual dimorphism of the aging kidney: role of nitric oxide deficiency. *Physiology (Bethesda).* 2008;23:142–150. doi:10.1152/physiol.00001.2008. Review. PubMed PMID: 18556467.
167. Baylis C. Sexual dimorphism in the aging kidney: differences in the nitric oxide system. *Nat Rev Nephrol.* 2009;5(7):384–396. doi:10.1038/nrneph.2009.90. [Epub 2009 Jun 2]; Review. PubMed PMID: 19488070.
168. Baylis C. Sexual dimorphism: the aging kidney, involvement of nitric oxide deficiency, and angiotensin II overactivity. *J Gerontol A Biol Sci Med Sci.* 2012;67(12):1365–1372. doi:10.1093/gerona/gls171. [Epub 2012 Sep 7]; Review. PubMed PMID: 22960474. PubMed Central PMCID: PMC3708515.
169. Gava AL, Freitas FP, Meyrelles SS, et al. Gender-dependent effects of aging on the kidney. *Braz J Med Biol Res.* 2011;44(9):905–913. [Epub 2011 Aug 19]; Review. PubMed PMID: 21956533.
170. Hollenberg NK, Adams DF, Solomon HS, et al. Senescence and the renal vasculature in normal man. *Circ Res.* 1974;34:309–316.
171. Hill C, Lateef AM, Engels K, et al. Basal and stimulated nitric oxide in control of kidney function in the aging rat. *Am J Physiol.* 1997;272:R1747–R1753.
172. Zhang XZ, Qiu C, Baylis C. Sensitivity of the segmental renal arterioles to angiotensin II in the aging rat. *Mech Ageing Dev.* 1997;97:183–192.
173. McLachlan MS, Guthrie JC, Anderson CK, et al. Vascular and glomerular changes in the ageing kidney. *J Pathol.* 1977;121:65–78.
174. Olson JL. Hyaline arteriolosclerosis: new meaning for an old lesion. *Kidney Int.* 2003;63:1162–1163.
175. Glassock RJ, Rule AD. The implications of anatomical and functional changes of the aging kidney: with an emphasis on the glomeruli. *Kidney Int.* 2012;82:270–277.
176. Mundel P, Shankland SJ. Podocyte biology and response to injury. *J Am Soc Nephrol.* 2002;13:3005–3015.

177. Pavenstadt H, Kriz W, Kretzler M. Cell biology of the glomerular podocyte. *Physiol Rev*. 2003;83:253–307.
178. Kriz W, Gretz N, Lemley KV. Progression of glomerular diseases: is the podocyte the culprit? *Kidney Int*. 1998;54:687–697.
179. Kim YH, Goyal M, Kurnit D, et al. Podocyte depletion and glomerulosclerosis have a direct relationship in the PAN-treated rat. *Kidney Int*. 2001;60:957–968.
180. Fukuda A, Chowdhury MA, Venkatareddy MP, et al. Growth-dependent podocyte failure causes glomerulosclerosis. *J Am Soc Nephrol*. 2012;23:1351–1363.
181. Wiggins JE, Goyal M, Sanden SK, et al. Podocyte hypertrophy, "adaptation," and "decompensation" associated with glomerular enlargement and glomerulosclerosis in the aging rat: prevention by calorie restriction. *J Am Soc Nephrol*. 2005;16:2953–2966.
182. Wharram BL, Goyal M, Wiggins JE, et al. Podocyte depletion causes glomerulosclerosis: diphtheria toxin-induced podocyte depletion in rats expressing human diphtheria toxin receptor transgene. *J Am Soc Nephrol*. 2005;16:2941–2952.
183. Luyckx VA, Perico N, Somaschini M, et al. A developmental approach to the prevention of hypertension and kidney disease: a report from the Low Birth Weight and Nephron Number Working Group. *Lancet*. 2017;390(10092):424–428. doi:10.1016/S0140-6736(17)30576-7. [Epub 2017 Mar 9]; PubMed PMID: 28284520. PubMed Central PMCID: PMC5884413.
184. Ikezumi Y, Suzuki T, Karasawa T, et al. Low birthweight and premature birth are risk factors for podocytopenia and focal segmental glomerulosclerosis. *Am J Nephrol*. 2013;38(2):149–157. doi:10.1159/000353898. [Epub 2013 Aug 6]; PubMed PMID: 23920104.
185. Ruggajo P, Svarstad E, Leh S, et al. Low birth weight and risk of progression to end stage renal disease in IgA nephropathy–a retrospective registry-based cohort study. *PLoS ONE*. 2016;11(4):e0153819. doi:10.1371/journal.pone.0153819. eCollection 2016. PubMed PMID: 27092556. PubMed Central PMCID: PMC4836690.
186. Epstein M, Hollenberg NK. Age as a determinant of renal sodium conservation in normal man. *J Lab Clin Med*. 1976;87:411–417.
187. Macias Nunez JF, Garcia Iglesias C, Bondia Roman A, et al. Renal handling of sodium in old people: a functional study. *Age Ageing*. 1978;7:178–181.
188. Bauer JH. Age-related changes in the renin-aldosterone system. Physiological effects and clinical implications. *Drugs Aging*. 1993;3:238–245.
189. Mimran A, Ribstein J, Jover B. Aging and sodium homeostasis. *Kidney Int Suppl*. 1992;37:S107–S113.
190. Luft FC, Grim CE, Fineberg N, et al. Effects of volume expansion and contraction in normotensive whites, blacks, and subjects of different ages. *Circulation*. 1979;59:643–650.
191. Brenner BM, Ballermann BJ, Gunning ME, et al. Diverse biological actions of atrial natriuretic peptide. *Physiol Rev*. 1990;70:665–699.
192. Leosco D, Ferrara N, Landino P, et al. Effects of age on the role of atrial natriuretic factor in renal adaptation to physiologic variations of dietary salt intake. *J Am Soc Nephrol*. 1996;7:1045–1051.
193. Pollack JA, Skvorak JP, Nazian SJ, et al. Alterations in atrial natriuretic peptide (ANP) secretion and renal effects in aging. *J Gerontol A Biol Sci Med Sci*. 1997;52:B196–B202.
194. Or K, Richards AM, Espiner EA, et al. Effect of low dose infusions of ile-atrial natriuretic peptide in healthy older males: evidence for a postreceptor defect. *J Clin Endocrinol Metab*. 1993;76:1271–1274.
195. Tan AC, Jansen TL, Termond EF, et al. Kinetics of atrial natriuretic peptide in young and older subjects. *Eur J Clin Pharmacol*. 1992;42:449–452.
196. Haller BG, Zust H, Shaw S, et al. Effects of posture and ageing on circulating atrial natriuretic peptide levels in man. *J Hypertens*. 1987;5:551–556.
197. Ohashi M, Fujio N, Nawata H, et al. High plasma concentrations of human atrial natriuretic polypeptide in aged men. *J Clin Endocrinol Metab*. 1987;64:81–85.
198. Frame AA, Wainford RD. Mechanisms of altered renal sodium handling in age-related hypertension. *Am J Physiol Renal Physiol*. 2018;315(1):F1–F6. doi:10.1152/ajprenal.00594.2017. [Epub 2018 Feb 14]; PubMed PMID: 29442548. PubMed Central PMCID: PMC6087788.
199. Luft FC, Miller JZ, Grim CE, et al. Salt sensitivity and resistance of blood pressure. Age and race as factors in physiological responses. *Hypertension*. 1991;17(1 suppl):I102–I108. PubMed PMID: 1846122.
200. Tank JE, Vora JP, Houghton DC, et al. Altered renal vascular responses in the aging rat kidney. *Am J Physiol*. 1994;266(6 Pt 2):F942–F948. PubMed PMID: 8023973.
201. Tian Y, Riazi S, Khan O, et al. Renal ENaC subunit, Na-K-2Cl and Na-Cl cotransporter abundances in aged, water-restricted F344 x Brown Norway rats. *Kidney Int*. 2006;69(2):304–312. PubMed PMID: 16408120.
202. Tiwari S, Li L, Riazi S, et al. Sex and age result in differential regulation of the renal thiazide-sensitive NaCl cotransporter and the epithelial sodium channel in angiotensin II-infused mice. *Am J Nephrol*. 2009;30(6):554–562. doi:10.1159/000252776. [Epub 2009 Oct 21]; PubMed PMID: 19844087. PubMed Central PMCID: PMC2853589.
203. Palmer BF, Clegg DJ. Physiology and pathophysiology of potassium homeostasis. *Adv Physiol Educ*. 2016;40:480–490.
204. Kehayias JJ, Fiatarone MA, Zhuang H, et al. Total body potassium and body fat: relevance to aging. *Am J Clin Nutr*. 1997;66:904–910.
205. Kyle UG, Genton L, Hans D, et al. Age-related differences in fat-free mass, skeletal muscle, body cell mass and fat mass between 18 and 94 years. *Eur J Clin Nutr*. 2001;55:663–672.
206. Noth RH, Lassman MN, Tan SY, et al. Age and the renin-aldosterone system. *Arch Intern Med*. 1977;137:1414–1417.
207. Mulkerrin E, Epstein FH, Clark BA. Aldosterone responses to hyperkalemia in healthy older humans. *J Am Soc Nephrol*. 1995;6:1459–1462.
208. Musso CG, Miguel R, Algranati L, et al. Renal potassium excretion: comparison between chronic renal disease patients and old people. *Int Urol Nephrol*. 2005;37:167–170.
209. Bengele HH, Mathias R, Perkins JH, et al. Impaired renal and extra-renal potassium adaptation in old rats. *Kidney Int*. 1983;23:684–690, 199.
210. Musso C, Liakopoulos V, De Miguel R, et al. Transtubular potassium concentration gradient: comparison between healthy old people and chronic renal failure patients. *Int Urol Nephrol*. 2006;38:387–390.
211. Minaker KL, Rowe JW. Potassium homeostasis during hyperinsulinemia: effect of insulin level, beta-blockade, and age. *Am J Physiol*. 1982;242:E373–E377.
212. Blaine J, Chonchol M, Levi M. Renal control of calcium, phosphate, and magnesium homeostasis. *Clin J Am Soc Nephrol*. 2015;10:1257–1272.
213. Saris NE, Mervaala E, Karppanen H, et al. Magnesium. An update on physiological, clinical and analytical aspects. *Clin Chim Acta*. 2000;294:1–26.
214. Barbagallo M, Belvedere M, Dominguez LJ. Magnesium homeostasis and aging. *Magnes Res*. 2009;22:235–246.
215. Yang XY, Hosseini JM, Ruddel ME, et al. Blood magnesium parameters do not differ with age. *J Am Coll Nutr*. 1990;9:308–313.
216. Gullestad L, Nes M, Ronneberg R, et al. Magnesium status in healthy free-living older Norwegians. *J Am Coll Nutr*. 1994;13:45–50.
217. Grober U, Schmidt J, Kisters K. Magnesium in Prevention and Therapy. *Nutrients*. 2015;7:8199–8226.
218. Rebholz CM, Tin A, Liu Y, et al. Dietary magnesium and kidney function decline: the healthy aging in neighborhoods of diversity across the life span study. *Am J Nephrol*. 2016;44(5):381–387. [Epub 2016 Oct 22]; PubMed PMID: 27771720. PubMed Central PMCID: PMC5130225.
219. Corman B, Roinel N. Single-nephron filtration rate and proximal reabsorption in aging rats. *Am J Physiol*. 1991;260:F75–F80.
220. Armbrecht HJ, Zenser TV, Gross CJ, et al. Adaptation to dietary calcium and phosphorus restriction changes with age in the rat. *Am J Physiol*. 1980;239:E322–E327.
221. Hanai H, Ishida M, Liang CT, et al. Parathyroid hormone increases sodium/calcium exchange activity in renal cells and the blunting of the response in aging. *J Biol Chem*. 1986;261:5419–5425.
222. Orwoll ES, Meier DE. Alterations in calcium, vitamin D, and parathyroid hormone physiology in normal men with aging: relationship to the development of senile osteopenia. *J Clin Endocrinol Metab*. 1986;63:1262–1269.
223. Nordin BE, Horowitz M, Need A, et al. Renal leak of calcium in post-menopausal osteoporosis. *Clin Endocrinol (Oxf)*. 1994;41:41–45.
224. Armbrecht HJ, Zenser TV, Bruns ME, et al. Effect of age on intestinal calcium absorption and adaptation to dietary calcium. *Am J Physiol*. 1979;236:E769–E774.
225. Armbrecht HJ, Boltz M, Strong R, et al. Expression of calbindin-D decreases with age in intestine and kidney. *Endocrinology*. 1989;125:2950–2956.

226. Johnson JA, Beckman MJ, Pansini-Porta A, et al. Age and gender effects on 1,25-dihydroxyvitamin D3-regulated gene expression. *Exp Gerontol*. 1995;30:631–643.
227. Armbrecht HJ, Forte LR, Halloran BP. Effect of age and dietary calcium on renal 25(OH)D metabolism, serum 1,25(OH)2D, and PTH. *Am J Physiol*. 1984;246:E266–E270.
228. Chapuy MC, Durr F, Chapuy P. Age-related changes in parathyroid hormone and 25 hydroxycholecalciferol levels. *J Gerontol*. 1983;38:19–22.
229. van Abel M, Huybers S, Hoenderop JG, et al. Age-dependent alterations in Ca2+ homeostasis: role of TRPV5 and TRPV6. *Am J Physiol Renal Physiol*. 2006;291:F1177–F1183.
230. Kuro-o M, Matsumura Y, Aizawa H, et al. Mutation of the mouse klotho gene leads to a syndrome resembling ageing. *Nature*. 1997;390:45–51.
231. Imura A, Tsuji Y, Murata M, et al. Alpha-Klotho as a regulator of calcium homeostasis. *Science*. 2007;316:1615–1618.
232. Portale AA, Lonergan ET, Tanney DM, et al. Aging alters calcium regulation of serum concentration of parathyroid hormone in healthy men. *Am J Physiol*. 1997;272:E139–E146.
233. Musso CG, Juarez R, Vilas M, et al. Renal calcium, phosphorus, magnesium and uric acid handling: comparison between stage III chronic kidney disease patients and healthy oldest old. *Int Urol Nephrol*. 2012;44:1559–1562.
234. Halloran BP, Lonergan ET, Portale AA. Aging and renal responsiveness to parathyroid hormone in healthy men. *J Clin Endocrinol Metab*. 1996;81:2192–2197.
235. Marks J, Debnam ES, Unwin RJ. Phosphate homeostasis and the renal-gastrointestinal axis. *Am J Physiol Renal Physiol*. 2010;299:F285–F296.
236. Mulroney SE, Woda C, Haramati A. Changes in renal phosphate reabsorption in the aged rat. *Proc Soc Exp Biol Med*. 1998;218:62–67.
237. Levi M, Jameson DM, van der Meer BW. Role of BBM lipid composition and fluidity in impaired renal Pi transport in aged rat. *Am J Physiol*. 1989;256:F85–F94.
238. Kiebzak GM, Sacktor B. Effect of age on renal conservation of phosphate in the rat. *Am J Physiol*. 1986;251:F399–F407.
239. Levi M, Baird BM, Wilson PV. Cholesterol modulates rat renal brush border membrane phosphate transport. *J Clin Invest*. 1990;85:231–237.
240. Breusegem SY, Halaihel N, Inoue M, et al. Acute and chronic changes in cholesterol modulate Na-Pi cotransport activity in OK cells. *Am J Physiol Renal Physiol*. 2005;289:F154–F165.
241. Chen ML, King RS, Armbrecht HJ. Sodium-dependent phosphate transport in primary cultures of renal tubule cells from young and adult rats. *J Cell Physiol*. 1990;143:488–493.
242. Sorribas V, Lotscher M, Loffing J, et al. Cellular mechanisms of the age-related decrease in renal phosphate reabsorption. *Kidney Int*. 1996;50:855–863.
243. De Vita F, Lauretani F, Bauer J, et al. Relationship between vitamin D and inflammatory markers in older individuals. *Age (Dordr)*. 2014;36(4):9694. doi:10.1007/s11357-014-9694-4. [Epub 2014 Aug 3]; PubMed PMID: 25086618. PubMed Central PMCID: PMC4150893.
244. Gallagher JC. Vitamin D and aging. *Endocrinol Metab Clin North Am*. 2013;42(2):319–332. doi:10.1016/j.ecl.2013.02.004. [Epub 2013 Apr 9]; Review. PubMed PMID: 23702404. PubMed Central PMCID: PMC3782116.
245. MacLaughlin J, Holick MF. Aging decreases the capacity of human skin to produce vitamin D3. *J Clin Invest*. 1985;76(4):1536–1538. PubMed PMID: 2997282. PubMed Central PMCID: PMC424123.
246. Ameri P, Canepa M, Milaneschi Y, et al. Relationship between vitamin D status and left ventricular geometry in a healthy population: results from the Baltimore Longitudinal Study of Aging. *J Intern Med*. 2013;273(3):253–262. doi:10.1111/joim.12007. [Epub 2012 Nov 12]; PubMed PMID: 23061475. PubMed Central PMCID: PMC3568460.
247. Liang CT, Barnes J, Cheng L, et al. Effects of 1,25-(OH)2D3 administered in vivo on phosphate uptake by isolated chick renal cells. *Am J Physiol*. 1982;242:C312–C318.
248. Kurnik BR, Hruska KA. Effects of 1,25-dihydroxycholecalciferol on phosphate transport in vitamin D-deprived rats. *Am J Physiol*. 1984;247:F177–F184.
249. Brandis M, Harmeyer J, Kaune R, et al. Phosphate transport in brush-border membranes from control and rachitic pig kidney and small intestine. *J Physiol*. 1987;384:479–490.
250. Brasitus TA, Dudeja PK, Eby B, et al. Correction by 1-25-dihydroxycholecalciferol of the abnormal fluidity and lipid composition of enterocyte brush border membranes in vitamin D-deprived rats. *J Biol Chem*. 1986;261:16404–16409.
251. Tareen N, Zadshir A, Martins D, et al. Alterations in acid-base homeostasis with aging. *J Natl Med Assoc*. 2004;96:921–925, quiz 5–6.
252. Vallet M, Metzger M, Haymann JP, et al. Urinary ammonia and long-term outcomes in chronic kidney disease. *Kidney Int*. 2015;88:137–145.
253. Berkemeyer S, Vormann J, Gunther AL, et al. Renal net acid excretion capacity is comparable in prepubescence, adolescence, and young adulthood but falls with aging. *J Am Geriatr Soc*. 2008;56:1442–1448.
254. Adler S, Lindeman RD, Yiengst MJ, et al. Effect of acute acid loading on urinary acid excretion by the aging human kidney. *J Lab Clin Med*. 1968;72:278–289.
255. Prasad R, Kinsella JL, Sacktor B. Renal adaptation to metabolic acidosis in senescent rats. *Am J Physiol*. 1988;255:F1183–F1190.
256. Rylander R, Remer T, Berkemeyer S, et al. Acid-base status affects renal magnesium losses in healthy, elderly persons. *J Nutr*. 2006;136:2374–2377.
257. Dawson-Hughes B, Harris SS, Ceglia L. Alkaline diets favor lean tissue mass in older adults. *Am J Clin Nutr*. 2008;87:662–665.
258. Amodu A, Abramowitz MK. Dietary acid, age, and serum bicarbonate levels among adults in the United States. *Clin J Am Soc Nephrol*. 2013;8:2034–2042.
259. Verove C, Maisonneuve N, El Azouzi A, et al. Effect of the correction of metabolic acidosis on nutritional status in elderly patients with chronic renal failure. *J Ren Nutr*. 2002;12:224–228.
260. Sebastian A, Harris ST, Ottaway JH, et al. Improved mineral balance and skeletal metabolism in postmenopausal women treated with potassium bicarbonate. *N Engl J Med*. 1994;330:1776–1781.
261. Dawson-Hughes B, Harris SS, Palermo NJ, et al. Treatment with potassium bicarbonate lowers calcium excretion and bone resorption in older men and women. *J Clin Endocrinol Metab*. 2009;94:96–102.
262. Moseley KF, Weaver CM, Appel L, et al. Potassium citrate supplementation results in sustained improvement in calcium balance in older men and women. *J Bone Miner Res*. 2013;28:497–504.
263. Knepper MA, Kwon TH, Nielsen S. Molecular physiology of water balance. *N Engl J Med*. 2015;372:1349–1358.
264. Hasler U, Leroy V, Martin PY, et al. Aquaporin-2 abundance in the renal collecting duct: new insights from cultured cell models. *Am J Physiol Renal Physiol*. 2009;297:F10–F18.
265. Kortenoeven ML, Pedersen NB, Rosenbaek LL, et al. Vasopressin regulation of sodium transport in the distal nephron and collecting duct. *Am J Physiol Renal Physiol*. 2015;309:F280–F299.
266. Moeller HB, Olesen ET, Fenton RA. Regulation of the water channel aquaporin-2 by posttranslational modification. *Am J Physiol Renal Physiol*. 2011;300:F1062–F1073.
267. Rector FC Jr. Sodium, bicarbonate, and chloride absorption by the proximal tubule. *Am J Physiol*. 1983;244:F461–F471.
268. Rocha AS, Kokko JP. Sodium chloride and water transport in the medullary thick ascending limb of Henle. Evidence for active chloride transport. *J Clin Invest*. 1973;52:612–623.
269. Burg MB, Green N. Function of the thick ascending limb of Henle's loop. *Am J Physiol*. 1973;224:659–668.
270. Fenton RA, Knepper MA. Mouse models and the urinary concentrating mechanism in the new millennium. *Physiol Rev*. 2007;87:1083–1112.
271. Knepper MA, Nielsen S, Chou CL, et al. Mechanism of vasopressin action in the renal collecting duct. *Semin Nephrol*. 1994;14:302–321.
272. Sands JM, Nonoguchi H, Knepper MA. Vasopressin effects on urea and H2O transport in inner medullary collecting duct subsegments. *Am J Physiol*. 1987;253:F823–F832.
273. Schafer JA, Andreoli TE. Cellular constraints to diffusion. The effect of antidiuretic hormone on water flows in isolated mammalian collecting tubules. *J Clin Invest*. 1972;51:1264–1278.
274. Sands JM. Urine concentrating and diluting ability during aging. *J Gerontol A Biol Sci Med Sci*. 2012;67:1352–1357.
275. Findley T. Role of the neurohypophysis in the pathogenesis of hypertension and some allied disorders associated with aging. *Am J Med*. 1949;7:70–84.
276. Cowen LE, Hodak SP, Verbalis JG. Age-associated abnormalities of water homeostasis. *Endocrinol Metab Clin North Am*. 2013;42:349–370.
277. Rolls BJ, Phillips PA. Aging and disturbances of thirst and fluid balance. *Nutr Rev*. 1990;48:137–144.
278. Carpenter CR. Geriatric emergency medicine. Preface. *Clin Geriatr Med*. 2013;29:xiii–xiv.

279. Sherman FT. Hyponatremia and SIADH in the elderly. Introducing the "water-offered test" for dehydration. *Geriatrics*. 2007;62: 8, 18.
280. Combet S, Gouraud S, Gobin R, et al. Aquaporin-2 downregulation in kidney medulla of aging rats is posttranscriptional and is abolished by water deprivation. *Am J Physiol Renal Physiol*. 2008;294:F1408–F1414.
281. Beck LH. The aging kidney. Defending a delicate balance of fluid and electrolytes. *Geriatrics*. 2000;55:26–28, 31–32.
282. Fried LF, Palevsky PM. Hyponatremia and hypernatremia. *Med Clin North Am*. 1997;81:585–609.
283. Faull CM, Holmes C, Baylis PH. Water balance in elderly people: is there a deficiency of vasopressin? *Age Ageing*. 1993;22:114–120.
284. Helderman JH, Vestal RE, Rowe JW, et al. The response of arginine vasopressin to intravenous ethanol and hypertonic saline in man: the impact of aging. *J Gerontol*. 1978;33:39–47.
285. Phillips PA, Rolls BJ, Ledingham JG, et al. Reduced thirst after water deprivation in healthy elderly men. *N Engl J Med*. 1984;311: 753–759.
286. Duggan J, Kilfeather S, Lightman SL, et al. The association of age with plasma arginine vasopressin and plasma osmolality. *Age Ageing*. 1993;22:332–336.
287. Davies I, O'Neill PA, McLean KA, et al. Age-associated alterations in thirst and arginine vasopressin in response to a water or sodium load. *Age Ageing*. 1995;24:151–159.
288. Abramow M, Beauwens R, Cogan E. Cellular events in vasopressin action. *Kidney Int Suppl*. 1987;21:S56–S66.
289. Catudioc-Vallero J, Sands JM, Klein JD, et al. Effect of age and testosterone on the vasopressin and aquaporin responses to dehydration in Fischer 344/Brown-Norway F1 rats. *J Gerontol A Biol Sci Med Sci*. 2000;55:B26–B34.
290. Combet S, Geffroy N, Berthonaud V, et al. Correction of age-related polyuria by dDAVP: molecular analysis of aquaporins and urea transporters. *Am J Physiol Renal Physiol*. 2003;284:F199–F208.
291. Phillips PA, Bretherton M, Johnston CI, et al. Reduced osmotic thirst in healthy elderly men. *Am J Physiol*. 1991;261:R166–R171.
292. Mack GW, Weseman CA, Langhans GW, et al. Body fluid balance in dehydrated healthy older men: thirst and renal osmoregulation. *J Appl Physiol*. 1994;76:1615–1623.
293. Crowe MJ, Forsling ML, Rolls BJ, et al. Altered water excretion in healthy elderly men. *Age Ageing*. 1987;16(5):285–293. PubMed PMID: 3687569.
294. Jung FF, Kennefick TM, Ingelfinger JR, et al. Down-regulation of the intrarenal renin-angiotensin system in the aging rat. *J Am Soc Nephrol*. 1995;5(8):1573–1580. PubMed PMID: 7756590.
295. Conti S, Cassis P, Benigni A. Aging and the renin-angiotensin system. *Hypertension*. 2012;60(4):878–883. doi:10.1161/HYPERTENSIONAHA.110.155895. [Epub 2012 Aug 27]; Review. No abstract available. PMID: 22926952.
296. Nanba K, Vaidya A, Rainey WE. Aging and adrenal aldosterone production. *Hypertension*. 2018;71(2):218–223. doi:10.1161/HYPERTENSIONAHA.117.10391. [Epub 2017 Dec 11]; Review. PubMed PMID: 29229745. PubMed Central PMCID: PMC5839673.
297. Mulkerrin E, Epstein FH, Clark BA. Aldosterone responses to hyperkalemia in healthy elderly humans. *J Am Soc Nephrol*. 1995;6(5):1459–1462. PubMed PMID: 8589323.
298. Ershler WB, Sheng S, McKelvey J, et al. Serum erythropoietin and aging: a longitudinal analysis. *J Am Geriatr Soc*. 2005;53(8):1360–1365. PubMed PMID: 16078962.
299. Guo W, Li M, Bhasin S. Testosterone supplementation improves anemia in aging male mice. *J Gerontol A Biol Sci Med Sci*. 2014;69(5): 505–513. doi:10.1093/gerona/glt127. [Epub 2013 Aug 23]; PubMed PMID: 23974081. PubMed Central PMCID: PMC3991143.
300. Cappellini MD, Motta I. Anemia in clinical practice-definition and classification: does hemoglobin change with aging? *Semin Hematol*. 2015;52(4):261–269. doi:10.1053/j.seminhematol.2015.07.006. [Epub 2015 Jul 17]; Review. PubMed PMID: 26404438.
301. Garimella PS, Katz R, Patel KV, et al. Association of serum erythropoietin with cardiovascular events, kidney function decline, and mortality: the health aging and body composition study. *Circ Heart Fail*. 2016;9(1):e002124. doi:10.1161/CIRCHEARTFAILURE.115.002124. PubMed PMID: 26721912. PubMed Central PMCID: PMC4698899.
302. National Kidney Foundation. K/DOQI clinical practice guidelines for chronic kidney disease: evaluation, classification and stratification. *Am J Kidney Dis*. 2002;39:S1–S266.
303. KDIGO 2012 Clinical practice guidelines for the evaluation and management of chronic kidney disease. *Kidney Int Suppl*. 2013;3:1–150.
304. Polkinghorne KR, Chadban SJ. A decade after the KDOQI CKD guidelines: a perspective from Australia. *Am J Kidney Dis*. 2012;60(5):725–726. doi:10.1053/j.ajkd.2012.08.005. PubMed PMID: 23067642.
305. National Clinical Guideline Centre (UK). *Chronic Kidney Disease (Partial Update): Early Identification and Management of Chronic Kidney Disease in Adults in Primary and Secondary Care*. London: National Institute for Health and Care Excellence (UK); 2014. PubMed PMID: 25340245.
306. Delanaye P, Glassock RJ, Pottel H, et al. An age-calibrated definition of chronic kidney disease: rationale and benefits. *Clin Biochem Rev*. 2016;37(1):17–26. PubMed PMID: 27057075. PubMed Central PMCID: PMC4810758.
307. Glassock R, Delanaye P, El Nahas M. An age-calibrated classification of chronic kidney disease. *JAMA*. 2015;314(6):559–560. doi:10.1001/jama.2015.6731. PubMed PMID: 26023760.
308. Levey AS, Inker LA, Coresh J. Chronic kidney disease in older people. *JAMA*. 2015;314(6):557–558. doi:10.1001/jama.2015.6753. PubMed PMID: 26023868.
309. Winearls CG, Glassock RJ. Classification of chronic kidney disease in the elderly: pitfalls and errors. *Nephron Clin Pract*. 2011;119(suppl 1):c2–c4. doi:10.1159/000328013. [Epub 2011 Aug 10]; Review. PubMed PMID: 21832853.
310. Winearls CG, Glassock RJ. Dissecting and refining the staging of chronic kidney disease. *Kidney Int*. 2009;75(10):1009–1014. doi:10.1038/ki.2009.49. [Epub 2009 Feb 25]; PubMed PMID: 19242501.
311. Moynihan R, Glassock R, Doust J. Chronic kidney disease controversy: how expanding definitions are unnecessarily labelling many people as diseased. *BMJ*. 2013;347:f4298. doi:10.1136/bmj.f4298. PubMed PMID: 23900313.
312. Ebert N, Jakob O, Gaedeke J, et al. Prevalence of reduced kidney function and albuminuria in older adults: the Berlin Initiative Study. *Nephrol Dial Transplant*. 2017;32(6):997–1005. doi:10.1093/ndt/gfw079. PubMed PMID: 27190381.
313. Okparavero A, Foster MC, Tighiouart H, et al. Prevalence and complications of chronic kidney disease in a representative elderly population in Iceland. *Nephrol Dial Transplant*. 2016;31(3):439–447. doi:10.1093/ndt/gfv370. [Epub 2015 Oct 31]; PubMed PMID: 26519958. PubMed Central PMCID: PMC4762399.
314. Schaeffner ES, Ebert N, Delanaye P, et al. Two novel equations to estimate kidney function in persons aged 70 years or older. *Ann Intern Med*. 2012;157(7):471–481. doi:10.7326/0003-4819-157-7-201210020-00003. PubMed PMID: 23027318.
315. Glassock RJ, Warnock DG, Delanaye P. The global burden of chronic kidney disease: estimates, variability and pitfalls. *Nat Rev Nephrol*. 2017;13(2):104–114. doi:10.1038/nrneph.2016.163. [Epub 2016 Dec 12]; Review. PubMed PMID: 27941934.
316. Rule AD, Glassock RJ. GFR estimating equations: getting closer to the truth? *Clin J Am Soc Nephrol*. 2013;8(8):1414–1420. doi:10.2215/CJN.01240213. [Epub 2013 May 23]; PubMed PMID: 23704300. PubMed Central PMCID: PMC3731897.
317. Pottel H, Delanaye P, Schaeffner E, et al. Estimating glomerular filtration rate for the full age spectrum from serum creatinine and cystatin C. *Nephrol Dial Transplant*. 2017;32(3):497–507. doi:10.1093/ndt/gfw425. PubMed PMID: 28089986. PubMed Central PMCID: PMC5837496.
318. Evangelopoulos AA, Vallianou NG, Bountziouka V, et al. Association between serum cystatin C, monocytes and other inflammatory markers. *Intern Med J*. 2012;42(5):517–522. doi:10.1111/j.1445-5994.2011.02500.x. PubMed PMID: 21470355.
319. Liu X, Foster MC, Tighiouart H, et al. Non-GFR determinants of low-molecular-weight serum protein filtration markers in CKD. *Am J Kidney Dis*. 2016;68(6):892–900. doi:10.1053/j.ajkd.2016.07.021. [Epub 2016 Sep 20]; PubMed PMID: 27663042. PubMed Central PMCID: PMC5123901.
320. Foster MC, Levey AS, Inker LA, et al. Non-GFR determinants of low-molecular-weight serum protein filtration markers in the elderly: AGES-Kidney and MESA-Kidney. *Am J Kidney Dis*. 2017;70(3):406–414. doi:10.1053/j.ajkd.2017.03.021. [Epub 2017 May 24]; PubMed PMID: 28549536. PubMed Central PMCID: PMC5572311.
321. Inker LA, Schmid CH, Tighiouart H, et al. Estimating glomerular filtration rate from serum creatinine and cystatin C. *N Engl J Med*. 2012;367(1):20–29. doi:10.1056/NEJMoa1114248. Erratum in: *N Engl J Med*. 2012;367(7):681. Erratum in: *N Engl J Med*. 2012;367(21):2060. PubMed PMID: 22762315. PubMed Central PMCID: PMC4398023.

322. Björk J, Grubb A, Gudnason V, et al. Comparison of glomerular filtration rate estimating equations derived from creatinine and cystatin C: validation in the Age, Gene/Environment Susceptibility-Reykjavik elderly cohort. *Nephrol Dial Transplant*. 2017;doi:10.1093/ndt/gfx272. [Epub ahead of print]; PubMed PMID: 29040701. PubMed Central PMCID: PMC6070032.
323. Meeusen JW, Rule AD, Voskoboev N, et al. Performance of cystatin C- and creatinine-based estimated glomerular filtration rate equations depends on patient characteristics. *Clin Chem*. 2015;61(10):1265–1272. doi:10.1373/clinchem.2015.243030. [Epub 2015 Aug 3]; PubMed PMID: 26240296. PubMed Central PMCID: PMC4726452.
324. Fan L, Levey AS, Gudnason V, et al. Comparing GFR estimating equations using cystatin C and creatinine in elderly individuals. *J Am Soc Nephrol*. 2015;26(8):1982–1989.
325. Nerpin E, Ingelsson E, Risérus U, et al. The combined contribution of albuminuria and glomerular filtration rate to the prediction of cardiovascular mortality in elderly men. *Nephrol Dial Transplant*. 2011;26(9):2820–2827. doi:10.1093/ndt/gfq848. [Epub 2011 Feb 18]; PubMed PMID: 21335440.
326. Grubb A, Lindström V, Jonsson M, et al. Reduction in glomerular pore size is not restricted to pregnant women. Evidence for a new syndrome: 'Shrunken pore syndrome'. *Scand J Clin Lab Invest*. 2015;75(4):333–340. doi:10.3109/00365513.2015.1025427. PubMed PMID: 25919022. PubMed Central PMCID: PMC4487590.
327. Lang J, Katz R, Ix JH, et al. Association of serum albumin levels with kidney function decline and incident chronic kidney disease in elders. *Nephrol Dial Transplant*. 2018;33(6):986–992. doi:10.1093/ndt/gfx229. PubMed PMID: 28992097.
328. Raman M, Middleton RJ, Kalra PA, et al. Estimating renal function in old people: an in-depth review. *Int Urol Nephrol*. 2017;49(11):1979–1988. doi:10.1007/s11255-017-1682-z. [Epub 2017 Sep 15]; Review. PubMed PMID: 28913589. PubMed Central PMCID: PMC5643354.
329. Delanaye P, Glassock RJ, De Broe ME. Epidemiology of chronic kidney disease: think (at least) twice! *Clin Kidney J*. 2017;10(3):370–374. doi:10.1093/ckj/sfw154. [Epub 2017 Feb 27]; PubMed PMID: 28617483. PubMed Central PMCID: PMC5466090.
330. Hallan SI, Matsushita K, Sang Y, et al. Age and association of kidney measures with mortality and end-stage renal disease. *JAMA*. 2012;308(22):2349–2360. PubMed PMID: 23111824. PubMed Central PMCID: PMC3936348.
331. Denic A, Glassock RJ, Rule AD. Structural and functional changes with the aging kidney. *Adv Chronic Kidney Dis*. 2016;23(1):19–28. doi:10.1053/j.ackd.2015.08.004. Review. PubMed PMID: 26709059. PubMed Central PMCID: PMC4693148.
332. Gansevoort RT, Correa-Rotter R, Hemmelgarn BR, et al. Chronic kidney disease and cardiovascular risk: epidemiology, mechanisms, and prevention. *Lancet*. 2013;382(9889):339–352. doi:10.1016/S0140-6736(13)60595-4. [Epub 2013 May 31]; Review. PubMed PMID: 23727170.
333. Malmgren L, McGuigan FE, Berglundh S, et al. Declining estimated glomerular filtration rate and its association with mortality and comorbidity over 10 years in elderly women. *Nephron*. 2015;130(4):245–255. doi:10.1159/000435790. [Epub 2015 Jul 15]; PubMed PMID: 26184510.
334. Delanaye P, Glassock RJ. Glomerular filtration rate and aging: another longitudinal Study–A long time coming! *Nephron*. 2015;131(1):1–4. doi:10.1159/000439147. [Epub 2015 Aug 21]; PubMed PMID: 26303019.
335. Rutkowski M, Mann W, Derose S, et al. Implementing KDOQI CKD definition and staging guidelines in Southern California Kaiser Permanente. *Am J Kidney Dis*. 2009;53(3 suppl 3):S86–S99. doi:10.1053/j.ajkd.2008.07.052. PubMed PMID: 19231766.
336. Warnock DG, Delanaye P, Glassock RJ. Risks for all-cause mortality: stratified by age, estimated glomerular filtration rate and albuminuria. *Nephron*. 2017;136(4):292–297. doi:10.1159/000455197. [Epub 2017 Jan 27]; Review. PubMed PMID: 28125814.
337. Goff DC Jr, Lloyd-Jones DM, Bennett G, et al. 2013 ACC/AHA guideline on the assessment of cardiovascular risk: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines. *J Am Coll Cardiol*. 2014;63(25 Pt B):2935–2959. doi:10.1016/j.jacc.2013.11.005. [Epub 2013 Nov 12]; Erratum in: *J Am Coll Cardiol*. 2014;63(25 Pt B):3026. PubMed PMID: 24239921. PubMed Central PMCID: PMC4700825.
338. O'Hare AM, Choi AI, Bertenthal D, et al. Age affects outcomes in chronic kidney disease. *J Am Soc Nephrol*. 2007;18(10):2758–2765. [Epub 2007 Sep 12]; PubMed PMID: 17855638.
339. Nichols LA, Slusarz A, Grunz-Borgmann EA, et al. α(E)-catenin regulates BMP-7 expression and migration in renal epithelial cells. *Am J Nephrol*. 2014;39(5):409–417. doi:10.1159/000362250. [Epub 2014 May 6]; PubMed PMID: 24818804. PubMed Central PMCID: PMC4086150.
340. Eriksen BO, Ingebretsen OC. The progression of chronic kidney disease: a 10-year population-based study of the effects of gender and age. *Kidney Int*. 2006;69(2):375–382. PubMed PMID: 16408129.
341. Shardlow A, McIntyre NJ, Fluck RJ, et al. Chronic kidney disease in primary care: outcomes after five years in a prospective cohort study. *PLoS Med*. 2016;13(9):e1002128. doi:10.1371/journal.pmed.1002128. eCollection 2016 Sep. PubMed PMID: 27648564. PubMed Central PMCID: PMC5029805.
342. Remuzzi G, Glassock R. Interplay between diagnostic criteria and prognostic accuracy in chronic kidney disease. *PLoS Med*. 2016;13(9):e1002129. doi:10.1371/journal.pmed.1002129. eCollection 2016 Sep. PubMed PMID: 27648826. PubMed Central PMCID: PMC5029803.
343. Rosner MH. Acute kidney injury in the elderly. *Clin Geriatr Med*. 2013;29(3):565–578. doi:10.1016/j.cger.2013.05.001. Review. PubMed PMID: 23849008.
344. Ponticelli C, Sala G, Glassock RJ. Drug management in the elderly adult with chronic kidney disease: a review for the primary care physician. *Mayo Clin Proc*. 2015;90(5):633–645. doi:10.1016/j.mayocp.2015.01.016. [Epub 2015 Mar 12]; Review. PubMed PMID: 25771152.
345. Glassock RJ. Glomerular disease in the elderly. *Clin Geriatr Med*. 2009;25(3):413–422. doi:10.1016/j.cger.2009.06.006. Review. PubMed PMID: 19765489.
346. Jin B, Zeng C, Ge Y, et al. The spectrum of biopsy-proven kidney diseases in elderly Chinese patients. *Nephrol Dial Transplant*. 2014;29(12):2251–2259. doi:10.1093/ndt/gfu239. [Epub 2014 Jul 17]; PubMed PMID: 25034755.
347. Perkowska-Ptasinska A, Deborska-Materkowska D, Bartczak A, et al. Kidney disease in the elderly: biopsy-based data from 14 renal centers in Poland. *BMC Nephrol*. 2016;17(1):194. PubMed PMID: 27884116. PubMed Central PMCID: PMC5123353.
348. Kirkland JL. The biology of senescence: potential for prevention of disease. *Clin Geriatr Med*. 2002;18(3):383–405. Review. PubMed PMID: 12424865.
349. Westendorp RG. What is healthy aging in the 21st century? *Am J Clin Nutr*. 2006;83(2):404S–409S. Review. PubMed PMID: 16470003.
350. Mungenast AE. Diacylglycerol signaling underlies astrocytic ATP release. *Neural Plast*. 2011;2011:537659. doi:10.1155/2011/537659. [Epub 2011 Jun 13]; PubMed PMID: 21826278. PubMed Central PMCID: PMC3151491.
351. Kanasaki K, Kitada M, Koya D. Pathophysiology of the aging kidney and therapeutic interventions. *Hypertens Res*. 2012;35(12):1121–1128. doi:10.1038/hr.2012.159. [Epub 2012 Oct 18]; Review. PubMed PMID: 23076406.
352. de Cavanagh EM, Inserra F, Ferder L. Angiotensin II blockade: how its molecular targets may signal to mitochondria and slow aging. Coincidences with calorie restriction and mTOR inhibition. *Am J Physiol Heart Circ Physiol*. 2015;309(1):H15–H44. doi:10.1152/ajpheart.00459.2014. [Epub 2015 May 1]; Review. PubMed PMID: 25934099.
353. Mazucanti CH, Cabral-Costa JV, Vasconcelos AR, et al. Longevity pathways (mTOR, SIRT, Insulin/IGF-1) as key modulatory targets on aging and neurodegeneration. *Curr Top Med Chem*. 2015;15(21):2116–2138. Review. PubMed PMID: 26059361.
354. Miles JM, Rule AD, Borlaug BA. Use of metformin in diseases of aging. *Curr Diab Rep*. 2014;14(6):490. doi:10.1007/s11892-014-0490-4. Review. PubMed PMID: 24752835. PubMed Central PMCID: PMC4209240.
355. Baar MP, Brandt RMC, Putavet DA, et al. Targeted Apoptosis of Senescent Cells Restores Tissue Homeostasis in Response to Chemotoxicity and Aging. *Cell*. 2017;169(1):132–147.e16. doi:10.1016/j.cell.2017.02.031. PubMed PMID: 28340339. PubMed Central PMCID: PMC5556182.

356. Krimpenfort P, Berns A. Rejuvenation by therapeutic elimination of senescent cells. *Cell*. 2017;169(1):3–5. doi:10.1016/j.cell.2017.03.014. PubMed PMID: 28340347.
357. Park SJ, Ahmad F, Philp A, et al. Resveratrol ameliorates aging-related metabolic phenotypes by inhibiting cAMP phosphodiesterases. *Cell*. 2012;148(3):421–433. doi:10.1016/j.cell.2012.01.017. PubMed PMID: 22304913. PubMed Central PMCID: PMC3431801.
358. Cufí S, Vazquez-Martin A, Oliveras-Ferraros C, et al. Metformin against TGFβ-induced epithelial-to-mesenchymal transition (EMT): from cancer stem cells to aging-associated fibrosis. *Cell Cycle*. 2010;9(22):4461–4468. [Epub 2010 Nov 15]; PubMed PMID: 21088486.

BOARD REVIEW QUESTIONS

1. You have been asked to see a man 78 years of age who has an estimated glomerular filtration rate (eGFR) of 45 mL/min/1.73 m^2 with a question about the need to start an angiotensin-converting enzyme (ACE) inhibitor. He has essentially been healthy apart from having two inguinal hernia repairs, and he has mild symptoms of prostatism with nocturia twice a night. Otherwise he feels well. Electrolyte levels are within normal limits, with a potassium level of 5.0 mmol/L. His urinary albumin-to-creatinine ratio is 12 mg/g of creatinine. Which outcomes are most likely in this patient?
   a. He will progress to end-stage kidney disease (ESKD) within the next 5 years.
   b. He will progress to ESKD within the next 10 years.
   c. He will die before the development of ESKD.
   d. Kidney function will remain stable.
   e. Kidney function will improve.

   Answer: c and d

   Rationale: In the absence of albuminuria, it is unlikely this patient has significant kidney disease. The GFR is low, most likely as a result of the loss of glomeruli with age. An ACE inhibitor in this patient is not indicated for his kidney function.

2. A 72-year-old woman is admitted after a fall. She was found 24 hours later at home. It is noted that her serum sodium level is 156 mmol/L. Her estimated GFR is 48 mL/min/1.73 m^2. She is on no medication and lives alone. Which age-related factors may have contributed to the hypernatremia?
   a. Reduced thirst
   b. Reduced urinary concentrating capacity
   c. Reduced total body water
   d. Increased vasopressin secretion
   e. High salt intake

   Answer: a, b, c

   Rationale: All these factors predispose to hypernatremia in older adults. Older kidneys have a reduced capacity to conserve water and solutes. Total body water is reduced (replaced with greater fat mass), and therefore older adults are more prone to volume depletion. Older adults have a reduced thirst response to increased osmolality and therefore are less likely to be able to correct hypernatremia spontaneously.

3. A 76-year-old woman wishes to donate a kidney to her 46-year-old son and is coming to you for a second opinion, given that she was declined as a donor elsewhere because her GFR is below 60 mL/min/1.73 m^2. Her body mass index (BMI) is 21. Her eGFR, calculated by the CKD-EPI formula, is 57 mL/min/1.73 m^2. She has no hypertension or diabetes, is a nonsmoker, and is a vegetarian. She exercises regularly, taking long walks three times a week with a seniors group. She is adamant that she wants to donate a kidney to her son—she sees him suffering on dialysis and wants to do everything possible to help him. Which of the following may assist in assessment of this woman as a potential donor?
   a. Determination of a cystatin C GFR
   b. Measurement of her GFR before and after a protein load (renal functional reserve)
   c. Determination of 24-hour urinary albumin excretion
   d. A renal biopsy showing the presence of 10% focal and segmental glomerulosclerosis
   e. Determination of kidney size by ultrasound

   Answer: All of the above

   Rationale: Given her relatively low BMI, use of a cystatin C GFR may assist in determining the accuracy of the creatinine-based eGFR value. Given that she is a vegetarian, the GFR may be lower than expected. Although not routinely used, detection of the capacity of her kidneys to respond to a protein load may indicate preservation of renal reserve. Any evidence of albuminuria or the presence of focal segmental glomerulosclerosis (FSGS) on the biopsy would indicate a primary renal pathology and likely exclude the woman from donation. Similarly significant renal asymmetry or small kidneys would indicate potential renal ischemia or renal disease and would be exclusion factors for donation.